Complex Relativity from P1 and P2

Complex Spacetime, the Common Null Cone, and Physical Dynamics

Charles K

2 November 2026

Abstract

Complex Relativity (CR) begins with a complex spacetime, a physical real slice, and a causal null cone. Its two physical postulates specify how observers and momenta relate to this geometry. P1 requires equivalence of local inertial descriptions and compatibility of every real null direction with the full complex interval. For an induced metric $\gamma = g + \mathrm{i}b$, this condition implies $b = \chi g$ and hence $\gamma = Fg$, where $F = 1 + \mathrm{i}\chi$. P2 supplies the real free-particle mass shell and its complex representation. With the natural momentum extension $P = p^{\mathbb{C}}$, Scheme A gives $C_A = F^{-1}C_{\mathrm{R}}$, a single regular real constraint. The canonical one-form pulls back to $p_\mu \mathrm{d}x^\mu$, yielding real geodesic motion in g.

The postulates admit explicit free scalar, Maxwell, and Dirac constructions, but leave interacting actions underdetermined. A restricted complex Einstein–Hilbert action produces a canonical geometric scalar $\phi = f \operatorname{arsinh} \chi$. Coupling this scalar to the response matrix G_c separates positive, critical, and negative kinetic sectors. We derive the exact target-space curvature for one normal sector and for several sectors sharing ϕ, then relate this geometry to mixed interactions, scattering, and dark-particle decays. Internal bundles and chiral representations provide candidate electroweak and strong sectors; the $c = 4$ response supplies a conditional dark-sector model. Their joint field equations, cosmological limits, and stability requirements are examined alongside the earlier orientation and microscopic constructions.

The final part concerns quantum structure and physical observation. It follows the construction of environments, records, composite systems, and internal clocks, distinguishing classical symplectic dynamics from a quantum algebra of measurable quantities. In particular, Hilbert-space representations, geometric phases, and measurement backreaction do not by themselves establish quantum state positivity or a universal action scale. The synthesis presents CR as a candidate fundamental unified theory, with explicit derivations, model assumptions, counterexamples, and unresolved problems kept distinct.

Keywords: P1/P2 v3.5; complex spacetime; common null cone; natural complexification of momentum; real symplectic structure; unified mass shell; response sectors; internal gauge structure; quantum positivity; physical records.

Contents

1 Introduction and Conventions 4

2 Complex Spacetime and the Null Cone 5

3 P1: Observers, Inertial Equivalence, and Null-Cone Compatibility 7

4 Flat Complex Geometry and Inertial Kinematics 8

5 Null-Cone Rigidity and Clock Calibration 9

6 P2: Real Momentum and the Complex Free Mass Shell 11

7 Mass-Shell Rigidity and Its Assumptions 12

8 Real Canonical Dynamics in Scheme A 14

9 Scheme B and Free-Particle Equivalence 15

10 Free Fields and the Variational Principle 16

11 Nonuniqueness of the Free-Field Couplings 18

12 Maxwell, Dirac, and Classical Field Phase Space 18

13 From the Postulates to Interacting Dynamics 20

14 Restricted Curvature and the Geometric Scalar 21

15 The Einstein–Scalar Core 22

16 The Response Matrix and the Parameter c 24

17 Target-Space Curvature of Normal Response 25

18 The Critical Sector $c = 1$ 27

19 Parent Actions, Rank Locking, and Quantum Corrections 30

20 Internal Bundles and Gauge Structure 31

21 Geometric Origins of the Internal Bundle 34

22 Composite Electromagnetic Fields and Maxwell Dynamics 35

23 The $c = 2$ Response and Electroweak Theory 37

24 The $c = 3$ Response and Color Dynamics 39

25 Chiral Matter and Dirac Structure 41

26 The Coupled Action and Field Equations 45

27 Mixed Curvature, Scattering, and Decay 48

28 The $c = 4$ Dark Sector 49

29 Shared-Mediator Interactions and Dark Stability 52

30 Characteristic Cones and the Critical Exception 53

31 Dark Cosmology and Background Stability 54

32 Bounces, Plateau Potentials, and Four-Form Cosmology **58**

33 The Earlier Orientation Model **60**

34 Microscopic Input, Higgs Production, and Vacuum Response **62**

35 Composite-Particle Spectra and Continuum Tests **64**

36 Six-Point Gravitational Amplitudes and Three-Body Kernels **67**

37 Quantum Effective Theory and Model Selection **70**

38 Pauli, Clifford, Dirac, and Schrödinger Structures **72**

39 Complex Fibers and Projective Reduction **73**

40 Probability Measures and First-Record Statistics **74**

41 Continuous Records and the Two-Channel Environment **76**

42 Operational States and Statistical Closure **79**

43 Physical Records and Latching Dynamics **81**

44 Nonlinear Feedback and Moment Closure **83**

45 Controlled Low-Energy and Long-Time Limits **84**

46 Parity, Dynamic References, and Readout **86**

47 Composite Systems, Contextuality, and Bell Bounds **87**

48 Self-Gravitating Clocks and Relational Readings **88**

49 Quantum Positivity, Action Scales, and Loop Phases **91**

50 Identifying the Physical Observable Algebra **93**

51 Conclusions and Open Problems **94**

A An Optimal Bound for Approximate Null-Cone Compatibility **97**

B Two Closure Theorems **98**

C Decay Rates and Sign Conventions **98**

D Source Map **99**

E Conventions, Revisions, and Computational Checks **100**

F Development of P1/P2 through v3.5 **101**

G Index of the Hundred Research Rounds in H **102**

H Notation and Model Assumptions **107**

Direct Sources **108**

1 Introduction and Conventions

1.1 Sources and scope

This article brings together the geometric, dynamical, and quantum branches of CR theory under the P1/P2 formulation of v3.5. Scheme A provides the momentum and reality conventions throughout. Four manuscripts supply the technical material.[U, R, V, H]

Code	Manuscript	Contribution to the synthesis
U	*Unified Geometry and the c Sectors*	Complex geometry, gravity, response spectra, internal $3+2$ structure, gauge and chiral matter, the dark universe, microscopic spectra, and amplitudes.
R	*Integrated Research Synthesis*, 2026-11-01	QB/QP; MST I–XXV; XXVI–XXXII; the null-cone study; and the statistics, records, and internal-observer work of XXXIII–XXXVIII.
V	*P1/P2 v3.5 Review*, 2026-11-02	A focused account of the new postulates, Schemes A/B, reality constraints, and free-field actions.
H	*P1/P2 One-Hundred-Round Comprehensive Report*, 2026-11-02	Evolution of the postulates, the hundred-round topic index, constructive results and counterexamples, normalized statements, and criteria for further research.

U and R collect the earlier geometric and statistical work; V and H revise the postulates and develop the free-field construction. Historical results are attributed through these sources. The screenshots and hundred rounds described in H are available here through H's account, rather than as independently examined conversation records. Its rounds 1–100, U's round indices, and R's Roman-numeral MST sequence are separate series.

1.2 Postulate conventions

In the original formulation, P1 concerns inertial equivalence preserving the complex interval, physical real slice, and time orientation; P2 concerns the complex free-particle mass shell. Some intervening syntheses used those names for "complex spacetime and real readout" and "the common null cone." We follow V and H in restoring the distinction: complex spacetime and the null cone are foundations, while P1 and P2 state physical compatibility requirements.

Version 3.5 also strengthens the original formulation. It specifies clock-carrying observers, compatibility of all real null directions, natural momentum extension, and real canonical dynamics. The derivations below use these explicit assumptions; they are not consequences of the weaker original statements alone.

1.3 Logical structure

Level	Content	Logical role
Mathematical and physical foundations	Complex spacetime and the real slice; causal null cones and clock/rod scales	Define the objects, reality conditions, and causal geometry.

Level	Content	Logical role
Basic physical postulates	P1: observers, inertial equivalence, common cone. P2: free mass shell, natural momentum, real symplectic structure	State the fundamental compatibility relations.
Rigorous consequences	$\gamma = Fg$; $C_A = F^{-1}C_{\mathrm{R}}$; one regular real constraint	Derive mathematical conclusions from explicit premises.
Dynamical realizations	Real-part actions, restricted curvature, G_c, internal bundles, and interactions	Supply calculable models that can be compared.
Physical and quantum tests	Propagation, spectra, cosmology, internal apparatus, operational states, and quantum positivity	Test whether models realize the objectives and agree with observation.

CR's proposed fundamental status is a research objective. The models developed here admit tree-level calculations, effective operators, and low-energy expansions; a high-energy definition, nonperturbative existence, and experimental confirmation remain open.

1.4 Notation and reality conventions

The spacetime signature is $(-+++)$; the speed of light is c_{light}, the response parameter is c, and cosmic time is τ. Natural units are used except where units are explicitly restored. Writing $\hbar = 1$ is a convention and does not imply that P1/P2 have derived a quantum action scale. We use

$$F = 1 + \mathrm{i}\chi, \quad \rho = |F| = \sqrt{1+\chi^2}, \quad \phi = f \operatorname{arsinh}\chi, \quad x = \phi/f, \quad M = mc_{\mathrm{light}}. \tag{1.1}$$

The gravitational scalar ϕ and reference matter scalar φ are different fields. Energy densities are written ρ_S, ρ_ϕ, and a candidate quantum density matrix is ϱ. We reserve G for the complex spacetime metric, G_c for the response kinetic matrix, P for complex momentum, and p for physical real cotangent momentum. Historical labels P1-c/P1-d refer to stages of the parent-theory construction, not clauses of the present P1.

2 Complex Spacetime and the Null Cone

2.1 Complex spacetime, a maximally real slice, and physical dimension

Let $\mathcal{Z}$ be a four-complex-dimensional manifold and G a nondegenerate holomorphic symmetric complex bilinear metric. Physical spacetime is carried by a four-real-dimensional maximally real embedding $\iota : M \hookrightarrow \mathcal{Z}$. At each point require

$$\mathrm{d}\iota(T_xM) \cap J\mathrm{d}\iota(T_xM) = \{0\}, \qquad T_xM \otimes_{\mathbb{R}} \mathbb{C} \simeq T^{1,0}_{\iota(x)}\mathcal{Z}. \tag{2.1}$$

Natural complex-linear extension gives the second isomorphism and fixes the momentum extension used in Scheme A. Although the underlying real manifold of $\mathcal{Z}$ has eight dimensions, physical events, clocks, and readouts lie on the four-dimensional slice M.

The pulled-back metric is

$$\gamma = \iota^* G = g + \mathrm{i}b, \qquad g = \Re\gamma, \tag{2.2}$$

where g is a time-oriented Lorentz metric and b a real symmetric tensor. The bilinear form G contains no complex conjugation and is distinct from a positive-definite Hermitian length. At this

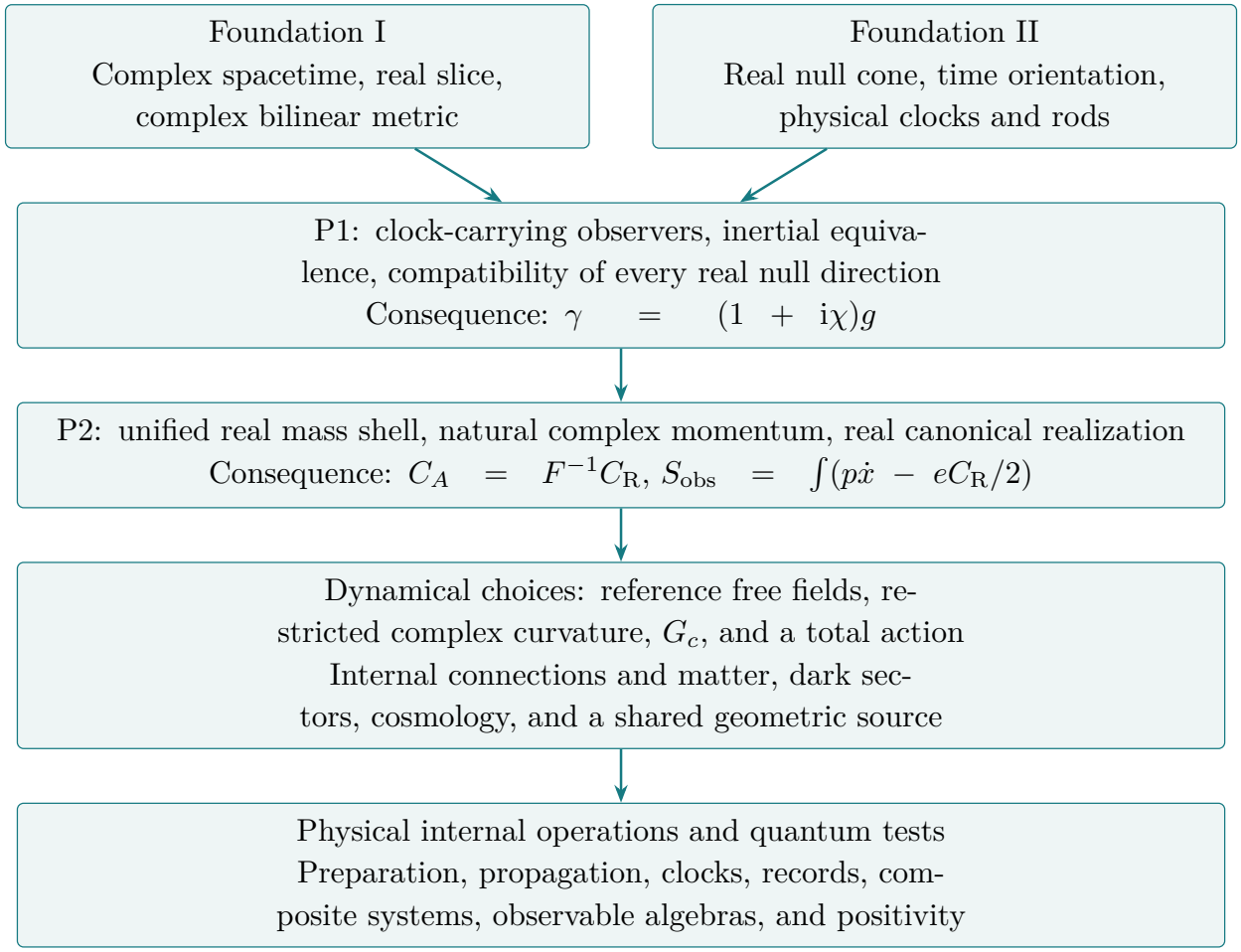


Figure 1: Dependence of the constructions on the foundations and postulates. The actions and interactions in the fourth row require additional dynamical choices.

stage the definition does not require b to be proportional to g; that restriction will follow from P1.2.

The maximally real slice is not required to be the fixed set of an isometric antiholomorphic involution. That stronger condition would make the pullback real and force $b = 0$. Allowing $\chi \neq 0$ instead requires the real readout $g = \Re\gamma$ used here.

2.2 Analytic extension, local existence, and global questions

Real-analytic g and χ can be complexified locally in real-analytic coordinates. Restricting the neighborhood and keeping the determinant nonzero then gives a compatible local holomorphic metric. General smooth functions need not admit such an extension. Obtaining a smooth solution of effective equations on the real manifold and constructing a global complex spacetime for every such solution are therefore different tasks.

Global extension requires compatible charts, topology, volume branches, spin structures, and a global real slice. The pointwise identities below do not depend on resolving all of these questions; statements about global complex geometry do.

2.3 Null cones, time orientation, and clock scale

Define $\mathcal{N}_x(g) = \{v \neq 0 : g(v,v) = 0\}$ in the real tangent space; time orientation selects its future half. The causal null cone determines the conformal class of a Lorentz metric, while a physical clock fixes the timelike scale through

$$\mathrm{d}\tau^2 = -c_{\mathrm{light}}^{-2} g_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu \tag{2.3}$$

A null curve has no nonzero proper time. A massless particle must not be assigned the same unit timelike four-velocity normalization as a massive particle.

P1.2 compares null directions in the real tangent space: physical signal directions must also annihilate the full complex interval. The larger null variety in the complex tangent space has no such direct signal interpretation.

3 P1: Observers, Inertial Equivalence, and Null-Cone Compatibility

3.1 Statement of P1

P1.1 Observation and inertial comparison. Physical observations are recorded by clock-carrying observers on real timelike worldlines. Event relations are established through the emission, reception, and directional identification of basic signals and comparison with local inertial frames. Observers may accelerate; an instantaneously comoving local inertial frame can be chosen at every worldline event. Basic free kinematics has the same form in Lorentz inertial frames preserving time and spatial orientation.

P1.2 Compatibility of real causality with the full complex interval. A sufficiently rich family of basic massless signals and clock/rod calibrations specifies the real Lorentz null cone and scale. For all real null directions at every event, require

$$\boxed{g(v,v) = 0 \implies \gamma(v,v) = 0, \qquad v \in T_x M \setminus \{0\}.} \tag{3.1}$$

This is an explicitly added compatibility principle connecting real causality and complex geometry.

P1.3 Locality and the free inertial limit. Free motion is straight in flat inertial coordinates. In curved spacetime, equivalence is implemented locally. Transport between points and full evolution are defined by real dynamics compatible with this structure.

These clauses distinguish an observer's physical worldline from the local inertial frame used to describe it. An observer may accelerate, and signal measurements require clock and rod calibration as well as a cone. The compatibility of that cone with the complex metric is the additional content of P1.2.

3.2 Synchronization conventions and operational data

Radar time requires a clock worldline, round-trip signals involving nearby events, and directional information. A single number does not measure a four-dimensional metric. Even with fixed two-way light speed, a synchronization convention can express one-way coordinate speeds in one dimension as

$$c_+ = \frac{c_{\mathrm{light}}}{1+\epsilon}, \qquad c_- = \frac{c_{\mathrm{light}}}{1-\epsilon}, \qquad |\epsilon| < 1, \tag{3.2}$$

while the round-trip time over length L is $L/c_+ + L/c_- = 2L/c_{\text{light}}$. One-way coordinate speeds therefore depend on synchronization. They neither establish physical anisotropy nor determine the independent components of b; the complex-interval condition remains a separate premise.

3.3 Form covariance and active symmetry of a fixed background

For $b = \varepsilon n \otimes n$, the tensor can transform covariantly under coordinate changes, but a fixed n generally breaks full active Lorentz symmetry. Preserving the form of equations under passive transformations differs from invariance of a given background under active boosts. If the latter is used to prove conformal rigidity, it must be stated as a strengthened premise. This report instead uses P1.2 directly.

3.4 Common geometry as a test of the complete dynamics

Each interacting model must realize P1's causal geometry in its propagating degrees of freedom. This requires a principal-symbol calculation after constraints and gauge freedom have been treated. Background media and higher derivatives can alter the result. We perform this check for the normal response sectors below and treat the degenerate critical sector separately.

An illustrative example uses $X = g^{\mu\nu}\partial_\mu\zeta\partial_\nu\zeta$ and

$$\mathcal{L} = -\frac{1}{2}X + \frac{\alpha}{4\Lambda^4}X^2. \tag{3.3}$$

On a locally flat background with constant timelike gradient, small perturbations obey

$$c_s^2 = \frac{1+a}{1+3a}, \qquad a = \frac{\alpha\dot{\zeta}^2}{\Lambda^4}. \tag{3.4}$$

For $-1/3 < a < 0$, both local kinetic coefficients are positive, yet the perturbation equations give $c_s > 1$. General covariance thus permits effective cones different from the metric cone. In an effective theory this low-frequency speed also need not equal the signal-front speed of a high-frequency completion; the causal test must specify its regime.

4 Flat Complex Geometry and Inertial Kinematics

4.1 Complex bilinear interval and real structure

The flat starting point takes the complex bilinear form $G_{AB} = \delta_{AB}$ on $\mathbb{C}^4$, rather than a positive-definite Hermitian length. Choose an antilinear real structure so that the physical slice is written as

$$z^A = (x^1, x^2, x^3, \mathrm{i}c_{\text{light}}t), \qquad D = \mathrm{diag}(1,1,1,\mathrm{i}), \qquad D^T G D = \mathrm{diag}(1,1,1,-1). \tag{4.1}$$

On this slice, $\mathrm{d}\Sigma^2 = \mathrm{d}\boldsymbol{x}^2 - c_{\text{light}}^2\mathrm{d}t^2$. The Lorentz real form is a choice within the complex bilinear space, which also admits other real forms.

If B preserves G, the physical real structure, and time orientation, then $\Lambda = D^{-1}BD$ is a real Lorentz transformation. This establishes the flat inertial model. Curved extensions require the local and analytic assumptions discussed in Section 2.

4.2 The original free shell in the reference vacuum

In this $\chi = 0$ reference model, momentum can be written as $P = (\boldsymbol{p}, \mathrm{i}E/c_{\text{light}})$, so $P_A\mathrm{d}z^A = \boldsymbol{p}\cdot\mathrm{d}\boldsymbol{x} - E\mathrm{d}t$ is the real canonical one-form. The original free mass shell gives

$$G^{AB}P_AP_B + m^2c_{\text{light}}^2 = 0 \quad \Longleftrightarrow \quad E^2 = c_{\text{light}}^2|\boldsymbol{p}|^2 + m^2c_{\text{light}}^4. \tag{4.2}$$

This relation includes both $m = 0$ and $m > 0$ free particles. Extending it to nonconstant F while preserving real momentum and the same physical mass is the Scheme A problem of Sections 6–8. Complex coordinates in this reference model place the time component last; elsewhere the report uses the time-first $(-+++)$ ordering. Reordering indices relates the two conventions without changing the physical signature.

5 Null-Cone Rigidity and Clock Calibration

5.1 Clock-and-signal data

Set $c_{\text{light}} = 1$ in this subsection. At a point, choose a future unit timelike vector u and let $H_u = \{s : g(u, s) = 0\}$. If $n \in H_u$ and $g(n, n) = 1$, then $u + n$ is future null. Thus

$$\mathcal{N}_p^+(g) = \{\alpha(u + n) : \alpha > 0,\ n \in H_u,\ g(n, n) = 1\}. \tag{5.1}$$

The observer's complete sphere of spatial directions therefore parameterizes the future null cone. This statement presupposes a Lorentz form, or equivalent clock-and-signal data; a clock reading alone supplies neither.

Emission and reception times τ_-, τ_+ on the same clock define radar time and distance by

$$T = \frac{\tau_+ + \tau_-}{2}, \qquad R = \frac{c_{\text{light}}}{2}(\tau_+ - \tau_-). \tag{5.2}$$

Full reconstruction still requires directional control, signals to sufficiently nearby events, regular propagation, and a suitable neighborhood; multiple light paths and synchronization conventions must also be distinguished. Within a fixed Lorentz model class, identical complete null cones imply that two metrics are positively conformally related. Applying the same clock normalization to the same nonzero timelike tangent fixes the ratio to one. At a single point this gives

$$10_{\text{real symmetric components}} = 9_{\text{conformal cone data}} + 1_{\text{clock scale}}. \tag{5.3}$$

The uniqueness is pointwise. Extending the calibration through spacetime also requires transport and signal-propagation assumptions.

5.2 Rigidity proof for the common null cone

Let $\gamma = g + \mathrm{i}b$, with b real symmetric and g a nondegenerate Lorentz metric. P1.2 supplies the compatibility condition

$$g(v, v) = 0 \implies \gamma(v, v) = 0 \quad \text{for every nonzero real tangent vector } v. \tag{5.4}$$

In a frame with $g = \mathrm{diag}(-1, 1, 1, 1)$, take $v = (1, \boldsymbol{n})$ and $|\boldsymbol{n}| = 1$, giving $b_{00} + 2b_{0i}n_i + b_{ij}n_in_j = 0$. Comparing $\boldsymbol{n}$ and $-\boldsymbol{n}$ gives $b_{0i} = 0$. The remaining spatial quadratic form is constant on the unit sphere, hence $b_{ij} = -b_{00}\delta_{ij}$. Therefore

$$\boxed{b = \chi g, \qquad \gamma = (1 + \mathrm{i}\chi)g, \qquad \chi = \frac{1}{4}g^{\mu\nu}b_{\mu\nu}.} \tag{5.5}$$

One-way inclusion of the real null cone suffices, even when b is degenerate. Since χ is real, F never vanishes. The trace formula also makes χ smooth whenever g and b are smooth. Its value may vary or vanish.

5.3 Nine angular multipoles and finite-direction tests

Let $B = (b_{ij})$ and define

$$\mathscr{F}(\boldsymbol{n}) = b_{00} + 2b_{0i}n_i + B_{ij}n_in_j = m + 2a_in_i + Q_{ij}n_in_j, \tag{5.6}$$

$$m = b_{00} + \tfrac{1}{3}\operatorname{Tr} B, \qquad a_i = b_{0i}, \qquad Q = B - \tfrac{1}{3}(\operatorname{Tr} B)I. \tag{5.7}$$

The monopole, dipole, and traceless quadrupole have $1, 3, 5$ coefficients, respectively. Writing the spherical average as $\langle\cdot\rangle$ gives

$$m = \langle\mathscr{F}\rangle, \quad a_i = \frac{3}{2}\langle n_i\mathscr{F}\rangle, \quad Q_{ij} = \frac{15}{2}\left\langle (n_in_j - \delta_{ij}/3)\mathscr{F}\right\rangle. \tag{5.8}$$

For $E = b - \chi g$, the components are

$$E_{00} = \frac{3}{4}m, \qquad E_{0i} = a_i, \qquad E_{ij} = Q_{ij} + \frac{1}{4}m\delta_{ij}. \tag{5.9}$$

Thus $\mathscr{F} \equiv 0$ is equivalent to $E = 0$, and the shared scalar χ disappears completely from the null-direction diagnostic. The six directions $(1, \pm e_i)$ and three directions $(1, (e_i + e_j)/\sqrt{2})$ give a rank-nine linear system whose kernel is precisely the span of g. This is the minimum number of independent finite-dimensional constraints when each direction supplies one scalar condition; it does not require nine observers.

For a fixed observer normalization, the source gives the coarse bound $\|E\|_F \le \sqrt{23}\,\varepsilon/2$. The synthesis in R strengthens it to

$$\sup_{|\boldsymbol{n}|=1} |\mathscr{F}(\boldsymbol{n})| \le \varepsilon \quad \Longrightarrow \quad \boxed{\|b - \chi g\|_F \le \frac{\sqrt{11}}{2}\varepsilon.} \tag{5.10}$$

The constant is attained; Appendix A gives the full proof. The norm is the positive-definite component Frobenius norm, so the numerical upper bound is not an observational bound invariant under arbitrary boosts.

5.4 Full complex inertial equivalence and its assumptions

Another sufficient condition requires transformations preserving g, γ at the same point to carry a unit observer into an open neighborhood on the velocity hyperboloid. With $u(v) = (1, v)/\sqrt{1 - |v|^2}$, invariance of the full complex interval gives

$$2b_{0i}v_i + (b_{ij} + b_{00}\delta_{ij})v_iv_j = 0. \tag{5.11}$$

Vanishing of this polynomial on an open set implies Eq. (5.5). Invariance under infinitesimal boosts in three independent spatial directions, $K_i^T b + bK_i = 0$, is also sufficient. The pointwise scalar $\chi(x)$ may vary; the complete field solution need not possess global Lorentz symmetry.

General coordinate covariance differs from active invariance of a fixed background. Covariance permits $b_{\mu\nu} = \chi g_{\mu\nu} + \beta\ell^2\nabla_\mu\chi\nabla_\nu\chi$, and a flat limit does not exclude this nonconformal term. More directly, if matter couples only to $g = \operatorname{diag}(-1, 1, 1, 1)$ while $b = \operatorname{diag}(\alpha, 0, 0, 0)$, all real clock-and-light behavior can remain standard even though $\gamma(1, n; 1, n) = \mathrm{i}\alpha$. A common cone for ordinary matter therefore does not establish a common cone for an independent complex metric.

> P1.2 is the link between measured real geometry and the full complex interval. The algebraic consequence is the conformal form (5.5); its realization by interacting signals is a dynamical question.

Null-cone priority on a continuous manifold must also be distinguished from the stronger pregeometric proposal that causal relations precede the event set. The latter still needs definitions of the primitive relation, dimension, topology, volume, and continuum limit. The sources discuss EPS and causal-set precedents to delimit historical originality and the scope of the present proofs.

6 P2: Real Momentum and the Complex Free Mass Shell

6.1 Statement of P2

P2.1 Real free mass shell. Each free-particle species has a real nonnegative invariant mass $m \geq 0$. Its actual momentum $p \in T_x^*M$ lies on the physical energy branch selected by time orientation and satisfies

$$\boxed{C_{\mathrm{R}}(x,p;m) = g^{\mu\nu}p_\mu p_\nu + m^2c_{\mathrm{light}}^2 = 0.} \tag{6.1}$$

P2.2 Natural complexification. Complex momentum is the natural complex-linear extension $P = p^{\mathbb{C}}$ determined by the maximally real slice. Pairing it with real displacements directly recovers real momentum, without an additional momentum real form rotating with χ.

P2.3 Reference limit and compatibility of the full complex shell. The original complex mass shell $G^{-1}(P,P) + m^2c_{\mathrm{light}}^2 = 0$ is recovered in the selected flat reference vacuum with $\chi = 0$. On general complex geometry, the quadratic free mass shell must be compatible with the same real mass shell and must not add a second independent condition on real momentum directions. The complex mass term is a compatibility coefficient in this geometric representation, not an additional independently measurable complex mass.

P2.4 Real dynamics. Free worldlines are generated by the canonical symplectic structure of real cotangent phase space and a real representative of the constraint. Massive particles admit proper-time normalization; massless particles use an appropriate affine or gauge parameter. The free mass shell alone does not fix complete field operators, interactions, or quantum states.

The reference vacuum fixes both $g = \eta$ and $\chi = 0$. Flat real geometry alone also permits $\chi = \chi_0 \neq 0$, where the compatible complex mass coefficient is still needed.

6.2 Existence and uniqueness of natural complexification; observer energy

Maximal reality means that every complex tangent vector is uniquely a complex-linear combination of real tangent vectors. For $p \in T_x^*M$, there is a unique complex-linear covector $p^{\mathbb{C}}$ satisfying

$$p^{\mathbb{C}}(\mathrm{d}\iota(v)) = p(v), \qquad v \in T_xM. \tag{6.2}$$

The embedding fixes this extension without a factor of $\sqrt{F}$. An observer with four-velocity u measures $E = -p_\mu u^\mu$; the real pairing and chosen time orientation select the positive-energy branch. This definition applies to every observer, unlike the real part of a particular coordinate component.

From $\gamma = Fg$, the natural identification gives

$$G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) = F^{-1}g^{-1}(p,p). \tag{6.3}$$

Consequently, within the class of momentum-independent quadratic mass coefficients, $C = G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + A(x)M^2$, vanishing on the complete positive-mass real shell immediately gives $A = F^{-1}$, hence

$$\boxed{C_A = G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + F^{-1}M^2 = F^{-1}C_{\mathrm{R}}.} \tag{6.4}$$

Uniqueness holds within the stated quadratic class with momentum-independent mass coefficient. Nonquadratic constraint functions and complete actions remain unrestricted. At $m = 0$, the representation still holds, but the shell fixes neither the unused mass coefficient nor the constraint's nonzero overall normalization.

6.3 Constraint rank

Since $F^{-1} = (1 - \mathrm{i}\chi)/(1 + \chi^2)$,

$$\Re C_A = \frac{C_{\mathrm{R}}}{1+\chi^2}, \qquad \Im C_A = -\frac{\chi C_{\mathrm{R}}}{1+\chi^2} = -\chi \Re C_A. \tag{6.5}$$

On-shell differentiation leaves the gradients of the real and imaginary parts collinear, and $\Re F^{-1} > 0$. Wherever $\mathrm{d}C_{\mathrm{R}} \neq 0$, the constraint map has real rank one. The positive-mass shell is regular everywhere; for the massless shell the vertex $p = 0$ must be excluded. Mechanically counting the complex equation as two independent real constraints would remove too many degrees of freedom.

Off shell, $\mathrm{d}(\Im C_A)$ includes $C_{\mathrm{R}}\mathrm{d}\chi$; this term vanishes on the constraint surface. The first-class mass-shell constraint and reparameterization gauge remove two phase-space dimensions, leaving six physical dimensions from the original eight-dimensional real cotangent space.

6.4 The obstruction for an unmodified natural complex mass term

If natural momentum $P = p^{\mathbb{C}}$ is retained while $+M^2$ is used unchanged for arbitrary F, then

$$G^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + M^2 = 0 \quad \Longleftrightarrow \quad g^{-1}(p,p) + FM^2 = 0. \tag{6.6}$$

For real p, m, the imaginary part requires $\chi M^2 = 0$. Thus this unmodified prescription is incompatible with positive mass wherever $\chi \neq 0$. Scheme A resolves the obstruction by changing the mass coefficient; Scheme B changes the momentum real form.

7 Mass-Shell Rigidity and Its Assumptions

7.1 A complete positive-mass shell determines the complex quadratic form

Fix a point and set $h = \gamma^{-1}$, $q_g(p) = g^{-1}(p,p)$. Suppose $M > 0$ and there is a complex number A, independent of momentum direction on the shell, such that the entire future real mass shell satisfies

$$q_g(p) = -M^2 \quad \Longrightarrow \quad h(p,p) = -AM^2. \tag{7.1}$$

The quadratic form $K = h - Ag^{-1}$ therefore vanishes on that shell. Any future timelike covector r can be rescaled by a positive real number onto it. Homogeneity implies $K(r,r) = 0$ throughout the open future timelike cone. A quadratic polynomial vanishing on an open set has every coefficient zero. Hence

$$h = Ag^{-1}, \qquad \gamma = A^{-1}g. \tag{7.2}$$

Nondegeneracy of h gives $A \neq 0$, and $\Re\gamma = g$ then gives $A^{-1} = 1 + \mathrm{i}\chi$. The conclusion also holds if only an open velocity patch of the shell is known within the same exact quadratic model: radial rescaling still produces a four-dimensional open set.

We call this sufficient condition **Scheme M**. It requires a complete positive-mass shell, a fixed quadratic form, and a direction-independent coefficient. It offers an alternative to the P1.2 derivation of conformality, rather than a consequence of the weaker original P2.

7.2 An exact test using ten mass-shell samples

A symmetric quadratic form in four dimensions has ten independent components. In units of M, choose one rest covector, six positive and negative axis momenta, and three momenta along coordinate-plane diagonals. More generally, fix $r > 0$ and take

$$p^{(0)} = (-M, 0, 0, 0), \quad p^{(\pm i)} = (-\sqrt{M^2 + r^2}, \pm r e_i), \quad p^{(ij)} = (-\sqrt{M^2 + 2r^2}, r(e_i + e_j)),\ i < j. \tag{7.3}$$

If $K(p, p) = 0$ at these ten points, the rest point gives $K^{00} = 0$; differences of the positive/negative axis equations give $K^{0i} = 0$, and their sums give $K^{ii} = 0$. The final three points give $K^{ij} = 0$. This proves that the sampling matrix has rank ten without treating a floating-point rank estimate as a proof.

This is an exact quadratic test. With noisy data, one must specify the sampling scale, condition number, and propagated errors; finitely many measurements do not exclude arbitrary nonquadratic models. The extra constraint relative to the nine-direction cone test fixes the quadratic normalization.

7.3 Massless-orbit bridge N and species consistency

Suppose one future real null direction is also null for the full complex interval, and additionally require the corresponding set to be closed under the full active Lorentz orbit. That orbit covers every future null direction, recovering P1.2. This is **Scheme N**. Full active-orbit closure is an additional premise and cannot be replaced by ordinary coordinate covariance.

For species sharing the same geometric quadratic form, the assumptions of M force a common A through Eq. (7.2). Modified dispersion and interacting effective masses fall outside this free quadratic argument.

7.4 A nonconformal inverse-metric counterexample

Take a g-null covector ℓ and set

$$\gamma = g + \mathrm{i}\varepsilon \ell \otimes \ell. \tag{7.4}$$

Since $g^{-1}(\ell, \ell) = 0$, quadratic and higher terms in the inverse-matrix update vanish, giving the exact result

$$\gamma^{-1} = g^{-1} - \mathrm{i}\varepsilon \ell^\sharp \otimes \ell^\sharp. \tag{7.5}$$

This satisfies both $\Re\gamma = g$ and $\Re\gamma^{-1} = g^{-1}$, yet $b = \varepsilon \ell \otimes \ell$ is generally not proportional to g. For most real null directions v, $\gamma(v, v) = \mathrm{i}\varepsilon \ell(v)^2 \neq 0$. Correct real parts of both a metric and its inverse therefore do not prove compatibility of the full complex null cone.

7.5 Direction-dependent compensation and off-shell nonuniqueness

If the mass coefficient may depend arbitrarily on on-shell direction, define

$$A(p) = 1 + \frac{q_g(p) - h(p, p)}{M^2} \tag{7.6}$$

Then $h(p,p) + A(p)M^2 = q_g(p) + M^2$ identically, compensating any quadratic form. Direction independence is therefore substantive in shell rigidity and cannot be omitted. It is needed only on shell; off-shell extension remains free.

Even for a completely fixed shell, $\widetilde{C} = C\exp(\beta C/\Lambda_C^2)$ has the same zero set as C, with Λ_C^2 carrying the same dimension as C. For real nonzero h, $X_{hC}|_{C=0} = hX_C|_{C=0}$, usually changing only the gauge parameter. A complex h requires an explicit real representative and multiplier reality conditions; a complex Hamiltonian flow cannot directly be interpreted as a real reparameterization.

Similarly, defining a complex phase $S_C = F(x)S_R$ does not generally prove that momentum is merely multiplied by F, because

$$\mathrm{d}S_C = F\mathrm{d}S_R + S_R\mathrm{d}F. \tag{7.7}$$

The extra term changes under an additive constant in S_R. Multiplying the action phase by $F(x)$ therefore does not implement natural momentum extension.

8 Real Canonical Dynamics in Scheme A

8.1 Direct pullback of the canonical one-form

The canonical one-form on the complex cotangent bundle is $\Theta_{\mathbb{C}} = P_A\mathrm{d}z^A$. Scheme A's natural embedding $j : T^*M \to T^*\mathcal{Z}$ satisfies

$$j^*\Theta_{\mathbb{C}} = p_\mu \mathrm{d}x^\mu = \Theta_{\mathrm{R}}, \qquad \omega_{\mathrm{R}} = \mathrm{d}\Theta_{\mathrm{R}} = \mathrm{d}p_\mu \wedge \mathrm{d}x^\mu. \tag{8.1}$$

The convention $\iota_{X_H}\omega_{\mathrm{R}} = -\mathrm{d}H$ gives $\dot{x} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial x$, with the usual $\{x^\mu, p_\nu\} = \delta^\mu{}_\nu$. The signs of the canonical one-form, two-form, and Hamilton equations are thus consistent.

Choosing the real constraint representative $C_{\mathrm{obs}} = \Re(FC_A) = C_{\mathrm{R}}$ gives the real free action

$$\boxed{S_{\mathrm{obs}} = \int \mathrm{d}\sigma \left[p_\mu \dot{x}^\mu - \frac{e}{2}\left(g^{\mu\nu}p_\mu p_\nu + M^2\right)\right], \quad e \in \mathbb{R}.} \tag{8.2}$$

Equivalently, set $e_{\mathbb{C}} = Fe$ in the complex expression, so that $e_{\mathbb{C}}C_A = eC_{\mathrm{R}}$. The real variables and allowed variations must be fixed before varying. Unrestricted complex variation defines a different problem and need not preserve the physical slice.

8.2 From a mass shell to physical trajectories

Variation gives

$$\dot{x}^\mu = eg^{\mu\nu}p_\nu, \qquad \dot{p}_\mu = -\frac{e}{2}\partial_\mu g^{\alpha\beta}p_\alpha p_\beta, \qquad C_{\mathrm{R}} = 0. \tag{8.3}$$

Eliminating p on a parameter patch with $e \neq 0$ gives

$$S_e = \frac{1}{2}\int \mathrm{d}\sigma \left[e^{-1}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu - eM^2\right]. \tag{8.4}$$

For $m > 0$, selecting the future timelike branch and eliminating positive e yields

$$S_m = -mc_{\mathrm{light}}\int \sqrt{-g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu} = -mc_{\mathrm{light}}^2\int \mathrm{d}\tau. \tag{8.5}$$

In the gauge $\sigma = \tau$, $p^\mu = mu^\mu$ and $e = 1/m$. For $m = 0$, retain the einbein action, obtain the null-curve constraint by varying e, and then choose an affine parameter. A massless gauge cannot be obtained by taking $m \to 0$ while continuing to use $e = 1/m$.

The trajectories are geodesics of g. At fixed g, m, Eq. (8.2) contains no χ, so reference free particles experience no direct scalar force from this action. Scalar gravitational energy, response-dependent masses, and additional portals can nevertheless affect their motion.

8.3 Real geodesics and the complex Levi–Civita connection

The conformal connections obey

$$\Gamma(\gamma)^{\mu}{}_{\nu\lambda} - \Gamma(g)^{\mu}{}_{\nu\lambda} = \frac{1}{2}\left(\delta^{\mu}_{\nu}\partial_{\lambda}\ln F + \delta^{\mu}_{\lambda}\partial_{\nu}\ln F - g_{\nu\lambda}\nabla^{\mu}\ln F\right). \tag{8.6}$$

Contracting with a tangent u, the difference is $(u\cdot \mathrm{d}\ln F)u^{\mu} - \frac{1}{2}g(u,u)\nabla^{\mu}\ln F$. For a timelike tangent there is generally a transverse component, so the two sets of geodesics cannot simply be identified. For a null tangent only the tangential term remains, formally preserving unparameterized null trajectories. Its coefficient may nevertheless be complex, so it need not be removable by a real parameter change. The physical parameter and real transport remain defined by Eq. (8.2).

8.4 Constraint representatives and measure

Using $\Re C_A$ rather than C_{R} gives the delta-function relation

$$\delta(\Re C_A) = (1+\chi^2)\delta(C_{\mathrm{R}}). \tag{8.7}$$

The two constraints define the same surface but different bare delta-function weights. A consistent gauge-fixed integral also transforms the Faddeev–Popov factor and multiplier, which can cancel this dependence. Quantum equivalence further requires compatible ordering, regularization, and functional measures.

9 Scheme B and Free-Particle Equivalence

9.1 A twisted momentum real form

Take $\lambda^2 = F$ and define $P_B = \lambda p^{\mathbb{C}}$. Then

$$G^{-1}(P_B, P_B) + M^2 = C_{\mathrm{R}}. \tag{9.1}$$

Scheme B retains the literal term $+M^2$, but actual momentum is no longer read directly from the original natural real form; it is recovered as $p = \lambda^{-1}P_B$. In a natural real frame, the new antilinear reality involution is

$$r_B(P) = \frac{\lambda}{\bar{\lambda}}\bar{P}, \qquad r_B^2 = 1, \qquad \mathrm{Fix}(r_B) = \lambda T^*_x M. \tag{9.2}$$

This specifies a complete reality condition rather than merely inserting a square root into the mass-shell formula.

Since $F = 1+\mathrm{i}\chi$ lies in the open right half-plane, its principal square root is smooth on the real slice and has $\Re\lambda > 0$. Zeros and branch obstructions may still arise on a larger complex extension.

9.2 Lorentz equivariance and calibration of the symplectic form

Between a fixed real Lorentz four-vector representation and its natural complexification, rotations first constrain a real-linear equivariant map to separate temporal and spatial scalars. Boosts then equate them, leaving only complex scalar multiplication $J = \lambda I$. Both representations must be fixed in advance. Defining $\Lambda_{\mathbb{C}} = J\Lambda J^{-1}$ instead makes equivariance tautological and excludes no J.

The original complex canonical one-form on the B slice gives

$$\Theta_{\mathbb{C}}|_B = \lambda p_{\mu}\mathrm{d}x^{\mu}, \qquad \mathrm{d}(\Theta_{\mathbb{C}}|_B) = \lambda \mathrm{d}p_{\mu}\wedge \mathrm{d}x^{\mu} + \mathrm{d}\lambda\wedge(p_{\mu}\mathrm{d}x^{\mu}). \tag{9.3}$$

Obtaining C_{R} as the mass shell therefore does not by itself recover standard real canonical dynamics. Matching A requires the explicit choice

$$\Theta_{\mathrm{phys}} = \lambda^{-1} P_{B,A} \mathrm{d}z^A, \qquad \Theta_{\mathrm{phys}}|_B = p_\mu \mathrm{d}x^\mu. \tag{9.4}$$

Here A in $P_{B,A}$ is a coordinate index. This choice calibrates the physical one-form; multiplying momentum by a position-dependent λ with positions fixed is not in itself a canonical transformation.

9.3 Transport, phase connection, and local conservation

If actual momentum is transported by the real connection, B momentum correspondingly uses

$$DP_B = (\nabla - \mathrm{d}\ln\lambda) P_B = \lambda \nabla p. \tag{9.5}$$

On the real slice, $\mathcal{A} = \mathrm{d}\ln\lambda$ is exact and $\mathrm{d}\mathcal{A} = 0$. It supplies no independent electromagnetic curvature. At a single point, $\sum P_B = \lambda \sum p$ preserves momentum conservation; comparisons between points require the specified transport.

9.4 Scope of the A/B equivalence

Item	Scheme A: main development	Scheme B: comparison
Momentum	$P = p^{\mathbb{C}}$	$P_B = \sqrt{F}\, p^{\mathbb{C}}$.
Complex mass term	$F^{-1}M^2$	M^2.
Complex constraint	$F^{-1}C_{\mathrm{R}}$	C_{R}.
Reality structure	Natural extension from the maximally real slice	A momentum real form rotating with F.
Canonical one-form	Natural pullback gives $p\mathrm{d}x$	Requires λ^{-1} calibration.
Matched free action	S_{obs}	Can be chosen as the same S_{obs}.
Unproved equivalence	Unconditional equivalence with arbitrary interactions, complete field measures, operator ordering, and quantum readout.	

With matched reality conditions and canonical forms, A and B give the same classical free-particle dynamics. We use A because it preserves the natural cotangent pairing directly. Distinguishing the schemes physically requires interactions or quantum structures beyond this free equivalence.

10 Free Fields and the Variational Principle

10.1 Mass shells and characteristic cones

In real spacetime, restoring units gives the reference scalar Klein–Gordon equation

$$\left(\Box_g - \frac{m^2 c_{\mathrm{light}}^2}{\hbar^2} \right) \varphi = 0. \tag{10.1}$$

Using the local complex WKB representation $\varphi = a \exp(\mathrm{i}S/\hbar)$ and collecting leading terms of order $\hbar^{-2}$ gives

$$g^{\mu\nu} \partial_\mu S \partial_\nu S + m^2 c_{\mathrm{light}}^2 = 0. \tag{10.2}$$

This is P2's massive Hamilton–Jacobi shell. The complex WKB form may be used as a calculation device and a real solution recovered afterward. Both the phase and the mass term must retain their explicit powers of $\hbar$ when taking the semiclassical limit.

The PDE principal symbol is instead $g^{\mu\nu}k_\mu k_\nu$ and has null characteristics. A massive wave packet's trajectory, dispersion, and wavefront therefore encode different information: the first two involve the mass shell, while the highest-order characteristic cone governs the local wavefront.

10.2 The additional constraint of a naive complex Klein–Gordon equation

In four dimensions, choose a branch compatible with the reference real volume. Then

$$\sqrt{-\gamma} = F^2\sqrt{-g}, \qquad \Box_\gamma\varphi = F^{-1}\left[\Box_g\varphi + \nabla^\mu \ln F\, \partial_\mu\varphi\right]. \tag{10.3}$$

The natural guess $(\Box_\gamma - F^{-1}m^2)\varphi = 0$ has the correct algebraic mass term but is equivalent to

$$(\Box_g - m^2)\varphi + \nabla^\mu \ln F\, \partial_\mu\varphi = 0. \tag{10.4}$$

For a real physical field φ, $\Im \mathrm{d}\ln F = \mathrm{d}\chi/(1+\chi^2)$, so the imaginary part additionally requires

$$\nabla^\mu\chi\, \partial_\mu\varphi = 0. \tag{10.5}$$

On the test background $g = \eta$, $\chi = kt$, $k \neq 0$, this requires $\partial_t\varphi = 0$ and excludes generic free oscillations. Although that background need not solve the full gravitational equations, it suffices to show that shell compatibility alone does not determine a suitable wave operator.

10.3 Exact recovery of the real free scalar action

Choose a real field φ, impose $\gamma = Fg$ first, and then define the real action

$$S_0 = -\frac{1}{2}\Re\int \mathrm{d}^4x\sqrt{-\gamma}\left[\gamma^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi + F^{-1}m^2\varphi^2\right]. \tag{10.6}$$

By Eq. (10.3), the bracket together with the volume element is exactly $F\sqrt{-g}[(\nabla\varphi)^2 + m^2\varphi^2]$. The bracket is real and $\Re F = 1$, yielding the off-shell identity

$$\boxed{S_0 = -\frac{1}{2}\int \mathrm{d}^4x\sqrt{-g}\left[(\nabla\varphi)^2 + m^2\varphi^2\right].} \tag{10.7}$$

Varying the real field gives $(\Box_g - m^2)\varphi = 0$ with no condition (10.5). Restriction to real configurations precedes variation. Varying unrestricted complex fields and then imposing both parts of the complex equation would yield a different system.

The identity is off shell. It establishes a compatible reference scalar using the four-dimensional volume weight, the mass coefficient, and real-part variation. It is one admissible action, rather than a uniqueness theorem.

10.4 Energy, conservation, and the role of the geometric scalar

The stress tensor of this reference field is

$$T_{\mu\nu} = \partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}g_{\mu\nu}\left[(\nabla\varphi)^2 + m^2\varphi^2\right]. \tag{10.8}$$

For constant m and on-shell fields, $\nabla^\mu T_{\mu\nu} = 0$. Its energy density in a local orthonormal frame is

$$T_{\hat{0}\hat{0}} = \frac{1}{2}\left[(\partial_{\hat{0}}\varphi)^2 + \sum_i(\partial_{\hat{i}}\varphi)^2 + m^2\varphi^2\right] \geq 0. \tag{10.9}$$

At fixed g, m, $\delta S_0/\delta\chi = 0$. Other terms in the total action can still give χ kinetic energy, gravitational stress, and direct matter sources through G_c, potentials, and portals.

11 Nonuniqueness of the Free-Field Couplings

11.1 A family of positive-weight actions

For $\alpha \geq 0$, let

$$h_\alpha(F) = 1 - \alpha(F-1)^2, \qquad w_\alpha(\chi) = h_\alpha(1+\mathrm{i}\chi) = 1 + \alpha\chi^2 > 0. \tag{11.1}$$

Multiplying Eq. (10.6) by h_α gives

$$S_\alpha = -\frac{1}{2}\int \sqrt{-g}\,\mathrm{d}^4x\, w_\alpha(\chi)\left[(\nabla\varphi)^2 + m^2\varphi^2\right]. \tag{11.2}$$

The equation is

$$w_\alpha(\Box_g - m^2)\varphi + \nabla^\mu w_\alpha\,\partial_\mu\varphi = 0. \tag{11.3}$$

Because $w_\alpha > 0$, the second-order principal cone is unchanged and the local kinetic energy on a fixed background remains normal. Restoring units in the WKB expansion also leaves the leading massive shell unchanged. Amplitude transport, background-gradient effects, and the source for χ nevertheless change. P1/P2 therefore do not uniquely fix the complete coupling between the reference field and geometric scalar.

11.2 Transport after field redefinition

To make the difference explicit, set $L = \ln w_\alpha$ and $q = \sqrt{w_\alpha}\,\varphi$. Direct differentiation gives

$$\left[\Box_g - m^2 - \frac{1}{2}\Box_g L - \frac{1}{4}(\nabla L)^2\right] q = 0. \tag{11.4}$$

The redefinition trades the first-order transport term for background-dependent mass terms. Observational equivalence would also require transforming the sources, probe couplings, and measured field amplitude. The new variable alone does not remove those differences.

If φ is the measured field and its equation must equal the reference free equation on every allowed χ background, then this family requires $w' = 0$. Normalization $w(0) = 1$ selects $w = 1$. This additional requirement restricts the weight family, but leaves curvature couplings, potentials, higher derivatives, and portals to be examined.

11.3 Energy exchange and the complete shared source

The explicit variational density of Eq. (11.2) with respect to the geometric variable contains

$$\frac{1}{\sqrt{-g}}\frac{\delta S_\alpha}{\delta\chi} = -\frac{1}{2}w'_\alpha(\chi)\left[(\nabla\varphi)^2 + m^2\varphi^2\right]. \tag{11.5}$$

For dynamical χ, this source accounts for energy exchange between matter and geometry; only their total stress is conserved. Curvature couplings such as $\xi R\varphi^2$, varying masses $m(\chi)$, and gauge portals must likewise be included in the shared source equation.

12 Maxwell, Dirac, and Classical Field Phase Space

12.1 Exact cancellation for the four-dimensional Maxwell reference action

For an independent real gauge potential A_μ, write $\mathcal{F}_{\mu\nu} = 2\partial_{[\mu}A_{\nu]}$ and choose

$$S_{\mathrm{Max}} = -\frac{1}{4}\Re\int \mathrm{d}^4x\sqrt{-\gamma}\,\gamma^{\mu\alpha}\gamma^{\nu\beta}\mathcal{F}_{\mu\nu}\mathcal{F}_{\alpha\beta}. \tag{12.1}$$

In four dimensions the two inverse metrics supply F^{-2} and the volume supplies F^2, giving exactly

$$S_{\mathrm{Max}} = -\frac{1}{4}\int \mathrm{d}^4x\sqrt{-g}\, g^{\mu\alpha}g^{\nu\beta}\mathcal{F}_{\mu\nu}\mathcal{F}_{\alpha\beta}. \tag{12.2}$$

Gauge reduction leaves two transverse photon polarizations with the cone of g. This construction assumes an independent one-form connection. The proposed composite field in the critical $c = 1$ sector requires its own degree-of-freedom analysis in Section 22.

A portal $Z(\chi)\mathcal{F}^2$ with $Z > 0$ can preserve the same second-order principal cone while changing coupling and transport. Non-Abelian connections admit conventional gauge kinetic terms by the same four-dimensional volume counting, but the cancellation formula does not uniquely produce the gauge group, couplings, or internal bundle.

12.2 The compatible Dirac mass

Use the consistent Clifford convention of the later chiral-matter section: $\{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}$ and $D_g = \mathrm{i}\Gamma^\mu\nabla_\mu$. On a region with a suitable local spin frame and conformal branch,

$$D_\gamma(F^{-3/4}\psi) = F^{-5/4}D_g\psi. \tag{12.3}$$

Choosing the Scheme A geometric mass $m_\gamma = mF^{-1/2}$ gives

$$\boxed{(D_\gamma - mF^{-1/2})\Psi = F^{-5/4}(D_g - m)\psi, \qquad \Psi = F^{-3/4}\psi.} \tag{12.4}$$

The map gives real mass m, consistent with the Hamilton–Jacobi shell. A constant complex-frame mass m_0 would instead give $m_0\sqrt{F}$ in the real frame. The later mass-phase analysis concerns this latter choice; Scheme A uses $m_\gamma = mF^{-1/2}$.

The local identity assumes a Spin or suitable Spin^c structure. A Hermitian Dirac action on g can be transformed with its conjugation and inner product. Spinor existence, Grassmann statistics, and quantum positivity require that additional structure; they do not follow from the operator identity.

12.3 Free-field constructions

Field	Established compatible construction	Structure still required
Real scalar	Real part of the complex action equals standard KG exactly off shell; positive energy	Reference-field choice, all extra couplings, and quantum states.
Maxwell	Exact four-dimensional conformal cancellation; common cone after gauge reduction	Independent connection, boundaries, and global gauge structure.
Dirac	Spinor weight and $mF^{-1/2}$ give a local map to real mass	Spin structure, conjugation, action, and quantum statistics.
G_c response	Calculable positive kinetic energy in the normal range; shared geometric source	Interaction model, critical reduction, and spectral interpretation.

These constructions establish compatible free theories for several spins. Selecting the observed masses, interactions, and internal representations remains a problem for the unified theory.

12.4 Field symplectic current, the global Cauchy problem, and quantum input

For two linearized solutions on the reference KG solution space, define

$$j^\mu(\delta_1\varphi, \delta_2\varphi) = \delta_1\varphi\nabla^\mu\delta_2\varphi - \delta_2\varphi\nabla^\mu\delta_1\varphi, \qquad \nabla_\mu j^\mu = 0. \tag{12.5}$$

With suitable Cauchy surfaces, boundary conditions, and decay conditions, $\Omega_\Sigma = \int_\Sigma j^\mu \mathrm{d}\Sigma_\mu$ is independent of the Cauchy surface and defines the symplectic structure of real classical field phase space. A common local null cone does not guarantee global hyperbolicity: periodically identifying the time direction in flat spacetime preserves the local cone but introduces closed timelike curves.

Quantization requires an observable algebra, positive states, probability rules, a representation, and an action scale. The MST and internal-observer studies below address these requirements. Classical symplectic structure and local energy positivity alone do not establish them, or the stability of an interacting quantum theory of gravity.

13 From the Postulates to Interacting Dynamics

13.1 Organization of the total action

A complete model built on the two foundations and P1/P2 can first be written schematically as

$$S_{\text{total}} = S_{\text{geom}}[g, \phi] + S_{\text{response}}[g, \phi, y_c] + S_{\text{gauge,fermion,Higgs}}[g, \phi, \ldots] + S_{\text{potential,portal}}. \tag{13.1}$$

All sectors share one $\phi = f \operatorname{arsinh} \chi$. A reference free field may also occur as the free limit of a matter sector; it is counted once in the total action. Environments, apparatus, and records are to be modeled by configurations of this matter, with explicit dynamics for any added bath or probe.

13.2 Free postulates and interacting response

P2 fixes the physical mass shell of free particles; it does not require every $G_c(\phi)$ interaction to vanish. On a normal branch with fixed constant background, first canonically normalize the positive-definite kinetic term and then solve the generalized mass eigenvalue problem. This defines local free mass eigenstates and their P2 shells. Mixing, geometric sources, and effective masses on varying backgrounds belong to an interacting or adiabatic approximation and require explicit conditions.

The source-free reference action S_0 and a sourced response action G_c describe different couplings. Their equations can coexist in the total model; each contributes its own variation with respect to χ.

13.3 Causal admissibility under P1 and reality admissibility under P2

Each candidate module must answer two questions. First, are its physical signals after gauge and constraint reduction compatible with P1's common causal structure? Second, are its real phase space, momentum reality conditions, and free limit compatible with P2? Normal $c = 2, 3, 4$ scalar modules allow direct checks of positive-definite target metrics and a common principal cone. Critical $c = 1$ requires background-dependent constraints, propagating degrees of freedom, and well-posedness at degeneracy surfaces to be examined.

The internal $3 + 2$ bundle, independent gauge fields, chiral matter, and Yukawa couplings introduced below form one possible realization. Four complex spacetime coordinates and integer response weights do not select it uniquely. The shared scalar does, however, produce definite mixed curvature, scattering coefficients, and energy exchange once this action is chosen.

13.4 Jacobi consistency of a variable quantum scale

P2.4 provides canonical symplectic structure for the real free system. Attempting to write a quantum scale as a field function $\kappa(\phi)$ additionally requires consistency of the entire algebra. For example, suppose

$$[q,p] = \mathrm{i}\kappa(\phi), \qquad [\phi,\Pi_\phi] = \mathrm{i}\kappa_0, \qquad [q,\Pi_\phi] = [p,\Pi_\phi] = 0, \tag{13.2}$$

and let q,p commute with ϕ, with constant κ_0. The Jacobi identity for (Π_ϕ, q, p) requires

$$\kappa_0\kappa'(\phi) = 0. \tag{13.3}$$

For $\kappa_0 \neq 0$, the assumed vanishing cross-commutators force a constant κ. A varying scale would need a different algebra or additional constraints. The scales introduced in projective reduction and measurement theory must satisfy the same consistency test.

13.5 Theory construction and low-energy calculation

The remaining sections examine concrete dynamical realizations: gravity and response, gauge and chiral matter, dark cosmology, microscopic spectra, and internal observation. Effective operators and tree amplitudes provide calculational tools within these models. Where several actions satisfy P1/P2, an additional selection principle or experiment is needed.

14 Restricted Curvature and the Geometric Scalar

14.1 Restrict configurations before variation

On configurations allowed by the mathematical foundations and strengthened P1, choose a two-derivative curvature action. P2 itself does not fix this gravitational action:

$$S_{\mathrm{geom}} = \frac{M_P^2}{2}\Re\int \mathrm{d}^4x\sqrt{-\det\gamma}\,R[\gamma], \qquad \gamma = Fg. \tag{14.1}$$

Allowed variations are $\delta\gamma = F\delta g + \mathrm{i}g\delta\chi$. On the complex volume branch continuous from $\chi = 0$,

$$\sqrt{-\det\gamma}\,R[\gamma] = \sqrt{-g}\left[FR[g] - 3\Box_g F + \frac{3}{2F}(\nabla F)^2\right]. \tag{14.2}$$

Treating compatible boundary terms and using $\Re[(\nabla F)^2/F] = -(\nabla\chi)^2/(1+\chi^2)$ gives exactly

$$S_{\mathrm{geom}} = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_P^2}{2}R - \frac{3M_P^2}{4(1+\chi^2)}(\nabla\chi)^2\right] = \int \mathrm{d}^4x\sqrt{-g}\left[\frac{M_P^2}{2}R - \frac{1}{2}(\nabla\phi)^2\right]. \tag{14.3}$$

The change of variable uses $f = \sqrt{3/2}M_P$ and is exact for the restricted action. If the real and imaginary spin-two modes were varied independently near flat space, the real-part action would instead have the schematic form $S_{\mathrm{FP}}[h] - S_{\mathrm{FP}}[b]$. The restriction removes these nonconformal imaginary tensor modes, providing a stability motivation without uniquely determining the remaining action.

14.2 Density identities and conformal information

In four dimensions,

$$\sqrt{-\det(Fg)}(Fg)^{\mu\nu} = F\sqrt{-g}\,g^{\mu\nu}, \qquad *_{Fg}\big|_{\Lambda^k} = F^{2-k}\,*_g\,. \tag{14.4}$$

Scalar and two-form kinetic terms therefore respond differently to the shared factor; the Hodge star on two-forms is conformally invariant in four dimensions. The auxiliary real modulus metric $\widehat{\gamma} = \rho g$ obeys the same formula with $F \to \rho$. It depends on complex conjugation and the reality structure and is not an independently generated set of gravitational degrees of freedom.

The complex conformal connection satisfies

$$\Gamma^{\rho}_{\mu\nu}[\gamma] - \Gamma^{\rho}_{\mu\nu}[g] = \frac{1}{2}\left(\delta^{\rho}_{\mu}\partial_{\nu}\ln F + \delta^{\rho}_{\nu}\partial_{\mu}\ln F - g_{\mu\nu}\nabla^{\rho}\ln F\right). \tag{14.5}$$

The connection difference is tangential along null directions but generally has a transverse gradient component along timelike directions. Physical clocks and free motion use the real action (8.2); the associated parameter and transport distinctions were discussed in Section 8.

15 The Einstein–Scalar Core

The restricted variational problem uses independent g, χ within Eq. (5.5). Imposing the unrestricted complex vacuum equations would add equations in deleted directions. For example, at linear order $\mathrm{Im}\, R_{\mu\nu}[\gamma] = -\partial_{\mu}\partial_{\nu}\chi - \eta_{\mu\nu}\Box\chi/2$, which need not vanish for a scalar wave satisfying $\Box\chi = 0$.[U]

Without a potential or other matter, Eq. (14.3) gives

$$M_P^2 G_{\mu\nu} = \nabla_{\mu}\phi\nabla_{\nu}\phi - \frac{1}{2}g_{\mu\nu}(\nabla\phi)^2, \qquad \Box_g\phi = 0. \tag{15.1}$$

Constant ϕ recovers vacuum GR; varying ϕ contributes canonical stress and backreaction. The exact equivalence is between the restricted complex action and Einstein gravity with a canonical scalar. The GR equations alone leave the complex factor, internal bundles, and response action undetermined.[U]

15.1 Degree-of-freedom count, energy conditions, and weak-field coupling

The ADM phase space of the restricted gravitational core contains six spatial metric components and their momenta, plus one scalar canonical pair: fourteen phase-space dimensions. Four first-class spacetime diffeomorphism constraints remove eight dimensions, leaving three local propagating configuration degrees of freedom: two tensor polarizations and one real scalar. Lapse and shift are multipliers and are not included in the fourteen. This count concerns regular, nonsingular classical backgrounds and excludes response or matter modules added later.[U]

For any null vector k^{μ}, the potential-free scalar obeys $T_{\mu\nu}k^{\mu}k^{\nu} = (k^{\mu}\partial_{\mu}\phi)^2 \geq 0$. With normalization $g = \eta + 2h/M_P$, the lowest-order interaction is

$$\mathcal{L}_{h\phi\phi} = \frac{1}{M_P}h^{\mu\nu}\left(\partial_{\mu}\phi\partial_{\nu}\phi - \frac{1}{2}\eta_{\mu\nu}(\partial\phi)^2\right), \qquad \gamma_{\mu\nu} = \eta_{\mu\nu} + \frac{2h_{\mu\nu}}{M_P} + i\sqrt{\frac{2}{3}}\frac{\phi}{M_P}\eta_{\mu\nu} + \cdots. \tag{15.2}$$

Gravitational amplitudes are organized in powers of E^2/M_P^2. The isolated core also has the shift symmetry $\phi \mapsto \phi + \text{constant}$, which forbids a potential if preserved by the quantum measure. Response terms $G_c(\phi)$ and portals can break it; their protection must be assessed in the coupled theory.

15.2 Exact plane waves and spatially homogeneous solutions

Take a Brinkmann metric and an arbitrary sufficiently smooth scalar depending on a null coordinate:

$$ds^2 = -2du\,dv + dx^2 + dy^2 + \mathcal{P}(u,x,y)du^2, \qquad \phi = \phi(u), \qquad \mathcal{P} = -\frac{\phi'(u)^2}{2M_P^2}(x^2+y^2). \tag{15.3}$$

Here $\Box\phi = 0$, $R_{uu} = \phi'^2/M_P^2$, and $R_{uxux} = R_{uyuy} = \phi'^2/(2M_P^2)$, giving exact scalar-induced tidal curvature. About constant ϕ, the scalar stress starts at second order in wave amplitude. A detector coupled minimally to g therefore has no extra linear breathing response from this argument. Background gradients or direct scalar couplings would require a new detector calculation.

A spatially flat universe containing only a free homogeneous scalar has

$$a(\tau) \propto \tau^{1/3}, \quad \phi = \phi_0 \pm \sqrt{\frac{2}{3}} M_P \ln\frac{\tau}{\tau_0}, \quad \chi = \sinh\left(\frac{\phi_0}{f} \pm \frac{2}{3}\ln\frac{\tau}{\tau_0}\right), \quad R = -\frac{2}{3\tau^2}. \tag{15.4}$$

The homogeneous kinetic solution has $w = 1$. An isotropic gas of massless particles has $w = 1/3$, cold dark matter has averaged $w \simeq 0$, and vacuum energy has $w = -1$. Obtaining the latter behaviors requires a potential, mass, or additional dynamical sector.

15.3 GR embedding, inverse reconstruction, and local sufficient conditions

In the isolated potential-free core, any Ricci-flat metric with any constant χ_0 is a solution. Conversely, a Ricci-flat real metric and $M_P^2 R_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi$ force $d\phi = 0$. With matter present, a complete GR subtheory requires all frozen-field equations to remain satisfied for every allowed matter configuration after freezing the scalar and responses. Vanishing of a source only at the Higgs vacuum, for example, does not establish a consistent GR subtheory with arbitrary Higgs excitations.[U]

Inverse reconstruction can be stated more precisely. Given a real metric, if there exists a real one-form v such that

$$M_P^2 R_{\mu\nu} = v_\mu v_\nu, \qquad dv = 0, \tag{15.5}$$

then $v = d\phi$ in a simply connected local region. The condition is a real outer-product factorization, not merely Ricci-matrix rank at most one; its sign and real factorization must also be compatible. Where $v \neq 0$, the Bianchi identity gives $(\Box\phi)\partial_\nu\phi = 0$, recovering the scalar equation. Gradient zeros, smooth matching of branches, and global first-cohomology obstructions require separate treatment.

With additional matter and a potential, subtract $T^{\mathrm{m}}_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T^{\mathrm{m}} + g_{\mu\nu}V$ from $M_P^2 R_{\mu\nu}$ before testing whether the remainder is $d\phi\otimes d\phi$, and verify all matter and scalar equations simultaneously. A constant potential $V_0 = M_P^2\Lambda$ already represents the corresponding cosmological constant and must not be counted twice in the Einstein equations. This is a conditional extension, not creation of an extra propagating scalar by pure GR itself. A classical reference truncation is also distinct from quantum heavy-field decoupling.

16 The Response Matrix and the Parameter c

16.1 A shared matrix and different values of c

On the common geometry admitted by the postulates, select the response constitutive relation

$$\mathcal{L}_c = -\frac{1}{2}\left[c\rho\,\partial_\mu\bar{\Psi}\partial^\mu\Psi + \Re(F\partial_\mu\Psi\partial^\mu\Psi)\right], \qquad \Psi = u + \mathrm{i}v. \tag{16.1}$$

This fixes the relative weights of complex bilinear and Hermitian-type response and is a dynamical choice. Expanding into two real components $y = (u, v)^T$ gives the unified kinetic matrix

$$G_c(\chi) = \begin{pmatrix} c\rho + 1 & -\chi \\ -\chi & c\rho - 1 \end{pmatrix}, \qquad \rho = \sqrt{1+\chi^2}. \tag{16.2}$$

Its traceless part squares to $\rho^2 I$, hence

$$\lambda_\pm(G_c) = (c \pm 1)\rho, \qquad \det G_c = (c^2-1)\rho^2, \qquad \mathrm{Tr}\, G_c = 2c\rho. \tag{16.3}$$

The classification requires no small-χ expansion:

Parameter range	Spectral signs	Meaning in the chosen real action
$c > 1$	Both positive	Normal positive-definite response with conventional two-component dynamics.
$c = 1$	One positive, one zero	Rank-one critical sector; the null eigen-direction requires constraint analysis.
$-1 < c < 1$	One positive, one negative	A negative-kinetic-energy direction relative to the common positive-energy convention.
$c = -1$	One negative, one zero	A negative-sign degenerate sector, distinct from the positive critical sector.
$c < -1$	Both negative	Overall negative kinetic energy; relabeling does not turn it into the same positive-energy matter.

The spectrum itself allows continuous c. Integers $c = 2, 3, 4$ are model choices emphasized in the sources. The common null cone, matrix positivity, and local Lorentz invariance do not individually reduce the continuous interval to these three integers. Nor can $c = 1$ be defined by setting a denominator in a normal-sector formula to zero; its phase space requires separate treatment.

At $c = 0$ the response has one positive and one negative eigenvalue. A pure-gravity truncation is instead specified by removing matter fields from the action. At $c = 1$, identifying the zero direction with electromagnetism would require Maxwell's constraint structure and two photon polarizations, which kinetic rank alone does not establish.

16.2 Integer carriers and the response involution

Let $J_{\mathrm{CR}} = \rho^{-1}\begin{pmatrix} 1 & -\chi \\ -\chi & -1 \end{pmatrix}$. Then

$$J_{\mathrm{CR}}^2 = I, \qquad G_c = \rho(cI + J_{\mathrm{CR}}). \tag{16.4}$$

All c therefore share the same eigendirections, while c changes their weights. For a complex rank-n carrier, the symmetric and antisymmetric parts of its tensor square have dimensions $n(n+1)/2$

and $n(n-1)/2$. For $\Phi = u\Pi_+ + v\Pi_-$, the reference point gives

$$\frac{2}{n}\,\mathrm{Tr}(\mathrm{d}\Phi)^2 = (n+1)\mathrm{d}u^2 + (n-1)\mathrm{d}v^2. \tag{16.5}$$

The construction explains the reference weights $n \pm 1$ in integer models. It leaves the ϕ-dependent off-diagonal response and the carrier's charged components unspecified. Connections and matter representations must still be supplied.

16.3 Vacuum normalization and shared-scalar vertices

Use $n = 2, 3, 4$ for the selected normal sectors and define

$$U_n = \sqrt{n+1}\,u_n, \quad V_n = \sqrt{n-1}\,v_n, \quad x = \phi/f. \tag{16.6}$$

The kinetic matrix is the identity at $\phi = 0$, while at arbitrary field values it is

$$\mathsf{G}_n = \begin{pmatrix} (n\cosh x + 1)/(n+1) & -\sinh x/\sqrt{n^2-1} \\ -\sinh x/\sqrt{n^2-1} & (n\cosh x - 1)/(n-1) \end{pmatrix}, \quad \det \mathsf{G}_n = \cosh^2 x. \tag{16.7}$$

Writing $K_n = \mathsf{G}_n^{-1}$ gives

$$K_n(0) = I, \qquad K_n'(0) = \lambda_n \sigma_x, \qquad \lambda_n = \frac{1}{f\sqrt{n^2-1}}. \tag{16.8}$$

The lowest-order coupling is

$$\mathcal{L}_{\phi UV} = \sum_n \lambda_n \phi\, \partial U_n \cdot \partial V_n, \qquad \lambda_2 : \lambda_3 : \lambda_4 = \frac{1}{\sqrt{3}} : \frac{1}{\sqrt{8}} : \frac{1}{\sqrt{15}}. \tag{16.9}$$

Using $f^2 = 3M_P^2/2$ gives, for example, $\lambda_2\lambda_3 M_P^2 = 1/(3\sqrt{6})$. This is a definite relation within the chosen action. If independent counterterms are permitted in different sectors, survival of the relation after renormalization is a stronger issue than its tree-level calculation.

The assignments of $c = 2, 3, 4$ to visible and dark sectors also require representations, masses, potentials, connections, and discrete symmetries. Neither spin nor stability follows from the two-component matrix alone. Its normalized form,

$$\frac{G_n}{\mathrm{Tr}\, G_n} = \frac{1}{2}(I + \boldsymbol{r}_n \cdot \boldsymbol{\sigma}), \qquad |\boldsymbol{r}_n| = \frac{1}{n}, \qquad \mathrm{Tr}\left(\frac{G_n}{\mathrm{Tr}\, G_n}\right)^2 = \frac{1}{2}(1 + n^{-2}) \tag{16.10}$$

lies inside the Bloch ball, with varying χ tracing a curve. It does not parameterize all pure states or specify their preparation and measurement.

17 Target-Space Curvature of Normal Response

17.1 Exact single-sector curvature

The positive response matrix defines a nonlinear sigma model for the shared scalar and a response pair. Its curvature measures interactions that cannot be removed by changing field coordinates. For one sector with $c > 1$, the target metric is

$$ds_{\mathcal{T}}^2 = d\phi^2 + (G_c)_{ij} dy^i dy^j. \tag{17.1}$$

Set $x = \phi/f$ and $A_c = G_c^{-1}G_c'$, with primes denoting ϕ derivatives. The scalar curvature of this three-dimensional field space is

$$\mathcal{R}_{\mathcal{T}} = -\operatorname{tr}(G_c^{-1}G_c'') + \frac{3}{4}\operatorname{tr}(A_c^2) - \frac{1}{4}(\operatorname{tr} A_c)^2. \tag{17.2}$$

From $\det G_c = (c^2-1)\rho^2$, $\operatorname{tr} A_c = 2\tanh x/f$. Substitution of matrix elements gives

$$\mathcal{R}_c = -\frac{3}{2f^2} - \frac{c^2}{2(c^2-1)f^2}\operatorname{sech}^2 x. \tag{17.3}$$

The curvature is negative for every real ϕ and depends on c. Hence no field redefinition can turn the full target into three free canonical scalars, although the one-dimensional geometric kinetic term alone is flattened by $\phi = f\operatorname{arsinh}\chi$. At $c = 4$, the result is $-M_P^{-2}[1 + (16/45)\operatorname{sech}^2 x]$, recovering the dark-sector expression.[U]

17.2 Additional curvature from a shared multisector base

For k normal sectors with $c_a > 1$ sharing ϕ, the target has dimension $1 + 2k$. Apply Eq. (17.2) to the full block matrix, replacing A_c by $\operatorname{diag}(A_{c_1}, \ldots, A_{c_k})$. The squared trace generates cross terms, giving

$$\mathcal{R}_{\text{all}} = \sum_{a=1}^{k} \mathcal{R}_{c_a} - \frac{k(k-1)}{f^2}\tanh^2 x. \tag{17.4}$$

or equivalently,

$$\mathcal{R}_{\text{all}} = -\frac{k(2k+1)}{2f^2} + \frac{1}{f^2}\left[k(k-1) - \frac{1}{2}\sum_{a=1}^{k}\frac{c_a^2}{c_a^2-1}\right]\operatorname{sech}^2 x. \tag{17.5}$$

For $c = 2, 3$, this gives $\mathcal{R}_{23} = -5/f^2 + [37/(48f^2)]\operatorname{sech}^2 x$. The seven-field case follows below.

Equation (17.10) remains everywhere negative. Sectors sharing a base coordinate therefore differ geometrically from an independent direct product. The critical $c = 1$ block is not everywhere invertible and cannot use this normal Riemannian curvature formula directly.

17.3 The seven-field target and its isometries

For the selected sectors $c = 2, 3, 4$, the target metric is

$$\mathrm{d}s_{\text{tar}}^2 = \mathrm{d}\phi^2 + \sum_{n=2}^{4} \mathrm{d}y_n^T G_n(\phi)\mathrm{d}y_n. \tag{17.6}$$

These seven coordinates are fields on four-dimensional spacetime. Write the six-dimensional response block as $H(\phi)$ and set $B = \frac{1}{2}H^{-1}H'$, one half of $\operatorname{diag}(A_2, A_3, A_4)$. The same curvature identity becomes

$$R_{\text{tar}} = -2\operatorname{Tr} B' - (\operatorname{Tr} B)^2 - \operatorname{Tr} B^2. \tag{17.7}$$

The traces give

$$\operatorname{Tr} B = \frac{3}{f}\tanh x, \tag{17.8}$$

$$\operatorname{Tr} B^2 = \frac{1}{2f^2}\left(3\tanh^2 x + \frac{21}{40}\operatorname{sech}^2 x\right), \tag{17.9}$$

$$R_{\mathrm{tar}} = -\frac{501 + 339\tanh^2 x}{80f^2} = -\frac{21}{2f^2} + \frac{339}{80f^2}\operatorname{sech}^2 x. \tag{17.10}$$

The scalar curvature varies with ϕ, excluding a homogeneous target with a freely translating ϕ direction. Any Killing field must preserve R_{tar} and therefore has zero ϕ component wherever $\phi \neq 0$; smoothness extends this restriction to the origin.

Normal coordinates remove the connection at one point, but leave curvature and the associated interactions. Section 27 derives the mixed Riemann components and on-shell amplitudes. The critical sector considered next has a degenerate metric and requires constraint reduction instead of this Riemannian analysis.

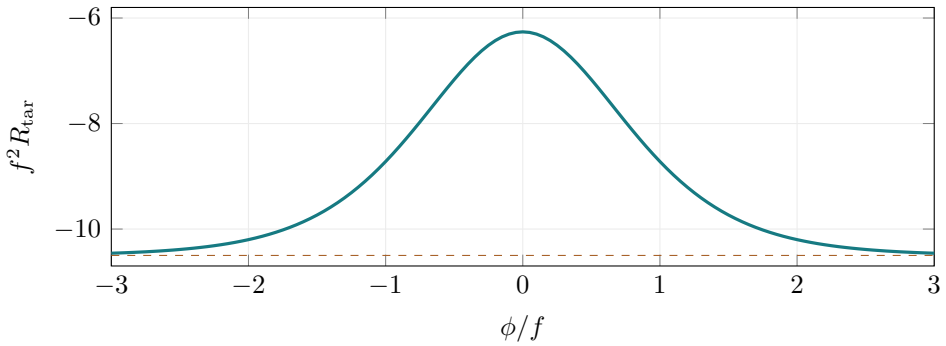


Figure 2: Scalar curvature for the shared $c = 2, 3, 4$ target, Eq. (17.10). The dashed line is the asymptote $f^2 R_{\mathrm{tar}} = -21/2$.

18 The Critical Sector $c = 1$

18.1 Rank-one kinetic energy and the regular branch

Let $a = \phi/(2f)$ and define

$$X = \cosh a\, u - \sinh a\, v, \qquad Y = \sinh a\, u - \cosh a\, v. \tag{18.1}$$

Then $w = \mathrm{d}X - Y\mathrm{d}\phi/(2f) = \cosh a\,\mathrm{d}u - \sinh a\,\mathrm{d}v$, giving exactly $\mathrm{d}y^T G_1 \mathrm{d}y = 2w^2$. The core Lagrangian of the critical sector and shared scalar is

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - w_\mu w^\mu, \qquad w = \mathrm{d}X - \frac{Y}{2f}\mathrm{d}\phi. \tag{18.2}$$

In a flat temporal decomposition, set $a = Y/(2f)$. Then

$$p_X = 2(\dot{X} - a\dot{\phi}), \quad p_\phi = \dot{\phi} - a p_X, \quad p_Y = 0. \tag{18.3}$$

Preservation of the primary constraint generates the secondary constraint

$$C_2 = N + \frac{Y}{2f}D = 0, \quad D = \frac{1}{2}p_X^2 + |\nabla\phi|^2, \quad N = \frac{1}{2}p_X p_\phi - \nabla X\cdot\nabla\phi. \tag{18.4}$$

The Poisson bracket of the two constraints is

$$\{p_Y(\boldsymbol{x}), C_2(\boldsymbol{x}')\} = -\frac{D(\boldsymbol{x})}{2f}\delta(\boldsymbol{x}-\boldsymbol{x}'). \tag{18.5}$$

For $D > 0$ the constraints are second class and give $Y = -2fN/D$. Reduction leaves two configuration degrees of freedom from the original three, including the shared scalar.

The reduced Hamiltonian density is

$$\mathcal{H}_{\rm red} = \frac{1}{2}p_\phi^2 + \frac{1}{4}p_X^2 + \frac{1}{2}|\nabla\phi|^2 + |\nabla X|^2 - \frac{N^2}{D}. \tag{18.6}$$

With $A = (p_X/\sqrt{2}, \nabla\phi)$ and $B = (p_\phi/\sqrt{2}, -\nabla X)$,

$$\mathcal{H}_{\rm red} = \frac{1}{2}|A|^2 + \left|B - \frac{A\cdot B}{|A|^2}A\right|^2 \geq 0. \tag{18.7}$$

The reduced energy is nonnegative on this branch, although the formula is singular at $D = 0$. Since $p_Y\delta Y = 0$ on the constraint surface, the reduced symplectic form is

$$\Omega_{\rm red} = \int (\delta X \wedge \delta p_X + \delta\phi \wedge \delta p_\phi)\, \mathrm{d}^3x. \tag{18.8}$$

Eliminating the spatial-gradient-dependent Y does not add a universal $\delta X\wedge\delta\phi$ term. For example, $\{\phi, Y\}_D = -fp_X D^{-1}\delta$ is state dependent; it does not derive a constant $\hbar$.

Independent variation of X, Y also gives the critical field equations directly:

$$\nabla_\mu w^\mu = 0, \qquad w^\mu\nabla_\mu\phi = 0. \tag{18.9}$$

If the other actions are independent of X, Y, then $X = X_0, Y = 0, w = 0$ is a consistent classical truncation. It supplies a calculable baseline for normal sectors without deleting critical quantum fluctuations from the full theory.

18.2 Contact structure and homogeneous reduction

On the three-dimensional field space,

$$w \wedge \mathrm{d}w = -\frac{1}{2f}\mathrm{d}X \wedge \mathrm{d}Y \wedge \mathrm{d}\phi \neq 0, \tag{18.10}$$

Thus w is a contact form. For spatially homogeneous fields and $r = p_X \neq 0$,

$$Y = -\frac{2fp_\phi}{r}, \quad \Theta = p_X\mathrm{d}X + p_\phi\mathrm{d}\phi = rw, \quad \Omega = -\mathrm{d}(rw), \quad H = \frac{r^2}{4}. \tag{18.11}$$

This is the symplectization of the contact manifold. Its Reeb vector is ∂_X, and the homogeneous dynamics gives $\dot\phi = 0$, $\dot X = r/2$. A Hilbert representation and probability rule require separate input. In particular, the source's homogeneous subsectors with $\{u_1, v_1\}_D = 0$ show that the two response components need not form a quantum canonical pair.

18.3 Ward transformations, allowed terms, and lattice tests

The following arbitrary-function transformation preserves w:

$$X \mapsto X + h(\phi), \quad Y \mapsto Y + 2fh'(\phi), \quad p_\phi \mapsto p_\phi - p_X h'(\phi). \tag{18.12}$$

The charges $Q_h = \int p_X h(\phi)$ commute, and $N \mapsto N - h'D$ preserves the constraint relation. The regular constraints remain second class, so this symmetry does not supply a local first-class gauge freedom. The alternative transformation $\delta u = \epsilon \sinh(\phi/2f)$, $\delta v = \epsilon \cosh(\phi/2f)$ gives $\delta w = \epsilon\, \mathrm{d}\phi/(2f)$ and generally fails to preserve the kinetic term.

The Ward symmetry allows terms such as

$$-\tfrac{1}{2}Z_\phi(\phi)(\partial\phi)^2 - Z_w(\phi)w^2 - Z_\times(\phi)w\cdot\partial\phi - V(\phi) \tag{18.13}$$

and the four-derivative term $(\mathrm{d}w)^2$. Low-degree generating functions test several two-derivative constraints but do not exclude counterterms at every order. The lattice edge variable

$$w_{ij} = X_j - X_i - \frac{Y_i + Y_j}{4f}(\phi_j - \phi_i) \tag{18.14}$$

preserves quadratic h exactly. For cubic $h = \phi^3$, the residual is $-\frac{1}{2}(\phi_j - \phi_i)^3$. A bipartite graph may additionally have a staggered Y zero mode. Discretization, measure, and continuous symmetry therefore require separate control.

18.4 Configuration and phase-space measures

With a flat configuration measure and Gaussian integration over Y, some lattice models give $\det(B^T B)^{-1/2}$, where $B = (4f)^{-1}\,\mathrm{diag}(\Delta\phi)$ multiplies the unsigned incidence matrix. Approaching constant ϕ along a full-rank branch and writing $\phi = \bar{\phi} + ru$, the source's radial count gives a divergence of type $r^{-2}\mathrm{d}r$. This conclusion concerns that measure and rank assumption, not every path-integral definition.

The regular Dirac phase-space measure instead contains

$$\delta(p_Y)\delta(C_2)\left|\frac{D}{2f}\right| \mathrm{d}X\,\mathrm{d}p_X\,\mathrm{d}\phi\,\mathrm{d}p_\phi\,\mathrm{d}Y\,\mathrm{d}p_Y. \tag{18.15}$$

The Jacobian from integrating Y exactly cancels the determinant, leaving the canonical Liouville measure. These procedures differ at their starting points. The latter holds only for $D > 0$ and does not determine weights of singular strata or transitions between them.

18.5 Linearization stability and singular strata

On the regular homogeneous background $X_0 = rt/2$, $\phi_0 = Y_0 = 0$, the quadratic Lagrangian is

$$\mathcal{L}^{(2)} = \dot{x}^2 - |\nabla x|^2 + \tfrac{1}{2}\dot{\varphi}^2 - \tfrac{1}{2}|\nabla\varphi|^2 - \frac{r}{2f}y\dot{\varphi}. \tag{18.16}$$

The y equation imposes $\dot{\varphi} = 0$, x is a wave degree of freedom, and y can undergo secular growth through $\dot{y} = -(2f/r)\Delta\varphi$. This differs from the apparent pair of free waves in the quadratic action near the zero background. On the entirely zero background, y first appears at cubic order, but the first nonvanishing order of the full constraint still requires

$$\dot{x}\,\dot{\varphi} - \nabla x\cdot\nabla\varphi = 0. \tag{18.17}$$

Generic counterpropagating free waves violate this condition. Consequently, a solution of the quadratic equations need not extend to a bounded analytic small-amplitude solution of the full critical system.

On the stratum $D = 0$, $p_X = 0$ and $\nabla\phi = 0$, while consistency further involves $p_\phi\Delta X - \nabla X\cdot\nabla p_\phi = 0$. On a three-torus, for example, if p_ϕ has the same sign everywhere, integration by parts

can imply $\nabla X = 0$. In the homogeneous model, the $p_X = 0$ branch may retain $H = p_\phi^2/2$, whereas every regular branch with nonzero $p_X = r$ has $H = r^2/4$. Taking $r \to 0$ at fixed nonzero p_ϕ makes Y diverge. The strata therefore cannot simply be identified with continuous energy levels of one quantum mechanics. Compactifying X, choosing a self-adjoint domain, or assigning a separate measure to a singular stratum adds information.

19 Parent Actions, Rank Locking, and Quantum Corrections

19.1 Rewriting, selection principles, and uniqueness

Charged compensators can rewrite structures such as $\rho = |F|$ in a formally covariant parent theory. For example, assign charges -2 to F, $+1$ to Ψ, and $+1$ to a unit-modulus compensator h. Then

$$Z = h^2 F, \qquad \widehat{\Psi} = h^{-1}\Psi \tag{19.1}$$

is gauge invariant, and fixing $h = 1$ recovers the original form. This establishes the existence of a Stueckelberg completion, not a unique UV completion. The same symmetry also permits

$$B_k = \frac{F^k h^{2(k-1)}}{|F|^{k-1}} \tag{19.2}$$

and other covariants and functions of real invariants. Unless an additional principle excludes them, the original response is not the symmetry's unique output.

Another proposed interpretation uses an exchange operator T. On a suitable tensor space, $\operatorname{Tr} I = n^2$ and $\operatorname{Tr} T = n$, so a normalized trace exhibits n and 1. Yet $aI + bT$ has the same exchange invariance. A general response with α_n multiplying the Hermitian term and β multiplying the real bilinear term may have spectrum $(\alpha_n \pm |\beta|)\rho$. Health imposes an inequality rather than a unique integer ratio.

19.2 Exact conditions for critical rank locking

Assume in advance that all sectors share the same real function A and complex bilinear coefficient $B = F$, with matrix structure $nAI + \text{traceless}(F)$. Rank one at $n = 1$ with a nonzero positive eigenvalue then requires

$$A = |F| = \rho. \tag{19.3}$$

Rank one therefore fixes A within the assumed common matrix ansatz. The argument does not establish universal sharing of A, preservation of rank under counterterms, or the absence of other invariants.

Curvature supplies a direct counterexample test. A parent theory constructed in the sources permits the coefficient $B_\epsilon = F(1 + \epsilon q^2)$, where $q = \Im Z/|Z|$. It preserves the vacuum matrix but changes the normal-sector curvature at the origin by

$$\Delta R_{\text{tar}}(0) = \frac{4\epsilon}{(n^2-1)f^2} + O(\epsilon^2) \tag{19.4}$$

(deforming one normal block and comparing scalar curvature of the complete target space). Similarly, the normal-matrix perturbation $\delta G_n = \eta\rho \tanh^2(\phi/f) I$ can leave the critical Ward structure untouched while changing sectional curvature at the origin. Critical symmetry therefore does not automatically exclude every independent normal-sector counterterm.

Quantum protection would have to control the higher-derivative counterterms of a four-dimensional nonlinear σ model, gravitational loops, and unequal matter couplings. Superpotential protection would not by itself fix a Kähler potential; the seven-real-dimensional target is not Kähler. The tree-level curvature remains calculable without such a protection theorem.

The heavy-scalar fluctuation examples in the historical stages P1-c/P1-d generate allowed response corrections and exhibit distinct compatible UV extensions. They are not a full one-loop calculation of CR. The density identities continue to hold even when the Wilsonian action contains further operators.

19.3 A five-dimensional parent candidate

The historical manuscripts considered a circle fiber

$$\Gamma_5 = Fg + R^2(\mathrm{d}\vartheta + A)^2, \qquad \widehat{\Gamma}_5 = \rho g + R^2(\mathrm{d}\vartheta + A)^2. \tag{19.5}$$

The horizontal density contains RF, recovering some coefficients in an enlarged model. A single circle contributes one $U(1)$, while its radius needs a field equation or stabilization. Nonzero KK modes also obey charge-selection rules in bilinear integrals. We retain this construction as a higher-dimensional parent candidate, with these additional assumptions.

19.4 A positive normal-sector deformation

H gives an explicit family useful for comparison:

$$\widetilde{G}_c = G_c + (c-1)\varepsilon\rho\tanh^2(\phi/f)I, \qquad \varepsilon \geq 0. \tag{19.6}$$

This deformation preserves the critical matrix at $c = 1$ and the vacuum kinetic term at $\phi = 0$. For $c > 1$, it stays positive definite but changes finite-background curvature. Its principal symbol is still an invertible internal matrix times $g^{\mu\nu}k_\mu k_\nu$. P1's cone and P2's free-shell requirements therefore admit this family, just as they admit the weighted free fields of Section 11.[H]

20 Internal Bundles and Gauge Structure

20.1 From a shared complex structure to internal fibers

The common complex factor F describes a restricted complexification of the spacetime metric. Internal gauge structure requires separate explicit data: a complex rank-five vector bundle E, a positive-definite Hermitian metric, a globally preserved splitting, and a trivialization of the total determinant bundle:

$$E = E_3 \oplus E_2, \qquad \operatorname{rank}_{\mathbb{C}} E_3 = 3, \quad \operatorname{rank}_{\mathbb{C}} E_2 = 2, \quad \det E \simeq \mathbf{1}. \tag{20.1}$$

The unitary transformations preserving these structures in each fiber form

$$G_{\mathrm{int}} = S\big(U(3)\times U(2)\big) \simeq \frac{SU(3)\times SU(2)\times U(1)}{\mathbb{Z}_6}. \tag{20.2}$$

This is the stabilizer of the specified internal structure. It organizes the Standard Model group, hypercharge, and matter representations in one bundle. The rank-five bundle and $3+2$ splitting are assumptions of the model, not consequences of the spacetime metric.[U]

Only splitting-preserving connections are varied. The model therefore has no off-diagonal $SU(5)$ gauge fields. Allowing the splitting to change would enlarge the field space and require a new analysis of the spectrum and interactions.

20.2 Hypercharge embedding and independent connections

Use Hermitian generators with $D = d - i\mathcal{A}$. A traceless hypercharge generator can be chosen as

$$Y_{\mathrm{hyp}} = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \qquad \mathrm{tr}\, Y_{\mathrm{hyp}}^2 = \frac{5}{6}. \tag{20.3}$$

The splitting-preserving connection is

$$\mathcal{A} = \begin{pmatrix} \mathcal{C} - \frac{1}{3}\mathcal{B} I_3 & 0 \\ 0 & \mathcal{W} + \frac{1}{2}\mathcal{B} I_2 \end{pmatrix}, \qquad \mathrm{tr}\, \mathcal{C} = \mathrm{tr}\, \mathcal{W} = 0. \tag{20.4}$$

Here $\mathcal{C}$, $\mathcal{W}$, and $\mathcal{B}$ are the color, weak-isospin, and hypercharge connections. Coupling constants are placed in the gauge kinetic coefficients rather than also inserted in the connection definition; canonically normalized gauge fields are introduced for particle masses below. Curvature is uniformly $\mathcal{F} = d\mathcal{A} - i\mathcal{A} \wedge \mathcal{A}$.

The internal algebra has $8 + 3 + 1 = 12$ generators. In the unbroken phase, with ordinary four-dimensional Yang–Mills dynamics and the usual constraint treatment, each massless generator has two physical transverse polarizations. This count relies on independent gauge connections and their dynamics, not the number of real response coordinates u_c, v_c.

20.3 Carriers for the different c sectors

Low-rank carriers compatible with the original architecture are

$$\mathcal{V}_1 = \det E_2, \qquad \mathcal{V}_2 = E_2, \qquad \mathcal{V}_3 = E_3. \tag{20.5}$$

Their complex ranks are one, two, and three, furnishing one construction relating $c = n$ to internal rank. The hypercharges of $\mathcal{V}_1$, E_2, and E_3 are respectively one, $1/2$, and $-1/3$. The same connection acts through induced representations, so these are not unrelated gauge structures.

The response coordinate $\Psi_n = u_n + iv_n$ has one complex component. A charged section of a rank-n carrier would require additional components, a representation, and a gauge-invariant contraction; $\mathrm{Re}[F(\partial\Psi)^2]$ is not invariant in an arbitrary complex representation. We take the response coordinates to be gauge singlets and introduce charged matter separately.

Rank four allows inequivalent carriers, including $E_2 \otimes E_2$ and $E_3 \oplus \det E_2$. Thus $c = 4$ fixes neither a microscopic carrier nor an $SU(4)$ force. The dark model below uses gauge-singlet response.

20.4 Gauge kinetic terms and the shared scalar

An allowed two-derivative gauge action is

$$\mathcal{L}_{\mathrm{gauge}} = -\sum_{a=Y,2,3} \frac{Z_a(\phi)}{4g_{a0}^2} \mathcal{F}^A_{a\mu\nu} \mathcal{F}_a^{A\mu\nu}, \qquad Z_a(\phi) > 0. \tag{20.6}$$

The adjoint index A is omitted for hypercharge. At fixed scalar background, the canonically normalized coupling is $g_a(\phi) = g_{a0}/\sqrt{Z_a(\phi)}$. Taking the same function for all three Z_a gives

identical fractional background dependence, but does not equate the three numerical couplings or guarantee preservation after quantum renormalization.

Minimal Yang–Mills kinetics on $\gamma = Fg$ has exact cancellation between the volume factor F^2 and the inverse metrics F^{-2}, giving constant Z_a. A choice such as $Z_a = \rho = \cosh(\phi/f)$ is an additional scalar-dependent coupling.[U]

We use $Z_a = 1$ as the baseline and retain general Z_a in the field equations. A common even function is compatible with the discrete symmetry introduced below.

20.5 The quotient group, global charges, and selectivity of the splitting

With integer-normalized hypercharge, the covering map is

$$(g_3, g_2, e^{i\alpha}) \longmapsto \mathrm{diag}(e^{-2i\alpha} g_3, e^{3i\alpha} g_2). \tag{20.7}$$

Its kernel is $\mathbb{Z}_6$. The structure group has twelve generators, consistent with $8 + 3 + 1$ local gauge-boson components. Global representations must obey the quotient condition as well. For color triality $\tau_3 \in \mathbb{Z}_3$ and weak-representation parity $s_2 \in \mathbb{Z}_2$, a convenient condition in standard hypercharge normalization is $\tau_3/3 + s_2/2 + Y \in \mathbb{Z}$. It restricts globally allowed charges without fixing the three independent dynamical couplings.[U]

A self-adjoint involution $\Gamma^2 = 1$, $\mathrm{Tr}\,\Gamma = 1$, gives $P_3 = (1 + \Gamma)/2$ and $P_2 = (1 - \Gamma)/2$. The trace condition already fixes their ranks. Among the twenty-five real parameters of a Hermitian 5×5 matrix, exact $3 + 2$ degeneracy has twelve orbit parameters and two eigenvalues: dimension fourteen, or codimension eleven. A spectral gap and $\mathrm{sign}\, K$ can stabilize such a splitting, but do not explain its selection.

For example, $V(K) = \lambda\,\mathrm{Tr}(K^2 - v^2 I)^2 + \kappa(\mathrm{Tr}\, K - v)^2$ can select a vacuum of type $\mathrm{diag}(vI_3, -vI_2)$. This is an explicit mechanism but encodes the desired rank in the potential. A fixed-rank projector cannot continuously become the zero projector to describe symmetry restoration. An amplitude variable must be added if the order parameter is to vanish.

20.6 Internal multiplicities and Lorentz spin

Complex dimensions three and two in a natural spacetime bundle do not establish color or weak isospin. For example, $\mathrm{Sym}^2 S_+ \oplus S_+$ has dimensions 3+2, but its Lorentz-commuting endomorphism algebra is $\mathbb{C} \oplus \mathbb{C}$, not $M_3(\mathbb{C}) \oplus M_2(\mathbb{C})$. Likewise, $\mathbb{C} \oplus S_+ \oplus S_-$ has only three scalar blocks in its commutant. The four-dimensional complex Clifford algebra is $M_4(\mathbb{C})$ and its even part is $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$; similar dimensions do not make this the Standard Model internal algebra.

The general decomposition

$$\mathcal{V} = \bigoplus_\alpha R_\alpha \otimes M_\alpha, \qquad \mathrm{End}_{\mathrm{Lorentz}}(\mathcal{V}) = \bigoplus_\alpha \mathrm{End}(M_\alpha) \tag{20.8}$$

shows that a genuine internal group acts on multiplicity spaces M_α of the same kinematic representation. Natural color M_3, for example, requires three copies of an identical Lorentz representation, not relabeling a spin-one representation's three components as color. A first metric jet may give a $1 + 4$ splitting, and a nonzero scalar gradient may select certain $3 + 2$ subspaces, but these constructions fail at $d\phi = 0$; spatial rotation $SO(3)$ is not internal $SU(3)$.[U]

20.7 A projective-tractor obstruction

A torsion-free projective structure in four dimensions has a natural rank-five tractor bundle, providing an $SL(5, \mathbb{C})$ geometric language but not automatically a positive-definite Hermitian metric

or $SU(5)$ connection. On the vacuum pp-wave $ds^2 = -2du\,dv + (x^2 - y^2)du^2 + dx^2 + dy^2$, a tractor-curvature component reported in the source is a nonzero nilpotent matrix with $\operatorname{rank}\mathcal{F} = 2$, $\operatorname{rank}\mathcal{F}^2 = 1$, and $\mathcal{F}^3 = 0$. If the same connection preserved positive-definite h, then $h^{1/2}\mathcal{F}h^{-1/2}$ would be anti-Hermitian. An anti-Hermitian matrix is unitarily diagonalizable and can be nilpotent only if zero, a contradiction.

The natural projective tractor connection therefore cannot preserve a positive-definite internal metric on every allowed background. Adding an inner product to construct a unitary connection introduces frame or scale choices. This projective tractor is also distinct from the orthogonal tractor of conformal geometry.[U]

21 Geometric Origins of the Internal Bundle

21.1 Self-dual bundles and Lorentz representations

With a spin structure, four-dimensional self-dual two-forms relate to $\operatorname{Sym}^2 S_+$, allowing the construction

$$E_{\text{geo}} = \operatorname{Sym}^2 S_+ \oplus S_+, \tag{21.1}$$

Its ranks are exactly $3 + 2$. The sources also use Casimir eigenvalues 2 and $3/4$ to construct $P_3 = \frac{4}{5}(C - \frac{3}{4})$ and $P_2 = \frac{4}{5}(2 - C)$, making the decomposition intrinsic. These are spacetime Lorentz spin representations. The even exterior algebra decomposes as

$$\Lambda^{\text{even}} E_{\text{geo}} \simeq \operatorname{Sym}^3 S_+ \oplus 2\operatorname{Sym}^2 S_+ \oplus 2S_+ \oplus 2I. \tag{21.2}$$

and contains spins $3/2, 1, 1/2, 0$, not sixteen copies of the same Lorentz chiral spinor carrying internal labels. Its Lorentz commutant consists only of scalars on two irreducible blocks, not $M_3(\mathbb{C}) \oplus M_2(\mathbb{C})$. If the Chern roots of S_+ are $\pm x$, those of E_{geo} are $2x, 0, -2x, x, -x$, yielding $c_1 = 0$, $c_2(E_{\text{geo}}) = 5c_2(S_+)$, and $c_3 = 0$, also displaying strong topological restrictions.

Noncompact Lorentz boosts do not preserve a positive-definite Hermitian form in a nontrivial finite-dimensional representation. For example, $\operatorname{diag}(e^{t/2}, e^{-t/2})$ cannot preserve the same positive-definite matrix for all t. Choosing a future unit time direction u defines a positive-definite spatial structure and can close $\mathfrak{so}(3) \oplus \mathrm{i}\operatorname{Sym}_0(3, \mathbb{R})$ into $\mathfrak{su}(3)$. But u is an additional geometric choice, $\nabla\phi$ need not be everywhere timelike or nonzero, and an allowed matrix algebra is not automatically an independent gauged connection.

21.2 Projectors and superconnections

U5.2 uses $P = P^\dagger$, $\operatorname{Tr} P = 3$ on a rank-five Hermitian bundle, together with

$$V_P = \lambda\mu^4 \operatorname{Tr}(P^2 - P)^2. \tag{21.3}$$

Its minima are rank-three projectors with vacuum manifold $\operatorname{Gr}_{\mathbb{C}}(3, 5)$. On the graded bundle $E \oplus E$, one can write a superconnection

$$\mathbb{A} = \nabla_A \otimes I + \mu(P - \tfrac{1}{2}I) \otimes \sigma_1, \tag{21.4}$$

whose curvature, after a specified central subtraction, contains F_A, $\mu D_A P\,\sigma_1$, and $\mu^2(P^2 - P)I$. This organizes the gauge and projector terms, but the sources establish no uniqueness theorem for the construction.

The degree-of-freedom count remains decisive: twelve ordinary massless gauge bosons carry twenty-four polarizations, whereas a metric and one scalar carry three local propagating modes. A local algebraic identification cannot supply the missing initial data.

21.3 Comparison of internal-geometry candidates

The composite field $\mathcal{F}_j = \mathrm{d}j$ obeys the Bianchi identity $\mathrm{d}\mathcal{F}_j = 0$. Its dynamical Maxwell equation requires a separate argument. Section 22 gives an exact non-Maxwell solution of the gravitational and critical equations.

Independent $SU(2)$ and $SU(3)$ connections both satisfy the same four-dimensional conformal density relation, so that relation does not select a gauge group. The sources also examine rank-five projective tractors, rank-six conformal tractors, and the finite spectral algebra $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$. The tractor ranks do not provide a positive-definite internal $SU(5)$ structure; the spectral construction requires its finite algebra and multiplicities as input.

22 Composite Electromagnetic Fields and Maxwell Dynamics

22.1 Closed field strengths, conserved response, and Maxwell equations

Defining a one-form j and setting $F_j = dj$ automatically gives $dF_j = 0$, not $\nabla_\mu F_j^{\mu\nu} = 0$ or an independently variable gauge connection. Even if the response equation gives $\nabla_\mu j^\mu = 0$, one can conclude only

$$\nabla_\mu F_j^{\mu\nu} = \Box j^\nu - R^\nu{}_\lambda j^\lambda \tag{22.1}$$

rather than set the right-hand side to zero. Naming it J^ν_{eff} defines an identically conserved effective current, but its charges, sources, and dynamics still need construction.[U]

For $c = 2$ without a potential, the two response currents are

$$\begin{aligned} j_u &= (2\cosh x + 1)du - \sinh x\, dv, \\ j_v &= -\sinh x\, du + (2\cosh x - 1)dv. \end{aligned} \tag{22.2}$$

Set $\eta_u = 2\sinh x\, u - \cosh x\, v$ and $\eta_v = -\cosh x\, u + 2\sinh x\, v$. Then

$$dj_u = f^{-1}d\phi \wedge d\eta_u, \qquad dj_v = f^{-1}d\phi \wedge d\eta_v, \qquad dw = (2f)^{-1}d\phi \wedge dY. \tag{22.3}$$

A constant-coefficient combination $\alpha w + \beta j_u + \gamma j_v$ therefore still generates a decomposable two-form, satisfying $F \wedge F = 0$. It cannot cover general electromagnetic configurations with $\boldsymbol{E}\cdot\boldsymbol{B} \neq 0$. At constant ϕ all these field strengths vanish, although critical X may still fluctuate, further distinguishing response scalars from photons.

The obstruction applies to these particular one-forms. A more general composite, such as $A = Y dX + u\, dv$, can have $F \wedge F \neq 0$. Its independent Maxwell dynamics and particle spectrum would still need to be established.

22.2 A non-Maxwell counterexample satisfying the gravitational and critical equations

To go beyond a test-field argument, consider the exact configuration

$$\phi = fku, \quad Y = bx^2, \quad X = 0, \quad w = -\frac{kbx^2}{2}du, \quad F_w = -kbx\, dx \wedge du, \tag{22.4}$$

together with

$$ds^2 = -2du\, dv + dx^2 + dy^2 + \mathcal{P}(x)du^2, \qquad \mathcal{P} = -\frac{f^2k^2}{M_P^2}x^2 - \frac{k^2b^2}{30M_P^2}x^6. \tag{22.5}$$

The curvature is $R_{uu} = (f^2k^2 + \frac{1}{2}k^2b^2x^4)/M_P^2$. All scalar, critical-response, and Einstein equations hold, yet $\nabla_\mu F_w^{\mu v} = kb \neq 0$. Thus the critical equations do not imply source-free Maxwell dynamics. Independent Maxwell fields and special on-shell test fields remain possible.[U]

22.3 Connection traces, projected curvature, and gauging

The trace of the complex Levi–Civita connection directly gives

$$A_{\mathrm{tr}} = \mathrm{Im}\big(\Gamma^{\nu}{}_{\mu\nu}(\gamma) - \Gamma^{\nu}{}_{\mu\nu}(g)\big)dx^{\mu} = 2d\arg F, \qquad dA_{\mathrm{tr}} = 0. \tag{22.6}$$

which is a pure gradient on the globally smooth real-χ branch $F = 1 + i\chi$. A projection route can instead yield nonzero curvature: a projector P in a positive-definite parent bundle has induced curvature $P(dP \wedge dP)P$. For example, the normalized line vector $z = (\cos(\vartheta/2), e^{i\varphi}\sin(\vartheta/2))$ gives $A = \sin^2(\vartheta/2)d\varphi$ and $F = \frac{1}{2}\sin\vartheta\, d\vartheta \wedge d\varphi$. Both angles, the parent bundle, and its Hermitian structure are additional data. An action and independent allowed initial data are still required.

Directly gauging response translations also need not produce a massless photon. At the reference point, replacing translation derivatives in dX, du, dv by a connection B produces a term proportional to

$$m_B^2 = 2q_X^2 + 3q_u^2 + q_v^2 \tag{22.7}$$

as a Stueckelberg mass term, rather than automatically yielding massless QED. Another identically conserved candidate, $j^{\mu}_{\mathrm{top}} = q\epsilon^{\mu\nu\rho\sigma}\partial_\nu\phi\partial_\rho u\partial_\sigma v/M_*^3$, can couple to an independent A_μ; that is precisely an addition of gauge theory. Under conventional smooth, decaying boundary conditions, its total charge may also reduce to a vanishing surface term.

22.4 Papapetrou test fields and FLRW duality

For a Killing vector ξ of the real metric, take $A = \xi^\flat$ and $F = dA$. Then $\nabla_\mu F^{\mu\nu} = -2R^{\nu}{}_{\rho}\xi^{\rho}$. On an Einstein–potential-free-scalar background with $\mathcal{L}_\xi\phi = 0$, the right-hand side vanishes. Geometry can therefore generate on-shell Maxwell test fields. Schwarzschild's time Killing vector yields Coulomb-type flux, but its amplitude is controlled by the original mass parameter. Including the test field's stress in the Einstein equations generally invalidates the original Ricci-flat metric. A derived flux is not a new independent charge.[U]

In the potential-free flat universe $a(\tau) = \tau^{1/3}$, choose $\xi = \partial_x$ and construct $F_E = d\xi^\flat$, $F_B = \iota_\xi \star d\phi$. With consistent orientation and scalar branch,

$$F_E = \frac{2}{3}\tau^{-1/3}d\tau \wedge dx, \qquad \star F_E = -\frac{2}{3}dy \wedge dz, \qquad F_B = -\sqrt{\frac{2}{3}}M_P dy \wedge dz = f \star F_E. \tag{22.8}$$

The equality holds on this isotropic background and follows from $\dot\phi/(2\mathcal{H})$ in Einstein–scalar theory. In Bianchi I, $a_i \propto \tau^{p_i}$ and $\phi = A\ln\tau$ obey $\sum p_i = 1$, $\sum p_i^2 + A^2/M_P^2 = 1$; the analogous ratio is $A/(2p_i)$ and depends on direction. The isotropic value is therefore background-specific.

22.5 Flux quantization and symplectic tests

Derived two- or three-forms on T^3 may have nonzero de Rham periods. Interpreting them as quantized curvature of a compact $U(1)$ bundle or gerbe additionally requires a charge unit and integral-period conditions. Conformal invariance of the four-dimensional two-form Hodge star preserves the same two-form. Redefining $A_\gamma = F\xi_g^\flat$ instead gives $dA_\gamma = F\, d\xi_g^\flat + dF \wedge \xi_g^\flat$; the extra term generally spoils the original source-free equation and cannot be omitted.

Independent Maxwell theory has three canonical pairs of homogeneous harmonic modes on T^3. Some Papapetrou-derived families instead have zero pullback of its symplectic form. Isotropic Einstein–scalar minisuperspace has only one physical pair after branch selection, expressible through $\Phi = \phi - \sqrt{6}M_P\ln a$ and the scalar momentum. Additional readouts cannot enlarge this

initial-data space. We accordingly retain an independent electromagnetic connection in the total action.

23 The $c = 2$ Response and Electroweak Theory

23.1 Positive-definite response and canonical coordinates

For $c = 2$, the response matrix is

$$G_2 = \begin{pmatrix} 2\rho + 1 & -\chi \\ -\chi & 2\rho - 1 \end{pmatrix}, \quad \operatorname{spec} G_2 = (3\rho, \rho), \quad \det G_2 = 3\rho^2. \tag{23.1}$$

This two-dimensional response kinetic term is positive definite for every real ϕ. At the common reference point $\phi = 0$, $U_2 = \sqrt{3}\, u_2$ and $V_2 = v_2$ turn it into the ordinary quadratic kinetic terms of two canonically normalized real scalars. Here U_n, V_n denote normalized real response fields, distinct from carrier bundles $\mathcal{V}_n$.

The lowest-order interaction with the shared scalar contains

$$\mathcal{L}_2 = -\frac{1}{2}(\partial U_2)^2 - \frac{1}{2}(\partial V_2)^2 + \frac{\phi}{\sqrt{3} f}\, \partial_\mu U_2 \partial^\mu V_2 + O(\phi^2/f^2). \tag{23.2}$$

The vertex couples scalar responses. Weak gauge bosons and the coupling g_2 belong to the independent connection $\mathcal{W}$ on E_2.

23.2 Higgs structure and electroweak mixing

In the internal geometry, choose the Higgs field H as a section of E_2, the standard representation $(\mathbf{1}, \mathbf{2}, +1/2)$. Define its kinetic term with the real readout metric g and select the potential

$$\mathcal{L}_H = -(D_\mu H)^\dagger D^\mu H - V_H, \qquad V_H = \lambda_H \left(H^\dagger H - \frac{v^2}{2} \right)^2. \tag{23.3}$$

For $\lambda_H > 0$, the vacuum in a chosen gauge is $H_0 = (0, v/\sqrt{2})^T$. Canonically normalized neutral fields mix according to

$$\begin{aligned} A_\mu &= \cos\theta_W B_\mu^{\mathrm{can}} + \sin\theta_W W_\mu^{3,\mathrm{can}}, \\ Z_\mu &= -\sin\theta_W B_\mu^{\mathrm{can}} + \cos\theta_W W_\mu^{3,\mathrm{can}}, \qquad \tan\theta_W = \frac{g_Y}{g_2} \end{aligned} \tag{23.4}$$

The unbroken generator is $Q = T_3 + Y_{\mathrm{hyp}}$, yielding

$$m_W = \frac{g_2 v}{2}, \qquad m_Z = \frac{v}{2}\sqrt{g_2^2 + g_Y^2}, \qquad e = g_2 \sin\theta_W = g_Y \cos\theta_W. \tag{23.5}$$

These mass relations follow from the chosen representations, gauge action, and Higgs vacuum. The photon remains massless because $U(1)_{\mathrm{em}}$ is unbroken.[U]

The associations of $c = 1$ with electromagnetism and $c = 2$ with weak interactions describe low-energy sectors of a coupled model. Hypercharge and weak isospin mix under symmetry breaking; the physical photon, $W^\pm$, and Z must be derived from the total action.

23.3 Undetermined electroweak parameters

Internal rank and hypercharge embedding do not fix v, λ_H, boundary values of the three gauge couplings, fermion Yukawa matrices, or the weak mixing angle. An added simple-group unification assumption with a single trace normalization could give a high-energy matching relation. The present splitting-preserving $3+2$ action allows independent gauge kinetic coefficients and does not itself supply that numerical prediction.

Response masses and portals such as $\phi^2 H^\dagger H$ add further parameters. Their effects must be separated from predictions of the shared geometric action when comparing with experiment.

23.4 The electromagnetic line bundle where the Higgs is nonzero

Where the Higgs section is everywhere nonzero, its normalization selects a trivial line in E_2, giving

$$E_2 \simeq K \oplus \mathbf{1}, \qquad K = \det E_2 \simeq (\det E_3)^*, \qquad c_1(K) = -c_1(E_3). \tag{23.6}$$

The unbroken structure group in this region is $U(3) \simeq (SU(3) \times U(1)_{\mathrm{em}})/\mathbb{Z}_3$. In a Higgs-adapted rank-five frame, $Q_{\mathrm{em}} = \mathrm{diag}(-1/3, -1/3, -1/3, 1, 0)$. This line-bundle description connects low-energy electromagnetic charges to the original $3+2$ determinant condition.[U]

A globally nonzero Higgs requires $c_2(E_2) = 0$. Weak bundles with other topology may have Higgs zeros or defects, where the full E_2 description is needed and the unitary-gauge splitting is only local.

For $e_L \in S_L \otimes K^*$ and $e^c \in S_L \otimes K$, the conjugate right-handed electron lies in $S_R \otimes K^*$. Weyl kinetics and Yukawa pairing then assemble a charged Dirac field. The line bundle supplies its charge representation; the action supplies its equation.

23.5 Weak parallelism and anomaly constraints

Identifying the response involution J_{CR} with a fixed self-adjoint weak-doublet direction and imposing $DJ_{\mathrm{CR}} = 0$ forces the connection to commute with σ_3 at $J_{\mathrm{CR}} = \sigma_3$. This removes W^1, W^2 and leaves only the stabilizer $U(1)$. It may describe a reduced structure but cannot simultaneously retain full unbroken $SU(2)$ weak dynamics. Response, spin, and weak internal spaces need distinct labels even when their dimensions coincide.

A spectrum of leptons without quarks is not closed: one lepton generation contributes $-1/4$ and $3/4$ to $SU(2)^2U(1)$ and $U(1)^3$, with an odd number of weak doublets. Quarks contribute $1/4$ and $-3/4$; color multiplicity gives four weak doublets per complete generation. Anomaly cancellation is a consistency test of the supplied representations, not a unique dynamical generation of that spectrum.

23.6 Exterior-algebra Higgs operators and mass degeneracy

On the thirty-two-dimensional internal space $\Lambda^\bullet E$, the Higgs defines $\Phi_H = \varepsilon_H + \iota_{H^\dagger}$, satisfying

$$\Phi_H^\dagger = \Phi_H, \qquad \Phi_H^2 = \|H\|^2 I_{32}, \qquad P_{\mathrm{even}} \Phi_H P_{\mathrm{even}} = 0. \tag{23.7}$$

It reverses exterior parity and cannot simply be restricted as an ordinary mass matrix within the even subspace. Using the total determinant volume form and reversal operator $\tau|_{\Lambda^k} = (-1)^{k(k-1)/2}$, construct

$$\mathcal{Y}_H(\psi, \psi') = \mathrm{vol}_E^{-1} \left[\tau(\psi) \wedge \Phi_H \psi' \right]_{\deg 5}. \tag{23.8}$$

This pairing can combine with Lorentz spinor contraction to form Yukawa terms. With only one common coefficient, however, all mass singular values in an orthonormal basis equal $\|H\|$. The source test $H = (1 + i, 2 - i)$ gives $\sqrt{7}$ for every value, so this minimal operator does not generate the observed mass hierarchy. Generation spaces, different allowed invariants, or other structure are needed. Four-dimensional conformal weight $4 - 3/2 - 1 - 3/2 = 0$ ensures formal compatibility of the massless Yukawa term but fixes no Yukawa values.[U]

24 The $c = 3$ Response and Color Dynamics

24.1 A three-dimensional internal carrier and positive-definite response

For $c = 3$,

$$G_3 = \begin{pmatrix} 3\rho + 1 & -\chi \\ -\chi & 3\rho - 1 \end{pmatrix}, \quad \operatorname{spec} G_3 = (4\rho, 2\rho), \quad \det G_3 = 8\rho^2. \tag{24.1}$$

Near $\phi = 0$, canonical coordinates are $U_3 = 2u_3$, $V_3 = \sqrt{2}v_3$, and the cubic derivative coupling is $\phi\, \partial U_3 \cdot \partial V_3/(\sqrt{8}f)$. This module has two real response degrees of freedom; the color connection has eight algebra directions. They coexist in the total theory but have different origins and counts.

The Hermitian structure and determinant condition of E_3 give an $SU(3)$ color connection. Specifying fundamental quarks, gauge kinetics, and the fermion action then gives the conventional color-covariant derivative and non-Abelian field strength

$$\mathcal{G}^A_{\mu\nu} = \partial_\mu \mathcal{C}^A_\nu - \partial_\nu \mathcal{C}^A_\mu + f^{ABC} \mathcal{C}^B_\mu \mathcal{C}^C_\nu. \tag{24.2}$$

The connection convention places the coupling in front of the kinetic term. Canonically normalizing the gauge field puts g_3 in the nonlinear term.

24.2 Projected connections and free gluons

A spacetime-varying internal projector can induce a Berry-type connection, but this alone does not give full Yang–Mills dynamics. Around a constant projector, induced curvature often starts at second order in projector perturbations, so curvature squared lacks the independent gauge field's free quadratic kinetic term. If the projector depends only on the single shared scalar, its curvature may even vanish identically. The original $c = 3$ study applied such tests to proposals deriving all gluons directly from local frame rotations.[U]

An independent splitting-preserving connection supplies the required gauge initial data. Its bundle fixes the transformation law, while the action fixes propagation and matter coupling.

24.3 QCD under standard quantum assumptions

QCD results are inherited only after taking the quark content, statistics, gauge kinetics, and quantization rules to be those of standard QCD. For N_f fundamental Dirac quarks, for example, the one-loop result is

$$\mu \frac{dg_3}{d\mu} = -\frac{g_3^3}{16\pi^2} \left(11 - \frac{2}{3} N_f \right) + O(g_3^5). \tag{24.3}$$

A positive bracket gives asymptotic freedom. Its coefficient depends on the group, representations, and quantum fields; it is not obtained by substituting $c = 3$ into the response matrix. Color-singlet response fields do not directly alter this one-loop gauge coefficient as extra colored particles would.

The tensors δ^a_b and ϵ_{abc} define meson- and baryon-type color singlets. They determine representation combinations, not confinement, the mass gap, or hadron spectra. This construction admits the QCD action; its nonperturbative solution remains a separate problem.

24.4 Degrees of freedom and weak-field expansion of projected connections

In a selected reference trivialization, a general connection preserving P is

$$\mathcal{A} = -[dP, P] + \mathcal{B}, \qquad [\mathcal{B}, P] = 0. \tag{24.4}$$

The block-diagonal part $\mathcal{B}$ remains independent connection data. For an orthonormal frame Q with $Q^\dagger Q = I$, $P = QQ^\dagger$, the induced connection and curvature are

$$\mathcal{A} = Q^\dagger dQ, \qquad \mathcal{F} = dQ^\dagger \wedge (I - QQ^\dagger) dQ. \tag{24.5}$$

Around constant Q_0, $\mathcal{F} = O(\epsilon^2)$ and $\operatorname{Tr} \mathcal{F}^2 = O(\epsilon^4)$, so the induced action has no free quadratic gluon term. A universal connection theorem represents a supplied connection in a larger bundle; it does not generate its independent dynamics from a few scalars. An $SU(3)$ Yang–Mills field has $8 \times 2 = 16$ transverse configuration degrees of freedom after reduction.[U]

On a Lorentz-invariant constant background, a covariant vector locally built from finite-order scalar jets is generally of type $\partial_\mu P(\Box)\phi$ at linear order and has zero linear curvature. This obstructs a particular local weak-field route, not nonlocal constructions, nonstandard backgrounds, or nonperturbative emergence. Those stronger routes require their own spectra and quantum state spaces.

24.5 Coupling normalization and running

An additional common spectral-trace kinetic assumption gives trace coefficients $C_3 = C_2 = 2$, $C_Y = 10/3$ for one even-exterior generation, allowing the matching relation

$$g_3^2 = g_2^2 = \frac{5}{3} g_Y^2, \qquad \sin^2 \theta_W = \frac{3}{8} \tag{24.6}$$

at a common matching scale. This is neither a low-energy experimental value nor a relation derived solely from $c = 1, 2, 3$. Using Standard Model one-loop coefficients $b = (41/10, -19/6, -7)$, the historical source inputs $\alpha_{\text{em}}^{-1} = 127.95$, $\sin^2 \theta_W = 0.23122$, and $\alpha_s = 0.1177$ give unified-normalization $\alpha_i^{-1} \simeq (59.02, 29.58, 8.50)$. When the first two meet near 1.03×10^{13} GeV, the three inverse couplings are approximately $(42.41, 42.41, 36.85)$, not simultaneous unification.

These numbers reproduce the historical source calculation, not a fit to current data. Thresholds, additional charged fields, and higher-order corrections can alter the running. Gauge-singlet responses do not directly change the one-loop color coefficient.

24.6 Conformal cancellation, trace anomaly, and the strong scale

Cancellation of volume and two inverse-metric factors in four-dimensional gauge kinetics supports $Z_3 = 1$. Retaining only $\operatorname{Re} F^{-2} = (1 - \chi^2)/(1 + \chi^2)^2$ as a proposed kinetic coefficient makes it negative for $|\chi| > 1$. Using $|F|^{-2}$ preserves positivity but adopts a different readout rule, not the same derivation.

For independently specified QCD, the renormalized trace anomaly contains

$$T^{\mu}{}_{\mu} = \frac{\beta(g_s)}{2g_s} G^a_{\mu\nu} G^{a\mu\nu} + \sum_q (1+\gamma_m) m_q \bar{q} q. \tag{24.7}$$

With an added $Z_3(\phi)$, fixed thresholds, and a one-loop approximation, one may obtain $\Lambda_{\mathrm{QCD}}(\phi) = \mu_0 \exp[-8\pi^2 Z_3(\phi)/(b_0 g_0^2)]$. This is a potentially sensitive channel from the geometric scalar to hadron masses, but depends on extra couplings and threshold treatment. Asymptotic freedom does not itself rigorously prove confinement. Dynamical quarks allow string breaking, so pure-gauge area-law statements cannot be transferred unconditionally.

24.7 Global topology, full SU(5), and generation number

The Whitney formula gives $c_1(E_3) + c_1(E_2) = 0$ and

$$c_2(E) = c_2(E_3) + c_2(E_2) + c_1(E_3)c_1(E_2). \tag{24.8}$$

Only the additional requirement that the entire E be trivial gives $c_2(E_3) + c_2(E_2) = c_1(E_3)^2$; a trivial total determinant is insufficient. The projector target $\mathrm{Gr}_{\mathbb{C}}(3,5)$ has real dimension twelve, $\pi_2 = \mathbb{Z}$, and $\pi_3 = 0$. These establish some topological classes but not finite-energy stable solitons.

Enlarging the structure group to full $SU(5)$ adds twelve off-diagonal generators, in $(\mathbf{3},\mathbf{2})_{-5/6}$ and its conjugate, potentially bringing heavy gauge bosons, monopoles, and proton decay. A native $S(U(3)\times U(2))$ bundle need not contain them. Representations can be organized as $E^* \oplus \Lambda^2 E$ plus a sterile singlet, but mass relations such as $M_d = M_e^{\mathsf{T}}$ still require particular full-unification Yukawa assumptions. The allowed operator $QQQL/\Lambda^2$ has $\Delta B = \Delta L = 1$, so the present group structure alone does not protect the proton.

For three generations, the untwisted Dirac index on a closed Euclidean spin four-manifold is $-\sigma/8$, while $\sigma \in 16\mathbb{Z}$. This simplest index route gives only even numbers. The twisted index

$$\mathrm{ind}\, D_E^+ = \int_M \left[\frac{1}{2}(c_1^2 - 2c_2) - \frac{\mathrm{rank}\, E}{24} p_1(TM)\right] \tag{24.9}$$

can be odd: a suitable $SU(2)$ bundle on S^4 with $c_2 = 3$ gives -3. This Euclidean net zero-mode count must still be connected to four-dimensional Lorentz propagation, chirality, and mass structure; it cannot directly be identified with three observed generations.

25 Chiral Matter and Dirac Structure

25.1 One generation organized by exterior algebra

Given $E = E_3 \oplus E_2$, take the internal even exterior algebra

$$\mathcal{R}_{\mathrm{gen}} = \Lambda^{\mathrm{even}} E = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \qquad \dim_{\mathbb{C}} \mathcal{R}_{\mathrm{gen}} = 1 + 10 + 5 = 16. \tag{25.1}$$

Writing all fermions as left-handed Weyl fields gives the decomposition in the table. u^c, d^c, e^c, N^c denote left-handed conjugate representations of the corresponding right-handed fields.[U]

Left-handed field	Internal origin	$SU(3) \times SU(2)$	Hypercharge
Q	$E_3 \otimes E_2 \subset \Lambda^2 E$	$(\mathbf{3}, \mathbf{2})$	$1/6$
u^c	$\Lambda^2 E_3$	$(\overline{\mathbf{3}}, \mathbf{1})$	$-2/3$
e^c	$\Lambda^2 E_2$	$(\mathbf{1}, \mathbf{1})$	1
d^c	$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$(\overline{\mathbf{3}}, \mathbf{1})$	$1/3$
L	$\Lambda^3 E_3 \otimes E_2$	$(\mathbf{1}, \mathbf{2})$	$-1/2$
N^c	$\Lambda^0 E$	$(\mathbf{1}, \mathbf{1})$	0

The generator fixes relative charges, for example $Q: -1/3+1/2 = 1/6$ and $L: 3(-1/3)+1/2 = -1/2$. The even exterior algebra supplies one generation. Three copies, their mixing, and their mass hierarchy require additional structure.

25.2 Representation consistency and anomaly cancellation

For one left-handed generation, the mixed gravitational-hypercharge and cubic hypercharge anomaly coefficients are

$$\begin{aligned} \operatorname{Tr} Y &= 6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 = 0, \\ \operatorname{Tr} Y^3 &= 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1 = 0. \end{aligned} \tag{25.2}$$

With fundamental index $T(\mathbf{N}) = 1/2$, the $SU(3)^2U(1)$ coefficient is $1/6 - 1/3 + 1/6 = 0$, the $SU(2)^2U(1)$ coefficient is $1/4 - 1/4 = 0$, and the $SU(3)^3$ fundamental-minus-antifundamental count is $2 - 1 - 1 = 0$. Three color copies of the Q doublet and one L doublet give four $SU(2)$ doublets per generation, satisfying the usual evenness condition.

The selected generation passes these anomaly checks. Additional charged responses or chiral representations require a new calculation.

25.3 Spin structure and statistics

The exterior decomposition acts on internal fibers. Physical fermions also require spacetime spin structure, Weyl spinor bundles, the corresponding covariant derivatives, and anticommuting quantum statistics. Internal degree parity is distinct from half-integer Lorentz spin.

Ordinary tensor constructions are invariant under the central spin-group element -1, while spinors change sign. They therefore cannot supply a nonzero equivariant spinor field by local tensor operations. Clifford factorization organizes the equations but does not provide fermionic statistics.[U]

To avoid signature mixing, the $(-+++)$ spacetime convention uses $\{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}$ and $D_g = i\Gamma^\mu \nabla_\mu$. The corresponding flat free operators satisfy $(D_g - m)(D_g + m) = \Box - m^2$. A different Clifford signature is possible only with the associated factors in the Dirac equation transformed as well.

25.4 The complex conformal Dirac operator and mass readout

On a region with a consistent local conformal branch and spin frame, the four-dimensional massless Dirac operator transforms as

$$D_\gamma = F^{-5/4} D_g F^{3/4}. \tag{25.3}$$

The field redefinition $\psi = F^{3/4}\Psi$ maps the massless equation to the real metric. If a constant complex-frame mass m is added, the transformed coefficient is $mF^{1/2}$, generally complex. A physical massive theory then needs a compatible conjugation, inner product, and action.

Scheme A instead uses geometric mass $mF^{-1/2}$, giving real mass m as in Eq. (12.4). The remaining global Spin and quantum-state assumptions are unchanged.

Choosing $\widehat{g} = \rho g$ as a real matter metric gives another conformal mass rule. A frame comparison must transform the full action, measuring standards, and observables together. Here the baseline fermion and Higgs actions use the physical metric g.

25.5 Yukawa and neutrino modules

For these representations, allowed Yukawa terms are

$$\begin{aligned}\mathcal{L}_{\rm Yuk} = -\big[&Y_u\, u^c (Q\cdot H) + Y_d\, d^c (H^\dagger Q) + Y_e\, e^c (H^\dagger L) \\ &+ Y_\nu\, N^c (L\cdot H) + \tfrac{1}{2} M_N N^c N^c + \text{h.c.}\big].\end{aligned} \tag{25.4}$$

Generation indices and spinor contractions are suppressed; the dot denotes antisymmetric $SU(2)$ contraction. Since N^c is a gauge singlet, a Majorana mass is allowed. Including it or a seesaw mechanism is a neutrino-module choice. Parameter matrices and phases require experimental constraints or a deeper mechanism.

These parameters are independent of ϕ in the baseline. Allowing $Y_f(\phi)$ or $M_N(\phi)$ adds shared-scalar sources and can affect discrete symmetry, equivalence-principle tests, and dark-sector stability. Such terms cannot be omitted from the total equations.

25.6 Local and topological routes to spin

If all inputs are ordinary scalars, the metric, and finitely many local derivatives, they are invariant under the central -1 of $\mathrm{Spin}(1,3)$, while Weyl and Dirac spinors change sign. A natural equivariant finite-jet map constructing ψ from these inputs would require $\psi = -\psi$, leaving only the zero map. This is a definite local obstruction. It does not exclude a supplied Spin structure, nonlocal framed constructions, topological sectors, or other added quantum degrees of freedom.[U]

The simple critical-plus-two-normal scalar target is locally $\mathbb{R}^5$, with $\pi_3(\mathbb{R}^5) = 0$, so it lacks the topology used in Skyrmion fermionization. Kähler–Dirac forms offer a different construction: sixteen complex components with $D = d - \delta$ and $D^2 = -(d\delta + \delta d)$. Their statistics, surplus components, and curved-space spectrum still require specification. This component count is independent of the sixteen internal labels of one chiral generation.

25.7 The response involution in a Clifford module

$J_{\rm CR}$ acts on a two-dimensional response factor and normally commutes with γ_5 on an independent spin factor; γ_5 anticommutes with Clifford multiplication. Matching eigenvalues ± 1 does not identify them. A similarity transformation can nevertheless be made in a separately constructed four-dimensional Clifford module. Take

$$T_3 = J_{\rm CR}, \quad T_2 = \sigma_2, \quad T_1 = -iT_2T_3, \qquad \Gamma^0 = T_1 \otimes I, \quad \Gamma^k = iT_2 \otimes \sigma_k. \tag{25.5}$$

The matrices satisfy $\{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}$, with spacetime signature $(-+++)$. Here $\Gamma_5 = -J_{\mathrm{CR}} \otimes I$; its ϕ dependence is a similarity transformation with flat connection $-(dU)U^{-1}$. This realizes the response involution inside a supplied Clifford module without deriving fermions from scalar response fields.

By contrast, raising an index on the metric's imaginary part gives χI, whose square is $\chi^2 I$. It is neither the complex structure nor the response involution. Chirality and hypercharge likewise act on their own specified spaces.

25.8 Spin, Spin^c, and the Lichnerowicz formula

On a Spin spacetime, a spin bundle S can be independently twisted by an internal bundle. For a Spin^c structure, the determinant line $L_{\det}$ must satisfy

$$c_1(L_{\det}) \equiv w_2(TM) \pmod 2. \tag{25.6}$$

Identifying it with $\det E_2$ adds a global pairing condition. On an already Spin manifold, a line specified as the Spin^c determinant must have even first Chern class. This does not restrict ordinary charged fields $S \otimes L$ to even-charge line bundles, because the corresponding Spin^c determinant is actually L^2.

In a Euclidean Riemannian test, the twisted Dirac square is

$$D_A^2 = \nabla_A^* \nabla_A + \frac{R}{4} + c(\mathcal{F}_E), \tag{25.7}$$

where $c(\mathcal{F}_E)$ is Clifford contraction with curvature. A Spin^c determinant connection adds $\frac{1}{2}c(F_{L_{\det}})$. The identity assumes the spinor bundle and connection. Relating its Euclidean spectrum to Lorentzian particles also requires reality and continuation conditions.

25.9 Mass readout, chiral phase, and strong CP

In the complex conformal relation $D_{Fg} = F^{-5/4} D_g F^{3/4}$,

$$d \ln F = \frac{\tanh x + i \operatorname{sech} x}{f}\, d\phi. \tag{25.8}$$

A massless conformal redefinition removes the corresponding gradient term without generating independent electromagnetic curvature. Insisting on constant complex-geometric mass m_0 gives generally complex $m_0\sqrt{F}$, not automatically a Hermitian physical mass. Choosing real matter metric $\tilde{g} = \rho g$ gives $m_{\mathrm{eff}} = m_0\sqrt{\cosh x}$ but changes the readout frame. Apparatus, units, and observables must transform together rather than only replacing a mass function.

A separately specified Hermitian extension is $-m\bar{\psi}e^{i\alpha(\phi)\gamma_5}\psi$. With $\alpha = \theta/2$, an axial rotation removing the mass phase produces, in the corresponding sign convention,

$$\frac{1}{2}(\partial_\mu \alpha)\bar{\psi}\gamma^\mu\gamma_5\psi = \frac{\operatorname{sech} x}{4f}(\partial_\mu \phi) j_5^\mu, \tag{25.9}$$

and transforms the measure through the Fujikawa anomaly. The physical strong-CP parameter is still $\bar{\theta} = \theta_{\mathrm{QCD}} + \arg\det M_q$. Identifying this phase with an axion would require periodicity, a potential, and a relaxation mechanism. The internal dark parity discussed below is not spacetime CP and does not alone forbid $\theta G\tilde{G}$.[U]

26 The Coupled Action and Field Equations

26.1 The total action

Let $y_n = (u_n, v_n)^T$, $n = 2, 3, 4$. Combine the modules as

$$\begin{aligned} S_{\rm tot} = \int d^4x\sqrt{-g}\,\Big[& \frac{M_P^2}{2}R - \frac{1}{2}(\nabla\phi)^2 - w_\mu w^\mu \\ & - \frac{1}{2}\sum_{n=2}^{4}(G_n)_{ij}\nabla_\mu y_n^i \nabla^\mu y_n^j - V(\phi, y_n, H^\dagger H) \\ & - \sum_{a=Y,2,3}\frac{Z_a(\phi)}{4g_{a0}^2}\mathcal{F}_a^2 - (D_\mu H)^\dagger D^\mu H \\ & + \mathcal{L}_{\rm Weyl} + \mathcal{L}_{\rm Yuk}\Big]. \end{aligned} \tag{26.1}$$

There is one geometric kinetic term and one shared ϕ. This fixes the normalization of every response coupling relative to M_P; summing separate gravity-plus-scalar actions would change the theory.

The potential is independent of X, Y, so the critical constraints and $w = 0$ truncation survive. Weyl and Yukawa parameters are also taken independent of ϕ. Extensions must be varied with their actual field dependence. Gauge indices are contracted as in Eq. (20.6).

A specific potential useful for spectra and stability is

$$\begin{aligned} V_{\rm bench} = V_0 + \frac{1}{2}M_\phi^2\phi^2 + \frac{5}{2}m_4^2u_4^2 + \lambda_H\left(H^\dagger H - \frac{v^2}{2}\right)^2 \\ + \frac{1}{2}\sum_{n=2}^{3}\left[(n+1)\mu_{Un}^2u_n^2 + (n-1)\mu_{Vn}^2v_n^2\right]. \end{aligned} \tag{26.2}$$

Choose nonnegative squared masses and $\lambda_H > 0$, with vacuum $\phi = y_n = 0$, $H^\dagger H = v^2/2$. The v_4 shift survives. This benchmark leaves symmetry-allowed portals and higher terms at zero at the matching scale; keeping them zero under running would need further protection.

26.2 The shared geometric source equation

Variation with respect to ϕ gives

$$\Box_g\phi = \mathcal{S}_\phi, \tag{26.3}$$

where

$$\begin{aligned} \mathcal{S}_\phi = \nabla_\mu\left(\frac{Y}{f}w^\mu\right) + \frac{1}{2}\sum_{n=2}^{4}(G_n')_{ij}\nabla_\mu y_n^i\nabla^\mu y_n^j \\ + V_{,\phi} + \sum_{a=Y,2,3}\frac{Z_a'}{4g_{a0}^2}\mathcal{F}_a^2. \end{aligned} \tag{26.4}$$

Primes denote differentiation with respect to ϕ. The four terms come from critical response, normal response, the potential, and scalar-dependent gauge kinetics, respectively. The last vanishes in the minimal baseline $Z_a = 1$.

In the original geometric coordinate χ, Eq. (26.3) is equivalent to

$$\Box_g \chi - \frac{\chi}{1+\chi^2}(\nabla\chi)^2 = \frac{\sqrt{1+\chi^2}}{f}\mathcal{S}_\phi, \tag{26.5}$$

or

$$\nabla_\mu \left(\frac{\nabla^\mu \chi}{\sqrt{1+\chi^2}} \right) = \frac{\mathcal{S}_\phi}{f}. \tag{26.6}$$

The total action fixes this source. The current $f\nabla^\mu\chi/\sqrt{1+\chi^2} = \nabla^\mu\phi$ is conserved precisely when $\mathcal{S}_\phi = 0$; interacting sectors generally exchange energy through it.[U]

26.3 Response, gauge, and matter equations

For $n = 2, 3, 4$, response fields satisfy

$$\nabla_\mu \left[(G_n)_{ij} \nabla^\mu y_n^j \right] = V_{,y_n^i}. \tag{26.7}$$

In terms of $\Psi_n = u_n + iv_n$, the same equation becomes

$$\nabla_\mu \left[n\rho\, \nabla^\mu \Psi_n + \overline{F}\, \nabla^\mu \overline{\Psi}_n \right] = 2\frac{\partial V}{\partial \overline{\Psi}_n}. \tag{26.8}$$

Here Ψ and $\overline{\Psi}$ are treated as independent in variational notation and finally satisfy complex conjugacy. The critical module is governed separately by Eq. (18.9) and cannot be rewritten with an everywhere invertible G_1^{-1}.

Independent gauge connections satisfy matter-sourced Yang–Mills equations

$$(D_\mu)^{AB} \left(\frac{Z_a}{g_{a0}^2} \mathcal{F}_a^{B\mu\nu} \right) = j_a^{A\nu}, \tag{26.9}$$

The current sign is defined as minus the variation of the corresponding matter action with respect to the connection, making Eq. (26.9) consistent with $D = d - i\mathcal{A}$. Higgs and Weyl equations follow by varying Eq. (26.1) in $H^\dagger$ and independent conjugate fermionic variables; they include potential derivatives and the Yukawa sources of Eq. (25.4). Gauge covariance depends on the full representation contractions. Response-scalar shift Noether currents are not color- or weak-charged matter currents.

26.4 Einstein equations and energy exchange

Variation with respect to the real metric gives

$$M_P^2 G_{\mu\nu} = T_{\mu\nu}^{\mathrm{tot}}. \tag{26.10}$$

The kinetic contributions of the shared scalar, critical module, and positive-definite responses are respectively

$$\begin{aligned} T_{\mu\nu}^{(\phi)} &= \nabla_\mu\phi\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2, \\ T_{\mu\nu}^{(1)} &= 2w_\mu w_\nu - g_{\mu\nu}w^2, \\ T_{\mu\nu}^{(n)} &= (G_n)_{ij}\nabla_\mu y_n^i \nabla_\nu y_n^j - \tfrac{1}{2}g_{\mu\nu}(G_n)_{ij}\nabla y_n^i \cdot \nabla y_n^j. \end{aligned} \tag{26.11}$$

The potential contributes $-g_{\mu\nu}V$ once, alongside gauge, Higgs, and fermion stress. This avoids counting the same potential in both $T^{(\phi)}$ and the individual sectors.

For a potential-free example retaining one normal response sector, its field equation gives

$$\nabla_\mu T_\nu^{(n)\mu} = -\frac{1}{2}(G_n')_{ij}\nabla y_n^i \cdot \nabla y_n^j \, \nabla_\nu \phi. \tag{26.12}$$

This cancels the corresponding scalar-stress divergence. Total energy-momentum is conserved even though individual sectors exchange energy. The Bianchi identity checks the combined equations, while the matter and constraint equations must still be solved.

26.5 Reference limits of GR and the Standard Model

Take $\phi = \phi_0$, $w = 0$, all responses at stationary points of the potential, and total source $\mathcal{S}_\phi = 0$. If visible matter is independent of ϕ, this yields Einstein gravity with the selected visible-matter theory and cosmological constant $\Lambda = V_0/M_P^2$. This classical truncation holds at the origin of Eq. (26.2) with baseline $Z_a = 1$.

For $Z_a' \neq 0$, nonzero gauge backgrounds can source ϕ again. Yukawa couplings depending on ϕ, portals, or masses similarly require source checks. Lifting every GR-plus-matter solution to CR therefore requires this independence or source cancellation for each solution.[U]

The truncation embeds classical solutions. Its quantum spectrum may still include massless responses, dark radiation, and virtual heavy fields. Recovering GR plus the Standard Model at low energy additionally requires suitable initial states, thresholds, and effective couplings.

26.6 Complex response equations, characteristic cones, and source linearization

The factor of two in Eq. (26.8) follows from Wirtinger variation of the real two-component action. The physical fields ultimately obey $\bar{\Psi}_n = \overline{\Psi_n}$. The singular critical block remains governed by its constraints.[U]

For $n > 1$, the second-order principal symbol of the response block on a given background is

$$\mathcal{P}_n(k) = G_n \, g^{\mu\nu}k_\mu k_\nu, \qquad \det \mathcal{P}_n = (n^2-1)\cosh^2 x \, (g^{\mu\nu}k_\mu k_\nu)^2. \tag{26.13}$$

The normal modes share the cone of g. Coupled gravity requires gauge fixing, and the critical block requires reduction before its characteristics can be interpreted.

Rewriting the shared geometric equation in χ introduces the nonlinear operator $\Box\chi - \chi(\nabla\chi)^2/(1+\chi^2)$. At fixed metric, setting $\chi = \bar{\chi} + \eta$ gives the linearization

$$\begin{aligned}\delta\mathcal{E}_\chi = \Box\eta &- \frac{2\bar{\chi}}{1+\bar{\chi}^2}\nabla_\mu\bar{\chi}\,\nabla^\mu\eta \\ &- \frac{1-\bar{\chi}^2}{(1+\bar{\chi}^2)^2}(\nabla\bar{\chi})^2\eta.\end{aligned} \tag{26.14}$$

The full perturbation equations must also include $\delta\mathcal{S}$, response mixing, and metric variations. This fixed-background operator alone does not determine stability of the entire system.

26.7 Local source strength, flux, and operator freedom

At the reference point $G_n'(0) = -\sigma_1/f$, normalization gives $\lambda_n = 1/[f\sqrt{n^2-1}]$, for example $\lambda_2/\lambda_3 = \sqrt{8/3}$. This is a derivative mixing ratio between response and shared scalar, not the gauge-coupling ratio g_2/g_3.

The scalar current is $f\nabla^\mu\chi/\sqrt{1+\chi^2} = \nabla^\mu\phi$, or $J^\mu_{\rm geom} = \nabla^\mu\phi - Yw^\mu/f$ with the critical contribution included. Integrating its equation relates boundary flux to the source. Compact charge quantization or an electromagnetic Gauss law would require additional structure.

Covariance also allows $\xi R\phi^2$, variable $Z_a(\phi)$, phase portals, and higher derivatives. The source equation consistently combines their contributions, but does not fix their coefficients. Their absence in a benchmark is a matching condition unless protected by symmetry.

27 Mixed Curvature, Scattering, and Decay

27.1 Local curvature of normal geometry and cross-sector transport

The scalar curvature derived in Section 17 establishes that the response manifold is curved. Its local Riemann tensor identifies how that curvature couples sectors. At $\phi = 0$, the normalized matrices satisfy

$$\mathsf{G}_n' = -\lambda_n\sigma_x, \qquad \mathsf{G}_n'' = \frac{1}{f^2}\,\mathrm{diag}\left(\frac{n}{n+1}, \frac{n}{n-1}\right). \tag{27.1}$$

For $\mathrm{d}s^2 = \mathrm{d}\phi^2 + H_{ab}(\phi)\mathrm{d}y^a\mathrm{d}y^b$,

$$R_{\phi a\phi b} = -\tfrac{1}{2}H''_{ab} + \tfrac{1}{4}H'_{ac}H^{cd}H'_{db}, \tag{27.2}$$

$$R_{abcd} = -\tfrac{1}{4}(H'_{ac}H'_{bd} - H'_{ad}H'_{bc}). \tag{27.3}$$

so that

$$R_{\phi U_n\phi U_n} = -\frac{n}{2f^2(n+1)} + \frac{\lambda_n^2}{4}, \quad R_{\phi V_n\phi V_n} = -\frac{n}{2f^2(n-1)} + \frac{\lambda_n^2}{4}, \quad R_{U_nV_nU_nV_n} = \frac{\lambda_n^2}{4}. \tag{27.4}$$

For $n \neq m$, even without a direct off-diagonal metric block,

$$R_{U_nU_mV_nV_m} = -\frac{\lambda_n\lambda_m}{4}. \tag{27.5}$$

The shared ϕ direction connects different sectors through nonflat geometry. At $f = 1$, the values for $(2,3)$, $(2,4)$, and $(3,4)$ are respectively $-0.0510310, -0.0372678, -0.0228218$. For equally normalized small loop areas, transport coefficients have ratio $|\delta\beta_{23}|/|\delta\beta_{24}| = \sqrt{15/8}$, $|\delta\beta_{23}|/|\delta\beta_{34}| = \sqrt{5}$. Measuring these transport coefficients would require a physical protocol that controls the field-space loop and reads the transported quantity.

27.2 Normal coordinates and on-shell amplitudes

A change to normal coordinates removes the cubic derivative term:

$$\phi_{\rm old} = \varphi - \tfrac{1}{2}\lambda uv, \quad U_{\rm old} = u + \tfrac{1}{2}\lambda\varphi v, \quad V_{\rm old} = v + \tfrac{1}{2}\lambda\varphi u \tag{27.6}$$

but the mass potential then produces

$$\mathcal{L}_3 = \frac{\lambda}{2}(M_\phi^2 - m_U^2 - m_V^2)\varphi uv. \tag{27.7}$$

The absolute on-shell decay amplitude is therefore $|\lambda(M_\phi^2 - m_U^2 - m_V^2)|/2$. For $M_\phi > m_U + m_V$,

$$\Gamma_{\phi\to UV} = \frac{\lambda^2(M_\phi^2 - m_U^2 - m_V^2)^2}{64\pi M_\phi}\sqrt{1 - \frac{(m_U+m_V)^2}{M_\phi^2}}\sqrt{1 - \frac{(m_U-m_V)^2}{M_\phi^2}}. \tag{27.8}$$

For equally normalized sectors with light final states and no competing couplings, the widths have ratio $1/3 : 1/8 : 1/15$.

Shared-scalar exchange produces a cross-sector amplitude proportional to

$$\frac{\lambda_n \lambda_m}{4} \frac{(s - m_{U_n}^2 - m_{V_n}^2)(s - m_{U_m}^2 - m_{V_m}^2)}{M_\phi^2 - s}. \tag{27.9}$$

After eliminating a heavy scalar at low energy, the derivative four-field coefficient $C_{nm} = \lambda_n \lambda_m / M_\phi^2$ has rank one. Such factorization is generic to exchange of one mediator. The particular λ_n ratios come from CR's chosen response geometry; all-order protection remains open in Section 19.

28 The $c = 4$ Dark Sector

28.1 Kinetic terms and a minimal mass potential

Continuing the same matrix family to $c = 4$ gives

$$G_4 = \begin{pmatrix} 4\rho + 1 & -\chi \\ -\chi & 4\rho - 1 \end{pmatrix}, \quad \operatorname{spec} G_4 = (5\rho, 3\rho), \quad \det G_4 = 15\rho^2. \tag{28.1}$$

At the reference point define $S = \sqrt{5} u_4$, $V_4 = \sqrt{3} v_4$. Here S is the dark candidate field, while S_{tot} with an integral denotes the total action. Choose gauge-singlet response and add

$$V_4^{\text{pot}} = \frac{1}{2} M_\phi^2 \phi^2 + \frac{5}{2} m_4^2 u_4^2, \qquad M_\phi^2 > 0, \quad m_4^2 > 0. \tag{28.2}$$

At $\phi = u_4 = 0$ and constant v_4, the spectrum contains masses M_ϕ and m_4, together with a massless scalar V_4 protected by the v_4 shift. Setting its background to zero does not remove this fluctuation from the spectrum.[U]

The kinetic matrix is positive definite. A viable dark-matter model must also meet abundance, lifetime, clustering, and visible-coupling constraints.

28.2 Effective mass at fixed geometric background

For approximately constant ϕ on the relevant scales, the generalized mass eigenvalue problem for u_4, v_4 gives the sole nonzero response mass

$$m_S^2(\phi) = m_4^2 \frac{4 \cosh x - 1}{3 \cosh^2 x}, \qquad x = \phi / f. \tag{28.3}$$

It is not simply $5m_4^2/(4\rho + 1)$, because the off-diagonal kinetic term must be diagonalized too. This defines the dimensionless mass response to the shared scalar

$$\beta_4(\phi) = M_P \frac{d \ln m_S}{d\phi} = \sqrt{\frac{2}{3}} \frac{\sinh x \, (1 - 2 \cosh x)}{\cosh x \, (4 \cosh x - 1)}. \tag{28.4}$$

At the origin, $\beta_4(0) = 0$, and for finite real x, $|\beta_4| < 1/\sqrt{6}$. There is no linear diagonal scalar charge at this vacuum, but the off-diagonal $\phi S V_4$ vertex remains. The mass-eigenstate interpretation requires an approximately constant or adiabatically varying background.

28.3 Stability conditions in the isolated dark module

For only the kinetics of ϕ, u_4, v_4 and Eq. (28.2), there are two discrete transformations:

$$\begin{aligned} P_\phi &: \quad (\phi, u_4, v_4) \mapsto (-\phi, u_4, -v_4), \\ P_{y_4} &: \quad (\phi, u_4, v_4) \mapsto (\phi, -u_4, -v_4). \end{aligned} \tag{28.5}$$

Under their product $P_D = P_\phi P_{y_4}$, S and ϕ are odd, while V_4 is even. If the vacuum, every interaction, and quantum corrections preserve these symmetries, and $m_4 < M_\phi$, then S is the lightest P_D-odd particle in this isolated module and cannot decay singly.

This proof uses the isolated module's particle spectrum. Adding the other response sectors changes the list of odd states and can open decay channels, as shown in Section 29.

28.4 Open decays and annihilation

Expansion at the reference vacuum gives

$$\mathcal{L}_{\phi S V_4} = \lambda_4 \phi\, \partial_\mu S \partial^\mu V_4, \qquad \lambda_4 = \frac{1}{\sqrt{15} f}. \tag{28.6}$$

For $m_4 > M_\phi$, the tree-level two-body width of $S \to \phi V_4$ is

$$\Gamma_{S\to\phi V_4} = \frac{m_4^3}{960\pi f^2}\left(1 - \frac{M_\phi^2}{m_4^2}\right)^3. \tag{28.7}$$

For $M_\phi > m_4$, interchanging the parent and massive-daughter masses gives the width of $\phi \to S V_4$. Thus closing the decay of S can leave a heavier scalar that later feeds dark matter and dark radiation.

At momenta well below M_ϕ, integrating out ϕ gives

$$\Delta\mathcal{L}_{\text{eff}} = \frac{(\partial S \cdot \partial V_4)^2}{30 f^2 M_\phi^2} + O(\partial^2/M_\phi^2). \tag{28.8}$$

Single-particle stability of S therefore permits pair annihilation. For $SS \to V_4 V_4$, the threshold result is

$$|\mathcal{M}|_{\text{thr}} = \frac{2m_4^4}{15 f^2 (M_\phi^2 + m_4^2)}, \qquad (\sigma v_{\text{rel}})_{\text{thr}} = \frac{m_4^6}{3600\pi f^4 (M_\phi^2 + m_4^2)^2}. \tag{28.9}$$

The result assumes tree level, massless V_4, nonrelativistic initial particles, and the identical-final-state phase-space factor. With signature $(-+++)$, define $s = -(p_1+p_2)^2$, $t = -(p_1-k_1)^2$, and $u = -(p_1-k_2)^2$. At threshold $t = u = -m_4^2$ and $p_1 \cdot k_1 = -m_4^2$. These consistent signs correct the earlier derivation without changing its cross-section magnitude (U, internal reference 7, Section 7.3).

28.5 Schur complement, zero mode, and a nonintegrable field redefinition

At fixed ϕ, masses follow from $\det(\mathcal{M}^2 - \omega^2 G_4) = 0$, not simply diagonal entries of the potential Hessian. The dark sector has one massive mode and one massless direction. For exact completion of the square, let

$$h(x) = \frac{\sinh x}{4\cosh x - 1}, \qquad C(x) = \frac{15\cosh^2 x}{4\cosh x - 1}. \tag{28.10}$$

The kinetic term becomes $C(x)du_4^2 + (4\cosh x - 1)(dv_4 - h du_4)^2$. However,

$$d(dv_4 - h du_4) = -\frac{4 - \cosh x}{f(4\cosh x - 1)^2}\, d\phi \wedge du_4 \tag{28.11}$$

is generally nonzero. The completed square therefore does not define a global scalar coordinate, and the massless direction persists on a varying background. Its shift current is a Noether current.[U]

28.6 Dark mass response and a finite-range fifth force

The quantity $\beta_4 = M_P\, d\ln m_S/d\phi$ of Eq. (28.4) satisfies $|\beta_4| < 1/\sqrt{6}$ and $\beta_4(0) = 0$. Near the reference point $m_S/m_4 = 1 - x^2/6 + O(x^4)$, so mass shifts only quadratically; at large $|x|$, β_4 approaches a finite limit. If dark particles are quasistatic nonrelativistic point sources and the shared scalar is a linear mediator of mass $M_{\rm med}$, their effective attraction is schematically

$$\frac{G_{SS}(r)}{G} \simeq 1 + 2\beta_4^2 e^{-M_{\rm med} r} < \frac{4}{3}. \tag{28.12}$$

The bound applies to the approximate dark interaction just specified. For $M_\phi = 3\times 10^{-19}\,\mathrm{eV}$, the reduced Compton wavelength is about 4.4 AU. At $m_4 = 10^{-19}\,\mathrm{eV}$ and speed 10^{-3}, the reduced de Broglie wavelength is 0.064 pc and the full wavelength $2\pi/(m_4 v)$ is 0.40 pc. Baryonic force bounds would depend on additional visible couplings.

For conserved number density and slowly background-varying mass,

$$\rho_S = n_0 a^{-3} m_S(\phi), \qquad \dot\rho_S + 3\mathcal{H}\rho_S = \frac{\beta_4}{M_P}\dot\phi\,\rho_S. \tag{28.13}$$

A fixed-mass cold-dark-matter approximation therefore requires a background near $\phi = 0$ or sufficiently small, slow mass variation. Adding ϕ portals to baryons or gauge fields requires renewed analysis of total energy exchange and equivalence-principle constraints.

28.7 Response sequences, mass protection, and carrier nonuniqueness

If each normal sector has the same form of mass potential, equal masses and phase space give isolated two-body widths with

$$\Gamma(U_c \to \phi V_c) = \frac{m_c^3}{64\pi f^2 (c^2-1)}\left(1 - \frac{M_\phi^2}{m_c^2}\right)^3. \tag{28.14}$$

with coefficient ratio $1/3 : 1/8 : 1/15$ for $c = 2, 3, 4$. The physical widths also depend on masses, portals, and available channels. Restoring the u_4 shift as $m_4 \to 0$ can protect a small m_4; it does not protect M_ϕ, since G_c depends explicitly on ϕ.

The gauge-singlet fields u_4, v_4 do not exhaust a rank-four carrier such as $E_2 \otimes E_2$ or $E_3 \oplus \det E_2$. Nor do the present assumptions impose $c \le 4$: further normal sectors would add spectra, parameters, and stability conditions.

29 Shared-Mediator Interactions and Dark Stability

29.1 Interactions induced by the common mediator

For any $n > 1$, define at the common origin

$$U_n = \sqrt{n+1}\, u_n, \qquad V_n = \sqrt{n-1}\, v_n, \qquad \lambda_n = \frac{1}{f\sqrt{n^2-1}}. \tag{29.1}$$

Every response action contains $\lambda_n \phi\, \partial U_n \cdot \partial V_n$. Thus

$$\lambda_2 : \lambda_3 : \lambda_4 = \frac{1}{\sqrt{3}} : \frac{1}{\sqrt{8}} : \frac{1}{\sqrt{15}}. \tag{29.2}$$

The ratio concerns scalar vertices normalized by the same f. The gauge couplings g_Y, g_2, g_3 are independent parameters of this action.

For a heavy ϕ and process energies far below M_ϕ, set $J = \sum_{n=2}^{4} \lambda_n \partial U_n \cdot \partial V_n$, with $U_4 = S$. Integrating out the shared mediator gives

$$\Delta\mathcal{L}_{\text{eff}} = \frac{J^2}{2M_\phi^2}. \tag{29.3}$$

Expanding the square produces both self-interactions and cross-sector terms, for example

$$\Delta\mathcal{L}_{4n} = \frac{\lambda_4 \lambda_n}{M_\phi^2} (\partial S \cdot \partial V_4)(\partial U_n \cdot \partial V_n), \quad n = 2, 3. \tag{29.4}$$

The shared mediator generates these terms even when no direct portal is added. They must therefore enter the stability analysis of the combined model.

29.2 Global dark parity and additional light particles

Extending P_ϕ in Eq. (28.5) to all response modules requires

$$P_\phi : \quad \phi \mapsto -\phi, \quad u_n \mapsto u_n, \quad v_n \mapsto -v_n, \quad X \mapsto X, \quad Y \mapsto -Y. \tag{29.5}$$

so that every off-diagonal kinetic term and w remain invariant. A symmetry of the total action further requires even visible gauge, Higgs, and ordinary matter fields and compatible even scalar-dependent functions. P_{y_4} still flips only the dark response pair.

The product $P_D = P_\phi P_{y_4}$ then has parity assignments:

Field	P_D parity	Role in stability
ϕ	Odd	Shared mediator; possibly a heavier odd state.
S	Odd	Dark-matter candidate to be tested.
V_4	Even	Can appear in decay or annihilation final states.
U_2, U_3	Even	Can enter cross-sector final states.
V_2, V_3	Odd	If light, can displace S as the lightest odd state.
X, Y	Even, odd	Critical constrained system; physical spectrum requires separate treatment.

Field	P_D parity	Role in stability
Standard Model matter and gauge fields	Even	Defined here as part of the selected symmetry.

If the $c = 2, 3$ responses are massless, V_2, V_3 are light odd states and S is no longer the lightest odd particle. The isolated condition $m_4 < M_\phi$ is then insufficient.

29.3 Off-shell three-body decay

Equation (29.4) allows

$$S \longrightarrow V_4 + U_n + V_n, \qquad n = 2, 3, \tag{29.6}$$

whenever the final masses sum to less than m_4. The mediator ϕ is off shell, so the process remains possible for $M_\phi > m_4$. Parity is conserved because V_n carries odd parity in the final state.

For a checkable estimate, take all three final particles massless, $m_4 \ll M_\phi$, and neglect interference from other operators. Equation (29.4) gives

$$\Gamma_{S\to V_4U_nV_n} = \frac{(\lambda_4\lambda_n)^2 m_4^9}{245760\pi^3 M_\phi^4}. \tag{29.7}$$

Appendix C derives the phase-space coefficient. For ultralight masses the width can be extremely small, but it is nonzero. Cosmological longevity must be checked using the actual masses and abundance; it is weaker than exact symmetry protection.

29.4 Conditions for a stable dark eigenstate

With the diagonal mass baseline of Eq. (26.2), classical $w = 0$ truncation, and no other light odd states in the low-energy content, one may impose

$$m_4 < M_\phi, \qquad m_4 < \mu_{V2}, \qquad m_4 < \mu_{V3}. \tag{29.8}$$

alongside unbroken P_ϕ and P_{y_4} in the full action and vacuum. Only then can S be the lightest odd state of that low-energy model. Allowed odd-state mixing requires Eq. (29.8) to compare actual mass eigenstates, not undiagonalized Lagrangian parameters.

The result is conditional on the low-energy spectrum. The full critical quantum spectrum is unresolved, so it does not establish stability in every CR completion. Models with open but sufficiently slow channels instead describe decaying dark matter.

30 Characteristic Cones and the Critical Exception

30.1 Normal-sector principal symbols

Let $q(k) = g^{\mu\nu}k_\mu k_\nu$. In original normal-sector coordinates,

$$\det P_n(k) = (n^2 - 1)\rho^2 q(k)^2. \tag{30.1}$$

Including the shared scalar, the principal determinant for seven normal real fields is

$$\det P_{\text{normal}}(k) = 360\,\rho^6 q(k)^7. \tag{30.2}$$

Reference canonical variables change the nonzero prefactor but not the characteristic zero set. With conventional gauge fixing and positive kinetic functions, Yang–Mills and Higgs principal parts are also set by $q(k)$; the Weyl determinant is $-q(k)$ and the Einstein equations have the same harmonic-gauge principal cone. Masses, ordinary potentials, and lower-derivative vertices do not alter this wavefront cone.

The selected normal dynamics thus realizes the cone required by P1.2. The critical sector is the remaining local propagation test.

30.2 The reduced critical principal part

On a critical background $\bar{w} = 0, \bar{Y} = 0$ with $n_\mu = \partial_\mu \bar{\phi}$, the quadratic critical term is

$$\mathcal{L}^{(2)}_{\text{crit}} = -\left(\partial_\mu x - \frac{y}{2f} n_\mu\right)^2 . \tag{30.3}$$

For $n^2 \neq 0$, eliminating y gives the projected principal part

$$P^{\mu\nu}_{\text{crit}} = g^{\mu\nu} - \frac{n^\mu n^\nu}{n^2}. \tag{30.4}$$

For timelike n, it may have only an elliptic spatial constraint structure; for spacelike n, one spatial direction is projected out. For example, $P_{\text{crit}} = \text{diag}(-1, 0, 1, 1)$, and the covector $k = (1, 2, 1, 0)$ obeys $P^{\mu\nu}_{\text{crit}} k_\mu k_\nu = 0$ while $q(k) = 4$. The normal-block determinant argument therefore cannot be applied to the critical block. This does not yet prove superluminal propagation of an independent physical signal: all constraints, remaining variables, and well-posedness must be treated. For zero or null n, the division fails and the original constrained system must be used.

30.3 Critical propagation and P1

After reduction, the critical directions must be classified as constraints, auxiliary variables, or propagating modes. Any physical mode incompatible with P1's cone would rule out that background or require changing the model. The consistent truncation $w = 0$ defines a normal classical subtheory, but leaves the critical quantum spectrum open.

Local principal-symbol compatibility does not settle the global initial-value problem. Gauge fixing, constraint propagation, boundaries, and global hyperbolicity remain necessary. The finite-dimensional clock and probe evolution below demonstrates existence on particular branches, not well-posedness of every inhomogeneous field.

31 Dark Cosmology and Background Stability

31.1 Free-scalar energy density

In spatially flat FRW, $ds^2 = -d\tau^2 + a(\tau)^2 d\boldsymbol{x}^2$, the minimal core's homogeneous free scalar obeys $\ddot{\phi} + 3\mathcal{H}\dot{\phi} = 0$, making $a^3\dot{\phi}$ constant, with energy density

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 \propto a^{-6}, \qquad p_\phi = \rho_\phi. \tag{31.1}$$

If it dominates, $a \propto \tau^{1/3}$ and $\phi = \phi_0 \pm \sqrt{2/3} M_P \ln(\tau/\tau_0)$. This is stiff matter, neither cold matter scaling as a^{-3} nor constant dark energy. A gas of free relativistic massless scalar particles usually scales as radiation, a^{-4}; it must be distinguished from a homogeneous kinetic background.[U]

Cold dark matter requires a mass or another mechanism that changes this scaling. Dark energy requires a constant or slowly varying potential, or a different source of negative pressure.

31.2 An exact reference dark-condensate solution

With baseline Eq. (26.2), a visible background not sourcing ϕ, and other responses zero, $\overline{\phi} = \overline{V}_4 = 0$ is a consistent classical solution and S obeys

$$\ddot{S} + 3\mathcal{H}\dot{S} + m_4^2 S = 0, \qquad \rho_S = \frac{1}{2}\dot{S}^2 + \frac{1}{2}m_4^2 S^2. \tag{31.2}$$

For $m_4 \gg \mathcal{H}$ and a background varying slowly during one oscillation, $S \simeq A(\tau)\cos(m_4\tau + \delta)$, $A \propto a^{-3/2}$. Period averaging gives $\langle p_S \rangle \simeq 0$, $\langle \rho_S \rangle \propto a^{-3}$, producing cold-matter-like background evolution.[U]

The condensate amplitude is initial-state data, so the branch does not predict its own abundance. Other relics, nonlinear conversion, and resonance must be included in the energy budget.

31.3 Initial misalignment and an abundance example

Suppose Hubble friction initially freezes S near S_i, and oscillations begin during radiation domination at $\mathcal{H}(a_{\rm osc}) \simeq m_4$. Ignoring changes in relativistic degrees of freedom, entropy injection, and exact onset coefficients,

$$a_{\rm osc} \simeq \left(\frac{H_0\sqrt{\Omega_r}}{m_4}\right)^{1/2}, \qquad \rho_{S0} \simeq \frac{1}{2}m_4^2 S_i^2 a_{\rm osc}^3. \tag{31.3}$$

If S supplies all dark matter,

$$\frac{S_i}{M_P} \simeq \sqrt{6\Omega_{\rm DM}}\left(\frac{H_0}{m_4}\right)^{1/4}\Omega_r^{-3/8}. \tag{31.4}$$

For the source's illustrative values $H_0 = 67.4\,{\rm km\,s^{-1}Mpc^{-1}}$, $\Omega_r = 9.2\times10^{-5}$, $\Omega_{\rm DM} = 0.265$, and $m_4 = 10^{-19}\,{\rm eV}$, the estimate gives $a_{\rm osc} \simeq 1.17\times10^{-8}$ and $S_i \simeq 0.0142 M_P$. These are benchmark inputs and outputs, not a joint cosmological fit.[U]

If the shared scalar also starts radiation-era oscillations from displacement ϕ_i, with quadratic potentials, separate onsets, and valid later cold-matter approximations for both fields,

$$\frac{\rho_\phi}{\rho_S} \simeq \left(\frac{\phi_i}{S_i}\right)^2\sqrt{\frac{M_\phi}{m_4}}. \tag{31.5}$$

For $M_\phi = 3m_4$, keeping this ratio below one percent requires $|\phi_i|/S_i \lesssim 0.076$. The shared scalar and the $c = 2, 3$ responses therefore need their own initial-state and decay histories.

31.4 Primordial isocurvature and dark radiation

If S is a light inflationary spectator, its perturbation is uncorrelated with the adiabatic mode, its potential is approximately quadratic, and misalignment supplies all dark matter, then $\delta S \simeq H_{\rm inf}/(2\pi)$ and $\rho_S \propto S_i^2$ give

$$\mathcal{P}_{\rm iso} \simeq \left(\frac{H_{\rm inf}}{\pi S_i}\right)^2, \qquad \beta_{\rm iso} = \frac{\mathcal{P}_{\rm iso}}{\mathcal{P}_{\mathcal{R}} + \mathcal{P}_{\rm iso}}. \tag{31.6}$$

As a specified data example, Planck 2018 gives a 95% upper bound $\beta_{\rm iso} < 0.038$ for a nearly scale-invariant uncorrelated cold-dark-matter isocurvature model. With $\mathcal{P}_{\mathcal{R}} = 2.1 \times 10^{-9}$ and the above S_i, this illustrative branch requires $H_{\rm inf} \lesssim 1.0 \times 10^{12}\,\mathrm{GeV}$. This uses a particular historical data combination and perturbation model, not a universal bound on all CR initial states. (Internal reference 17 of U, Section 9.4.1.)

If S is heavy during inflation, supplies only part of dark matter, is produced afterward, or correlates significantly with other modes, Eq. (31.6) and its numerical bound change. The geometric core does not uniquely specify inflation.

The existence of massless V_4 does not determine its density. Only if it has a thermal distribution and a definable temperature T_{V_4} does one real scalar contribute to effective neutrino number as

$$\Delta N_{\rm eff} = \frac{4}{7}\left(\frac{T_{V_4}}{T_\nu}\right)^4 . \tag{31.7}$$

If it never thermalizes and is produced by late decay, its distribution or injection history must be computed. Planck-suppressed interactions may make thermalization weak, but reheating temperature and allowed portals still matter.

31.5 Coupled transverse perturbations and resonance

The reference solution $\overline{\phi} = \overline{V}_4 = 0$ does not eliminate transverse perturbations. With $x = \delta\phi$, $v = \delta V_4$, their quadratic action on fixed FRW is

$$\begin{aligned} S^{(2)}_{xv} = \int d\tau\, d^3x\, a^3 \Bigg[&\frac{1}{2}\dot{x}^2 - \frac{(\boldsymbol{\nabla} x)^2}{2a^2} + \frac{1}{2}\dot{v}^2 - \frac{(\boldsymbol{\nabla} v)^2}{2a^2} \\ &- \frac{\dot{\overline{S}}}{\sqrt{15}f} x\dot{v} + \frac{\dot{\overline{S}}^2}{5f^2}x^2 - \frac{1}{2}M_\phi^2 x^2 \Bigg]. \end{aligned} \tag{31.8}$$

The transverse modes have zero linear stress source on this background, allowing a separate linear stability test. Their equations are

$$\begin{aligned} \ddot{x}_k + 3\mathcal{H}\dot{x}_k + \left(\frac{k^2}{a^2} + M_\phi^2 - \frac{2\dot{\overline{S}}^2}{5f^2}\right) x_k + \frac{\dot{\overline{S}}}{\sqrt{15}f}\dot{v}_k = 0, \\ \ddot{v}_k + 3\mathcal{H}\dot{v}_k + \frac{k^2}{a^2}v_k = \frac{1}{\sqrt{15}fa^3}\frac{d}{d\tau}\left(a^3\dot{\overline{S}}x_k\right). \end{aligned} \tag{31.9}$$

For generic $k \neq 0$, v_k cannot simply be set to zero or eliminated algebraically. Periodic $\dot{\overline{S}}$ may cause mixing and parametric resonance. Growth rates require the full coupled system, including expansion sweeping through resonance bands. (Internal reference 7 of U, Section 8.)

Only at $k = 0$, on the special branch with zero conserved perturbation momentum, does $\dot{v} = \dot{\overline{S}}x/(\sqrt{15}f)$ give

$$\ddot{x} + 3\mathcal{H}\dot{x} + \left(M_\phi^2 - \frac{\dot{\overline{S}}^2}{3f^2}\right) x = 0. \tag{31.10}$$

For $\overline{S} \simeq A\cos(m_4\tau)$, the condition $M_\phi^2 > m_4^2 A^2/(3f^2)$ keeps the instantaneous squared mass positive in this special mode equation. Stability at nonzero momentum requires the coupled Floquet problem, independently of particle lifetimes and annihilation rates.

31.6 Structure formation and observable deviations

For nonrelativistic weak gravity, rapid oscillations, and negligible extra couplings, a massive real scalar has effective sound speed $c_s^2 \simeq k^2/(4m_4^2a^2)$ and schematic density equation

$$\ddot{\delta} + 2\mathcal{H}\dot{\delta} + \left(\frac{k^4}{4m_4^2a^4} - 4\pi G_N\rho_S\right)\delta \simeq 0. \tag{31.11}$$

Wave pressure suppresses structure below a mass-dependent scale. This feature is shared by other light-scalar dark-matter models.[U]

On the reference branch $\overline{\phi} = \overline{V}_4 = 0$ with initially zero transverse perturbations, linear S and metric perturbations can reduce to ordinary GR plus a massive scalar. More distinctive signals should involve correlations induced by the common mediator: cross-sector energy conversion, background-dependent masses, transverse-mode production, dark radiation, and several processes tied to the common f.

On nonzero slowly varying ϕ, the scalar charge of mass eigenstates follows from Eq. (28.4). In a valid static weak-field limit, two dark bodies may have a Yukawa correction $2\beta_4^2 e^{-M_{\rm eff}r}$ relative to Newtonian gravity. Baseline minimally coupled visible matter with ϕ-independent masses has no analogous tree-level diagonal charge; this dark formula is not a universal fifth force for all matter.

31.7 Abundance scaling and additional shared-scalar relics

The radiation-era misalignment estimate gives $S_i \propto m_4^{-1/4}$ at fixed dark-matter fraction. For masses 10^{-21}, 10^{-20}, 10^{-19}, 10^{-18} eV, the required S_i/M_P is approximately 0.0451, 0.0253, 0.0143, 0.00802. These values illustrate scaling without early entropy injection, background-dependent mass, or a detailed structure-formation fit.

The scalar relic abundance follows conditionally from Eq. (31.5). A heavy-enough mediator can forbid S decay while remaining cosmologically long lived itself. If portals affect decay or oscillation onset, coupled energy-transfer equations must replace the independent misalignment estimates.[U]

31.8 Local effective masses and full Floquet stability

In the short-time approximation $S = A\cos(m_4\tau)$, the zero-momentum condition above gives a threshold $M_\phi > m_4A/(\sqrt{3}f)$, about $0.0067m_4$ for the sample amplitude. A full stability calculation must include derivative mixing, the kinetic and frequency matrices, all wavenumbers, and expansion.

A counterexample outside the principal heavy-mediator baseline is matter domination with $M_\phi \ll \mathcal{H}$. There $m_{x,\rm eff}^2 \simeq -2\mathcal{H}^2/3$, and $a \propto \tau^{2/3}$ gives

$$x \propto \tau^{p_+}, \qquad p_+ = \frac{-1+\sqrt{59/27}}{2} \simeq 0.239, \qquad x \propto a^{3p_+/2} \simeq a^{0.359}. \tag{31.12}$$

The exponent changes by 3/2 when expressed in a rather than τ. This growth occurs in the extremely light-mediator regime, outside the main baseline $M_\phi > m_4$.

31.9 Primordial conditions, linear degeneracy, and observational interfaces

Primordial isocurvature bounds depend on a light field during inflation, negligible correlation, dark-matter fraction, and the historical Planck inputs. Under these assumptions, the source gives the corresponding $H_{\rm inf}$ bound and illustrative $r \lesssim 1.5\times10^{-5}$. They are not updated universal bounds on all CR models. Inflationary mass, initial states, and late dilution can change them.

With $\bar{\phi} = V_4 = 0$ and no initial transverse perturbations, the linear adiabatic system can coincide with ordinary real-scalar dark matter. Distinguishing effects require nonzero backgrounds, transverse modes, nonlinear conversion, finite-range forces, or portals, followed through their coupled evolution.

32 Bounces, Plateau Potentials, and Four-Form Cosmology

32.1 Conditions and branches for a positive-energy bounce

The original free scalar satisfies the null energy condition. In spatially flat FLRW, $\dot{\mathcal{H}} = -(\rho + p)/(2M_P^2) \leq 0$, so it alone cannot produce a conventional smooth contraction-to-expansion bounce. Positive curvature $k = +1$ and a positive cosmological constant change this. With conserved $P = a^3\dot{\phi}$,

$$\mathcal{H}^2 = H_\Lambda^2 - a^{-2} + Ca^{-6}, \qquad C = \frac{P^2}{6M_P^2}, \quad x = H_\Lambda^2 a^2, \quad \mu = CH_\Lambda^4. \tag{32.1}$$

The turning points satisfy $x^3 - x^2 + \mu = 0$. For $0 < \mu < 4/27$, the larger positive root has $2/3 < x_b < 1$ and $\dot{\mathcal{H}}_b/H_\Lambda^2 = 3 - 2/x_b > 0$, giving a large-radius bounce with positive kinetic energy. The small-radius branch may remain singular. Completeness is a property of the selected real-slice solution; complex extension and perturbative stability require separate analysis.[U]

Curvature and vacuum energy thus allow exact bouncing solutions. Connecting them to a hot cosmological history requires anisotropic and inhomogeneous stability tests. On the GR reference branch, standard weak-field results are inherited: for example, $\Delta\varpi = 6\pi GM/[a(1-e^2)c_{\text{light}}^2]$ gives the historical Mercury value of 42.98 arcseconds per century.

32.2 A bounded invariant, plateau potentials, and earlier mass potentials

The shared scalar defines the bounded dimensionless invariant

$$I(\phi) = \frac{\chi^2}{1+\chi^2} = \tanh^2(\phi/f). \tag{32.2}$$

The earlier plateau–quartic model uses $U = \rho_\Lambda + V_0 I^2$. It approaches a constant at large field and a quartic potential near zero. Rapid homogeneous oscillations can therefore have a radiation-like average equation of state, although thermalization into Standard Model radiation would require interactions. Its curvature is

$$U'' = \frac{4V_0}{f^2} I(1-I)(3-5I) \tag{32.3}$$

which becomes negative in part of the field range. Corrections proportional to $[U'']^2$ also leave the original two-parameter family, requiring further operators.

Another early branch uses $U_5 = \frac{1}{2}m_\chi^2 f^2 \tanh^2(\phi/f)$, expanding as $\frac{1}{2}m_\chi^2\phi^2 - m_\chi^2\phi^4/(3f^2) + \cdots$ near zero and bounded at large field. Its $U_5'' = m_\chi^2(1-I)(1-3I)$ is negative for $|\phi/f| > \operatorname{artanh}(1/\sqrt{3}) \simeq 0.65848$. Negative local potential curvature does not alone establish instability of a complete cosmological background; kinetics, Hubble damping, and coupled perturbations matter. This also differs from the exactly quadratic two-flux potential below.

32.3 A single four-form and the geometric-volume identification obstruction

An ordinary three-form potential A_3 in four dimensions has field strength $F_4 = dA_3$ without ordinary local propagating modes. Fixed flux contributes vacuum energy $\rho_F = q^2/2$. Identifying

it with the imaginary complex volume, $\mathrm{Im}\,\Omega_\gamma = 2\chi\ \mathrm{vol}_g$, via $F_4 = \mu_*^2\,\mathrm{Im}\,\Omega_\gamma$, makes the source-free equation $d\star F_4 = 0$ require $d\chi = 0$. It freezes the desired oscillating scalar, so the identification cannot simultaneously describe free four-form flux and a dynamical geometric scalar.

A field-dependent kinetic factor $Z(\phi) = 1 - \zeta I$, $0 < \zeta < 1$, instead gives fixed-flux energy $q^2/[2Z(\phi)]$ and $\delta m_\phi^2 = \zeta q^2/f^2$ near zero. At the source's dark-energy flux scale, $q/f \sim 2.378 \times 10^{-33}\,\mathrm{eV}$ is far below the illustrative $10^{-18}\,\mathrm{eV}$ dark-matter mass. That single flux cannot account for both scales within this prescription.

32.4 Orthogonal decomposition of two four-forms

Adding two independent three-form potentials and linear couplings $c_i\phi F_{4i}$, with the proper Legendre boundary term for fixed-flux variation, gives

$$U_{\mathrm{DU}}(\phi) = \rho_{\mathrm{ren}} + \frac{1}{2}\sum_{i=1}^{2}(q_i + c_i\phi)^2. \tag{32.4}$$

Here c_i have mass dimension one and are not response weights c; q_i have dimension two. With $\boldsymbol{c} = (c_1, c_2)$, $\boldsymbol{q} = (q_1, q_2)$,

$$\begin{aligned} m^2 &= |\boldsymbol{c}|^2, \qquad \phi_* = -\frac{\boldsymbol{c}\cdot\boldsymbol{q}}{|\boldsymbol{c}|^2}, \\ \boldsymbol{q}_\perp &= \boldsymbol{q} - \frac{\boldsymbol{c}\cdot\boldsymbol{q}}{|\boldsymbol{c}|^2}\boldsymbol{c}, \\ U_{\mathrm{DU}} &= \frac{1}{2}m^2(\phi - \phi_*)^2 + \rho_{\mathrm{ren}} + \frac{1}{2}|\boldsymbol{q}_\perp|^2. \end{aligned} \tag{32.5}$$

The parallel flux fixes the equilibrium point; the orthogonal flux fixes the residual vacuum energy. The mass and vacuum constant are independent unless further matching is imposed. At fixed flux and identical initial data, local low-energy evolution coincides with GR coupled to a massive real scalar and a cosmological constant.

In conformal time η, with $\mathcal{H}_\eta = a'/a$, Newtonian-gauge scalar perturbations obey

$$\delta\phi'' + 2\mathcal{H}_\eta\delta\phi' + (k^2 + a^2U'')\delta\phi = \bar{\phi}'(\Psi' + 3\Phi') - 2a^2U'\Psi. \tag{32.6}$$

with $\delta\rho = \bar{\phi}'\delta\phi'/a^2 - \bar{\phi}'^2\Psi/a^2 + U'\delta\phi$. Observational differences require joint evolution with all other species and metric perturbations, not just comparison of background potentials.

32.5 Membrane transitions, energy accounting, and historical numbers

The source retains an example with $m = 10^{-18}\,\mathrm{eV}$, $c_1 = c_2 = m/\sqrt{2}$, $\boldsymbol{q} = (0, Q)$, $\rho_{\mathrm{ren}} = 0$, and $Q \simeq 1.003 \times 10^{-5}\,\mathrm{eV}^2$. These parameters assign dark-matter amplitude and vacuum energy; they do not derive constants of nature.

A membrane transition from (e, Q) to $(0, Q)$ shifts equilibrium by $|\Delta\phi_*| = |e|/(\sqrt{2}m)$. The oscillation energy scale around the new minimum is $e^2/4$, while the instantaneous potential drop is $eQ/2$. For the historical $e \simeq 1.905 \times 10^7\,\mathrm{eV}^2$, these are respectively $9.07 \times 10^{13}\,\mathrm{eV}^4$ and $95.5\,\mathrm{eV}^4$. Heating must be assessed from the latter energy balance.

In a particular flat Euclidean trial model with zero intrinsic membrane tension, the scalar wall width is about m^{-1} and tension $e^2/(8m)$. Source values $mR \simeq 2.0522 \times 10^{-6}$, $B_4 \simeq 2.7866 \times 10^{51}$ imply extreme tunneling suppression, not a completed gravitational vacuum-decay rate. At $\mathcal{H} \sim m$ during radiation domination, historical estimates give about 329 seconds and a comoving causal scale of 1.72 kpc. A local bubble cannot thereby reset the entire universe instantly. Compact shifts

would additionally require $c_i\Delta\phi/e_i \in \mathbb{Z}$; the current noncompact real scalar does not automatically supply axion periodicity.

32.6 Data benchmarks and temperature scales

The source's compressed DESI DR2 BAO calculation gives $\chi^2 \simeq 32.07$ at fixed Planck parameters, 10.46 after fitting H_0, Ω_m, and 16.18 with compressed Planck information. These values depend on its reconstructed covariance and compression scheme; no official full likelihood is rerun here. Fixed-flux and ordinary scalar cosmologies with the same parameters have the same local predictions. The estimate $k_{P/2} \sim 275\,\mathrm{Mpc}^{-1}$ at $m = 10^{-18}\,\mathrm{eV}$ is semianalytic, while the very large thermal action S_3/T near 63.3 keV leaves reheating unresolved.[U]

The following historical benchmarks distinguish unit conversions from model-dependent transition temperatures.

Scale or process	Temperature equivalent	Interpretation
$1\,\mathrm{GeV}/k_B$	1.16045×10^{13} K	Unit conversion, not a phase transition.
Order 10^{15} degrees Celsius	About 86.17 GeV	Celsius and kelvin differ negligibly here; not a precise electroweak crossover temperature.
Historical QCD crossover	156.5 ± 1.5 MeV, about 1.82×10^{12} K	Uses specified strong-interaction thermodynamic input.
Historical electroweak crossover	159 ± 1 GeV, about 1.85×10^{15} K	Requires Standard Model parameters and a thermal-field-theory definition.
Assumed unification scale	10^{15}–10^{16} GeV	About 10^{28}–10^{29} K; model dependent.
Reduced Planck scale	2.435×10^{18} GeV	About 2.83×10^{31} K, distinct from the unreduced Planck temperature 1.42×10^{32} K; not a measured gravitational-transition temperature.

33 The Earlier Orientation Model

33.1 The orientation target and its assumptions

The earlier theoretical-physics paper uses an internal orientation variable distinct from the present common-null-cone scalar. Its core consists of a positive-definite Hermitian form, a compatible complex bilinear form, and a real antilinear involution. Orientation is represented by $A = I - 2nn^{\mathsf{T}}$, with $n^{\mathsf{T}}n = 1$ and $n \sim -n$, giving target

$$\mathbb{RP}^3 \simeq O(4)/(O(1) \times O(3)). \tag{33.1}$$

The orientation n is an internal scalar variable, not a spacetime rest direction. The construction supplies Einstein and Standard Model sectors, global internal symmetry, sigma-model kinetics, a breaking direction n_*, and a potential. Its scale f_{o} is independent of the present normalization $f = \sqrt{3/2}M_P$.[U]

An internal complex frame $E_\mu{}^A = in^A e_\mu{}^0 + s_a{}^A e_\mu{}^a$ can satisfy $B_{AB}E_\mu{}^A E_\nu{}^B = g_{\mu\nu}$ independently of orientation. The induced real metric therefore does not supply n dynamics; a sigma-model action corresponding to $\mathrm{Tr}(\partial A\,\partial A) = 8(\partial n)^2$ must be added.

33.2 Local positive kinetics, constraints, and a degenerate mass triplet

Take $\mathcal{L} = -\frac{1}{2}f_o^2(\partial n)^2 - V(n)$ with $V = \frac{1}{2}f_o^2 m^2[1-(n\cdot n_*)^2]$. The projected equation is $(I - nn^{\mathsf{T}})(f_o^2\Box n - \partial_n V) = 0$. Constraints $n^2 = 1$, $n\cdot p = 0$ have on-shell bracket two, forming a second-class pair and leaving three scalar modes. The global identification $n \sim -n$ removes no local mode.

Near $n_* = e_4$, set $n^a = \pi^a/f_o$, $n^4 = \sqrt{1-\pi^2/f_o^2}$. The target metric is

$$G^o_{ab} = \delta_{ab} + \frac{\pi_a\pi_b}{f_o^2-\pi^2}, \qquad (G_o^{-1})^{ab} = \delta^{ab} - \frac{\pi^a\pi^b}{f_o^2}, \qquad \det G_o = \frac{f_o^2}{f_o^2-\pi^2}. \tag{33.2}$$

The target has curvature $6/f_o^2$ and three equal local masses m. Its scattering theory differs from that of the negatively curved normal-response manifold.

33.3 Scattering, radial completion, and effective-theory range

At tree level in the stated normalization, the four-point amplitude is

$$\mathcal{M}_{ab\to cd} = \frac{s\,\delta_{ab}\delta_{cd} + t\,\delta_{ac}\delta_{bd} + u\,\delta_{ad}\delta_{bc}}{f_o^2}. \tag{33.3}$$

The identical-component amplitude is $4m^2/f_o^2$ and the singlet amplitude $(2s+4m^2)/f_o^2$. At high energy, $a_0 \simeq s/(16\pi f_o^2)$ reaches the perturbative-unitarity boundary near $\sqrt{s}\sim\sqrt{8\pi}f_o$.

One may add a linear radial field $\Phi = (f_o+\sigma)n$ with potential

$$U = \frac{\lambda}{4}(\Phi^2 - f_o^2)^2 + \frac{m^2}{2}\big[\Phi^2 - (\Phi\cdot n_*)^2\big], \qquad M_\sigma^2 = 2\lambda f_o^2. \tag{33.4}$$

Eliminating σ at low energy gives

$$\Delta\mathcal{L} = \frac{f_o^2}{2M_\sigma^2}\left[(\partial n)^2 + m^2\big(1-(n\cdot n_*)^2\big)\right]^2 + \cdots. \tag{33.5}$$

The radial field, potential, and matching scale are new input. The linear Φ theory does not globally impose $\Phi\sim-\Phi$; exact recovery of $\mathbb{RP}^3$ needs an additional $\mathbb{Z}_2$ quotient.

33.4 Higgs portals, nonminimal coupling, and naturalness

The orientation portal $\kappa f_o^2[1-(n\cdot n_*)^2]H^\dagger H/2$ locally equals $\kappa\pi^2 H^\dagger H/2$. After electroweak breaking,

$$m_\pi^2 = m^2 + \frac{\kappa v^2}{2}, \qquad \Gamma(h\to\pi\pi) = \frac{3\kappa^2 v^2}{32\pi m_h}\sqrt{1-\frac{4m_\pi^2}{m_h^2}}. \tag{33.6}$$

The factor three counts orientation modes and cannot be transferred to the $c=4$ baseline with one massive dark mode. The limit $m,\kappa\to 0$ can restore internal Goldstone protection; a finite portal generally has matching sensitivity $\delta m^2 \sim \kappa M_{\rm match}^2/(16\pi^2)$.

With nonminimal gravitational coupling $F_{\rm grav} = M_P^2 + \xi\pi^2$, the Einstein-frame target metric is

$$G^E_{ab} = \frac{M_P^2}{F_{\rm grav}}G^o_{ab} + \frac{3M_P^2}{2F_{\rm grav}^2}(\partial_a F_{\rm grav})(\partial_b F_{\rm grav}), \tag{33.7}$$

For $F_{\rm grav} > 0$, positivity of the original target metric makes this metric positive as well. The coefficient ξ remains an extension parameter.

33.5 A single-circle truncation, nonrelativistic attraction, and density perturbations

Along one great circle, use $\varphi_{\rm o} = f_{\rm o}\vartheta$ with potential

$$V = \frac{m^2 f_{\rm o}^2}{2}\sin^2(\varphi_{\rm o}/f_{\rm o}) = \frac{1}{2}m^2\varphi_{\rm o}^2 - \frac{m^2}{6f_{\rm o}^2}\varphi_{\rm o}^4 + \cdots . \tag{33.8}$$

At small amplitude A, the average equation of state is $\langle w\rangle \simeq -A^2/(8f_{\rm o}^2)$. The negative quartic implies weak-amplitude attraction; the full periodic potential remains bounded and is not a negative-kinetic ghost. In conventional nonrelativistic normalization, $\lambda_4 = -4m^2/f_{\rm o}^2$, $g_{\rm NR} = -1/(2f_{\rm o}^2)$, and density perturbations satisfy schematically

$$\ddot{\delta} + 2\mathcal{H}\dot{\delta} + \left[\frac{k^4}{4m^2a^4} - \frac{\rho k^2}{2m^2 f_{\rm o}^2 a^2} - 4\pi G\rho\right]\delta = 0. \tag{33.9}$$

The terms represent quantum pressure, attractive self-interaction, and gravity. This applies to the single-circle truncation and its averaging approximation, not a completed transverse-stability proof for the three-orientation model.

33.6 Rank-five and spectral-action extensions

An earlier construction takes $W = V \oplus (\det V)^*$ from rank-four V, making $\det W$ trivial. Assigning $(-1/3, -1/3, -1/3, 1/2)$ to the original directions gives hypercharge $1/2$ to the added direction. This organizes rank-five labels, but an induced $U(4)$ connection alone does not provide full weak mixing between the fourth and fifth directions; new connection components are needed.

The spectral action $\mathrm{Tr}\, f_{\rm cut}(D_E^2/\Lambda^2)$ can generate gauge kinetic terms once its Dirac operator, internal bundle, and connection are given. Four-form vacuum stress and totally geodesic real slices likewise require their own field content and reality assumptions. These constructions remain distinct from the normal-response action.

34 Microscopic Input, Higgs Production, and Vacuum Response

A microscopic realization requires choices beyond P1 and P2. The source groups them into six levels, reviewed here before considering particle production and vacuum response.

34.1 Six testable levels of microscopic input

The microscopic-postulate document identifies quantum input missing from classical geometry. Its six levels are: select a common-null-cone restricted complex metric and real readout; add a positive-definite Hermitian $3+2$ splitting and total determinant condition; organize internal representations by even exterior algebra while separately supplying Spin structure, Grassmann fields, and fermion anticommutators; add the Higgs, Hermitian matter actions, and symmetry-allowed EFT counterterms; impose internal Gauss constraints and gravitational gauge reduction while testing the physical inner product; and specify quantum states with suitable short-distance behavior alongside independent long-wavelength cosmological initial data.[U]

These inputs define a conditional microscopic model. The sixteen dimensions of $\Lambda^{\rm even}E$ count internal labels, not the physical Hilbert space or generation number. Positivity of an internal Hermitian metric also leaves the stability of complex spacetime fluctuations to be checked.

34.2 Minimum requirements for BRST, Gauss constraints, and physical states

Formally one may write

$$\mathcal{H}_{\rm phys} = \frac{\ker Q_{\rm B}|_{\rm gh=0}}{\mathrm{im}\, Q_{\rm B}|_{\rm gh=-1}}, \qquad Q_{\rm B}^2 = 0. \tag{34.1}$$

A physical quotient requires anomaly-free constraints, suitable operator domains, and a positive inner product. Representation anomaly cancellation supplies only part of this argument. BRST ghosts remove gauge redundancy; they do not cure propagating negative kinetic modes.

Gauss constraints require dressing, boundary data, or singlet combinations for colored states. The Hadamard condition constrains short-distance singularities, but leaves inflationary states, dark-field displacements, and topological weights undetermined.

34.3 A bounded Higgs portal and nonadiabatic zero crossings

A microscopic baseline uses $V_{\rm port} = \kappa f^2 I(\phi) H^\dagger H$ with plateau potential $V_0 I^2$, defining an explicit early-universe extension. Near $\phi = 0$, $I \simeq \phi^2/f^2$. For a short crossing with $\phi \simeq \dot\phi_c(\tau - \tau_c)$, each real Higgs component obeys

$$\ddot h_p + \left[p^2 + m_0^2 + \kappa \dot\phi_c^2 (\tau - \tau_c)^2\right] h_p = 0. \tag{34.2}$$

Neglecting expansion and backreaction during this event gives

$$n_p = \exp\left[-\frac{\pi(p^2 + m_0^2)}{k_*^2}\right], \qquad k_*^2 = \sqrt{\kappa}\,|\dot\phi_c|, \qquad n_H = \frac{k_*^3}{2\pi^3} \tag{34.3}$$

for $m_0 = 0$ and all four real Higgs components. Here n_H is a number density from one crossing, not an already thermalized temperature.

For the historical benchmark $\kappa = 10^{-6}$ and $V_0/M_P^4 = 3.05475\times10^{-11}$, the source places the first crossing at dimensionless time 181.48695, with $\dot\phi_c/\sqrt{V_0} \simeq -0.276894$ and $\mathcal{H}_c M_P/\sqrt{V_0} \simeq 0.113041$. It estimates $k_* \simeq 9.5 \times 10^{13}$ GeV, $\mathcal{H}_c/k_* \simeq 0.016$, $n_H \simeq 1.4 \times 10^{40}\,\mathrm{GeV}^3$, and displacement $0.032 f$. These support the local crossing approximation. Repeated resonance, decay, plasma effects, backreaction, and the final reheating temperature are not calculated by that estimate.

34.4 Portal quantum corrections versus ultralight dark matter

When the field-dependent Higgs mass dominates other masses, the one-loop potential contains

$$\Delta U_H \simeq \frac{\kappa^2 f^4 I^2}{16\pi^2}\left[\ln\frac{\kappa f^2 I}{\mu^2} - \frac{3}{2}\right]. \tag{34.4}$$

The historical prefactor-to-V_0 ratio is about 4.66×10^{-4}. Smallness does not remove counterterms or keep logarithms small throughout the background range. Electroweak breaking also contributes

$$\Delta m_\phi^2 = \kappa v^2. \tag{34.5}$$

For $\kappa = 10^{-6}$, the corresponding mass is about 0.246 GeV. Keeping the same ϕ at 10^{-18} eV without extra cancellations needs $\kappa \lesssim 1.65 \times 10^{-59}$. The strong-portal early-production baseline therefore cannot directly be combined with an ultralight two-flux dark-matter baseline for that same field. Although $c = 4$ places dark matter in S, the shared ϕ mass and portals must still match all spectral conditions.

34.5 Vacuum response through additional four-form constraints

The microscopic manuscript also studies an optional vacuum response distinct from the two-flux potential. Using local variables $\mathcal{K}(x)$, $\Lambda(x)$ and two four-forms, its action has structure

$$\begin{aligned} S_{\rm vac} = & \int d^4x\sqrt{-g}\left[\frac{\mathcal{K}}{2}R - \Lambda + \mathcal{L}_{\rm dyn}(\mathcal{K})\right] \\ & + \int \sigma(\Lambda/\mu_*^4)F_4 + \int \hat{\sigma}(\mathcal{K}/M_*^2)\hat{F}_4. \end{aligned} \tag{34.6}$$

For nondegenerate function derivatives and suitable boundaries, the three-form equations make $\Lambda, \mathcal{K}$ spacetime constants. The Einstein equation is $\mathcal{K}G_{\mu\nu} = T_{\mu\nu} - \Lambda g_{\mu\nu}$, whose averaged trace yields

$$\Lambda = \frac{1}{4}\langle T\rangle + \Delta\Lambda, \qquad \Delta\Lambda = \frac{\mathcal{K}}{4}\langle R\rangle. \tag{34.7}$$

Here $\langle\cdot\rangle$ requires a well-defined global volume and boundary prescription and need not exist on arbitrary infinite spacetimes.

The original-variable CR kinetic term contains $-3\mathcal{K}(\partial\chi)^2/[4(1+\chi^2)]$, so $\mathcal{L}_{\rm dyn}$ explicitly depends on $\mathcal{K}$ and its variation cannot be omitted. The residual term has form

$$\Delta\Lambda = -\frac{\mathcal{K}}{2}\left\langle\frac{\partial\mathcal{L}_{\rm dyn}}{\partial\mathcal{K}}\right\rangle - \frac{\mathcal{K}\mu_*^4}{2M_*^2}\frac{\hat{\sigma}'}{\sigma'}\frac{\int\hat{F}_4}{\int F_4}. \tag{34.8}$$

A $\mathcal{K}$-independent vacuum constant cancels in $T_{\mu\nu} - \frac{1}{4}\langle T\rangle g_{\mu\nu}$. Fluxes, $\mathcal{K}$-dependent corrections, and the global average remain independent data. This extension therefore adds global constraints beyond both the two-flux potential and the minimal geometric core.

35 Composite-Particle Spectra and Continuum Tests

35.1 Compatible matter and emergent particles

Two microscopic questions must be distinguished. One asks whether the common geometry admits consistent gauge and matter fields; the preceding construction supplies such fields explicitly. The other asks whether those particles and their statistics emerge from the complex metric or Einstein–scalar core alone. The first construction does not answer the second.[U]

The tests below address the stronger question through free spectra, operator matrix elements, and candidate bound-state channels. Exact identities and source numerical benchmarks are separated from nonperturbative conclusions.

35.2 Finite-cutoff Hilbert space and quantum limits of holomorphy

On T_L^3, remove zero-momentum modes and retain a finite momentum set $\mathcal{K}$. After solving linearized gravitational constraints, define the positive-definite free Fock baseline

$$\mathcal{H}_{\rm phys}^{(0)} = \mathcal{F}(\ell^2(\mathcal{K})\otimes\mathbb{C}^2)\otimes\mathcal{F}(\ell^2(\mathcal{K})), \qquad H_0 = \sum_{\boldsymbol{k}\in\mathcal{K}}|\boldsymbol{k}|\left(a_{+2}^\dagger a_{+2} + a_{-2}^\dagger a_{-2} + b^\dagger b\right). \tag{35.1}$$

This gives a positive free quantization of two tensor helicities and one scalar at finite cutoff. The nonlinear gravitational physical space, compact-space constraints, and possible linearization instabilities remain to be constructed.

Removing the cutoff gives equal-time Gaussian covariance $\mathbb{E}|\phi_{\boldsymbol{k}}|^2 = (2|\boldsymbol{k}|L^3)^{-1}$. Three-dimensional mode counting gives $\mathbb{E}\|\phi\|_{H^s}^2 \sim \int^\infty K^{2s+1}dK$. Typical samples lie only in the

corresponding Sobolev spaces with $s < -1$: they are distributions, not pointwise analytic functions. A Borel–Cantelli argument over independent modes also excludes a nonzero analytic strip. Analyticity at finite Fourier cutoff does not establish holomorphic continuum quantum samples.

Correlation functions offer another place to impose analyticity. The free Euclidean correlator $1/[4\pi^2(\tau_E^2+|\boldsymbol{x}|^2)]$ for $\tau_E > 0$ has a distributional Lorentzian boundary value. Such analyticity can coexist with distribution-valued quantum fields, though it does not yet define a complex-metric functional measure.

35.3 Two-point functions and finite-volume tests of an earlier composite vector

For a free Euclidean scalar, define the normal-ordered candidate $\mathcal{O}_\mu =: (\partial\phi)^2\partial_\mu\phi :$. With $r = |x|$, $n = x/r$, Wick contraction gives

$$K_{\mu\nu} = \frac{\delta_{\mu\nu} - 4n_\mu n_\nu}{2\pi^2 r^4}, \qquad \langle \mathcal{O}_\mu(x)\mathcal{O}_\nu(0)\rangle = \frac{28\delta_{\mu\nu} - 208 n_\mu n_\nu}{8\pi^6 r^{12}} \quad (x \neq 0). \tag{35.2}$$

After local contact terms are removed, the nonlocal Fourier contribution is

$$\widetilde{C}^{\mathrm{nl}}_{\mu\nu}(p) = \frac{(p^2)^4}{552960\pi^4} \ln\frac{p^2}{\mu^2}\left(-\delta_{\mu\nu} + 13\frac{p_\mu p_\nu}{p^2}\right). \tag{35.3}$$

The transverse spectrum begins at a three-particle continuous threshold, with $\rho_T(s) \propto s^4$, and has no isolated photon spectral atom. The candidate $\partial_\mu\phi$ may have $p_\mu p_\nu/p^2$, but this pole is a longitudinal scalar and its transverse projection vanishes.

For $0 < |\boldsymbol{n}|^2 \leq 8$ and zero total momentum, the source reports 642 unordered triples, 618 with nonzero candidate response. The lowest response energy is $(2+\sqrt{2})2\pi/L$, and fixed-mode weights scale as L^{-9}. A massless particle requires a finite spectral residue in the continuum limit, not only levels tending to zero as L^{-1}. The enumeration is retained from the source.

35.4 Bose selection rules and the massless little group

In a positive-invariant-mass two-body rest frame, Bose symmetry excludes odd total angular momentum for identical real scalars. A scalar and helicity-two graviton require $J \geq 2$; identical same-helicity gravitons exclude odd J, and opposite helicities require $J \geq 4$. These are on-shell partial-wave statements, not universal no-go theorems for every nonperturbative massless composite. Three bodies have more relative momenta and admit a nonzero timelike $J = 1$ channel.

Timelike spin one nevertheless differs from a massless photon's two helicities. Assume a Poincaré-invariant vacuum, positive physical space, ordinary finite-helicity particles, and an operator J_μ transforming strictly as an ordinary four-vector on that space. For $p = E(1,0,0,1)$, little-group null rotations force its one-particle matrix element to be proportional to p_μ, while rotations about momentum require a helicity phase. Only zero satisfies both. Thus

$$\langle 0|J_\mu|\gamma_{\pm1}\rangle = 0, \qquad \langle 0|(dJ)_{\mu\nu}|\gamma_{\pm1}\rangle = 0. \tag{35.4}$$

The restriction concerns these operator assumptions. A photon potential transforms with gauge compensation, and field strength belongs to a suitable two-form representation. Gravitational dressing with anchors or boundary directions also needs its complete transformation law. Calling a construction nonlocal does not by itself establish an evasion.

35.5 A nine-derivative Bose-symmetric axial vector

To exhibit a timelike three-body spin-one structure, let $P = k_1 + k_2 + k_3$, $u_i = P \cdot k_i$, $\Delta = (u_1 - u_2)(u_2 - u_3)(u_3 - u_1)$, and define

$$W^\mu = \epsilon^{\mu\nu\rho\sigma} P_\nu k_{1\rho} k_{2\sigma} \Delta, \qquad P_\mu W^\mu = 0. \tag{35.5}$$

Both the Levi–Civita and Vandermonde factors change sign under scalar exchange, so W is Bose symmetric. In the timelike rest frame it is a spatial axial vector. The example $k_1 = (1, 1, 0, 0)$, $k_2 = (2, 0, 2, 0)$, $k_3 = (\sqrt{5}, -1, -2, 0)$ has a nonzero third component. This establishes an allowed spin-one tensor structure, without establishing binding.

A point-split expression is $J^\mu \propto \epsilon^{\mu\nu\rho\sigma} \partial_{3\nu} \partial_{1\rho} \partial_{2\sigma} D_{12} D_{23} D_{31} : \phi(x_1)\phi(x_2)\phi(x_3) :|_{x_a = x}$, where $D_{ab} = (\partial_1 + \partial_2 + \partial_3) \cdot (\partial_a - \partial_b)$. It has nine derivatives and engineering dimension twelve. Its normalization must include six identical-field contractions so that the three-particle matrix element is exactly W^μ. Curved backgrounds also require covariant point splitting, parallel transport, and composite-operator renormalization.

35.6 Gram identity, Dalitz integral, and an exact spectral coefficient

To avoid confusing a three-body invariant with response weight c, write $a_{12} = -(k_1 + k_2)^2$, $a_{23} = -(k_2 + k_3)^2$, $a_{31} = -(k_3 + k_1)^2$, and $s = a_{12} + a_{23} + a_{31} = -P^2$. The Gram identity gives

$$W_\mu W^\mu = \frac{a_{12} a_{23} a_{31} (a_{12} - a_{23})^2 (a_{23} - a_{31})^2 (a_{31} - a_{12})^2}{256}. \tag{35.6}$$

Labeled massless three-body phase space is $d\Phi_3 = da_{12}\, da_{23}/(128\pi^3 s)$. Setting $x = a_{12}/s$, $y = a_{23}/s$, $z = 1 - x - y$ gives the dimensionless integral

$$\int_{x,y \geq 0,\ x+y \leq 1} xyz(x - y)^2 (y - z)^2 (z - x)^2 \, dx\, dy = \frac{1}{277200}. \tag{35.7}$$

Expand the polynomial and sum exactly using the Dirichlet monomial integral $\int x^a y^b z^c dx dy = a! b! c!/(a + b + c + 2)!$.

With the source convention $\rho^{(3\phi)}_{\mu\nu} = \frac{1}{3!} \int d\Phi_3 W_\mu W^*_\nu$ and a further factor of three from rest-frame rotational averaging,

$$\rho^{(3\phi)}_{\mu\nu}(P) = \frac{s^{10}}{163499212800\pi^3} \left(g_{\mu\nu} + \frac{P_\mu P_\nu}{s} \right), \qquad s > 0. \tag{35.8}$$

The denominator is $256 \times 128 \times 6 \times 3 \times 277200$. A convention absorbing $2\pi\theta(P^0)$ changes the overall spectral normalization but preserves positivity, the continuous threshold, and the absence of a $\delta(s)$ pole. Nine derivatives contribute s^9 after squaring, and phase space supplies another s. The result describes a free composite continuum.

35.7 Improvement currents, the collinear boundary, and critical resummation

The axial current is expressible as $J^\mu = \partial_\nu S^{\mu\nu}$, $S^{\mu\nu} = -S^{\nu\mu}$, and is identically conserved. With standard decay at the boundary, total charge is a vanishing surface term, not automatically electric charge. For finitely many future null momenta,

$$-P^2 = 2 \sum_{i<j} \omega_i \omega_j (1 - \cos\theta_{ij}) \geq 0, \tag{35.9}$$

a massless total momentum occurs only at the fully collinear boundary, where $W = 0$. In a specified near-collinear family, $W = O(\theta^8)$, further suppressing the free threshold.

An auxiliary toy model takes $\rho(s) = 5s^4$ on $0 \le s \le 1$, $\Pi(q) = \int \rho(s)/(s+q)\,ds$, and defines $D_\lambda = \Pi/(1-\lambda\Pi)$ by hand. Since $\Pi(0) = 5/4$ and $\Pi'(0) = -5/3$, at $\lambda_c = 4/5$ it formally gives $D \sim 15/(16q)$. Adding contact terms $c_0 + c_1 q$ changes the critical coupling to $(5/4 + c_0)^{-1}$ and changes the residue. A continuum can formally reorganize with added interactions, but the original CR core has not supplied the critical coupling, matching prescription, or photon gauge structure.

35.8 Dressing, twistors, and topological candidates

In linearized gravity, $C_\nu = \partial^\mu h_{\mu\nu} - \frac{1}{2}\partial_\nu h$ obeys $\delta_\xi C_\nu = \Box\xi_\nu$. With an inverse operator's boundaries and zero modes specified, $V^\mu = -\Box^{-1}C^\mu$ makes $\phi(x+V) = \phi(x) + V^\mu\partial_\mu\phi + \cdots$ invariant at the corresponding order. Vectors and two-forms additionally need frame or pullback terms. This starting point neither completes nonlinear Dirac observables nor fixes soft gravitational boundary information.

Line-bundle section counts $(a+1)(b+1)$ on the complex projective null cone $\mathbb{CP}^1 \times \mathbb{CP}^1$ organize various dimensions, and celestial polarization bundles with $|c_1| = 2|s|$ organize helicity. Selecting weights does not produce particles. Twistor reconstruction of self-dual gauge fields also requires holomorphic gauge bundles and transition data. The two infinite-dimensional E/B representations of celestial one-forms cannot yield a Lorentz-invariant finite particle spectrum by an arbitrary finite spherical-harmonic truncation.

Topological geons, jet projectors, and solitons offer other routes, but rotation and exchange statistics depend on topology and quantum rules. A projector from normalized Euclidean scalar jets fails where its parent vector vanishes, and the original dynamics need not protect its topological number. For a regular static potential-free system with no inner boundary and vanishing surface terms, $\int N|D\phi|^2 = 0$ forces constant scalar. A minimum in a finite trial family may decrease along omitted amplitude directions. Such conditional results call for testing specific proposals, not excluding every nonperturbative route with one local counterexample.

36 Six-Point Gravitational Amplitudes and Three-Body Kernels

36.1 Complete classification of leading tree diagrams

U's rounds 791–820 compute the leading six-scalar amplitude and correct an index-contraction error in its seagull vertex. We use the corrected values below. The components are quoted from U; algebraic sums can be checked directly, but the original diagram-generation calculation is not reproduced here.[U]

Take $g = \eta + \kappa_g h$, $\kappa_g = 2/M_P$. The leading connected six-scalar tree amplitude is $O(\kappa_g^4)$. If x, y, z count $h\phi\phi$, $hh\phi\phi$, and hhh vertices, internal scalar lines number $I_\phi = x + y - 3$ and graviton lines $I_h = (x + 2y + 3z)/2$. The tree condition gives $x + 2y + z = 4$, allowing three classes:

Class	Vertices (x, y, z)	Labeled diagrams	Internal structure
C	$(4, 0, 0)$	90	Two graviton lines and one scalar bridge.
B	$(2, 1, 0)$	45	One two-graviton–two-scalar seagull and two graviton lines.

Class	Vertices (x, y, z)	Labeled diagrams	Internal structure
A	$(3, 0, 1)$	15	A three-graviton center with three graviton lines.

There are 150 labeled diagrams at this coupling order.

36.2 Stress vertices and correct indices in seagull contractions

Define a lower-index stress vertex and the source generated by a pair of external legs:

$$T_{\mu\nu}(p,q) = p_\mu q_\nu + p_\nu q_\mu - \eta_{\mu\nu}(p\cdot q), \qquad H_{\mu\nu} = \frac{T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\,\mathrm{tr}_\eta\, T}{(p+q)^2}. \tag{36.1}$$

The seagull comes from the quadratic expansion of $\sqrt{-g}g^{\mu\nu}$. For lower-index perturbation matrices H_1, H_2, set $t_a = \mathrm{tr}(\eta H_a)$, $t_{12} = \mathrm{tr}(\eta H_1 \eta H_2)$. The bilinear cross term is

$$\begin{aligned} Q_2(H_1,H_2) = &\left(\frac{t_1 t_2}{4} - \frac{t_{12}}{2}\right)\eta - \frac{1}{2}(t_1 \eta H_2 \eta + t_2 \eta H_1 \eta) \\ &+ \eta H_1 \eta H_2 \eta + \eta H_2 \eta H_1 \eta. \end{aligned} \tag{36.2}$$

If momentum columns are stored with upper indices p^μ, the contraction must be

$$B_{(ab);(ij),(kl)} = p_a^{\mathsf{T}} \eta Q_2(H_{ij}, H_{kl}) \eta p_b, \tag{36.3}$$

rather than $p_a^{\mathsf{T}} Q_2 p_b$, which omits both index lowerings. The three-graviton center follows from cubic order in $\sqrt{-g}g^{\mu\nu}(\Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\nu\rho} - \Gamma^\rho_{\mu\nu}\Gamma^\sigma_{\rho\sigma})$. Individual expressions depend on gauge and variables; consistent relative coefficients and the full amplitude carry the meaningful check.

36.3 Two kinematic configurations and final corrected values

The first incoming momenta are $k_1 = (3,3,0,0)$, $k_2 = (4,0,4,0)$, $k_3 = (5,-3,-4,0)$, totaling $(12,0,0,0)$. Outgoing momenta use specified rational spatial rotations and a minus sign in all-incoming notation. The first configuration uses $R_x R_z$ with sine/cosine pairs from $(5/13, 12/13)$ and $(3/5, 4/5)$; the second uses the source-defined $R_y R_z$. External legs are on shell and relevant internal denominators nonzero.

Strip the factor $i\kappa_g^4/16$ and define $\widehat{\mathcal{M}}_6 = C + 4B + 4A$. Components of the first configuration are

$$\begin{aligned} C &= -\frac{28266384203933989977 9983}{45345560418603759 60}, \\ B &= \frac{13898199988405204}{15934518501525}, \qquad A = -\frac{11890177326402307}{35410041114500}. \end{aligned} \tag{36.4}$$

The final total is

$$\begin{aligned} \widehat{\mathcal{M}}_{6,1} &= -\frac{68233526622404100362731 51}{113363901046509399000} \simeq -60189.81879814649, \\ \widehat{\mathcal{M}}_{6,2} &= -\frac{13986346982802291072751913 68}{62413566574661128900125} \simeq -22409.14555983813. \end{aligned} \tag{36.5}$$

These amplitudes are stripped of the common coupling and evaluated at fixed kinematics. Computing a cross section requires restoring that normalization and integrating over phase space.

Quantity	Earlier incorrect contraction	Corrected value
First seagull sum B	58.402313565	872.207088471
First total amplitude	−63445.037897772	−60189.818798146
Second total amplitude	−33327.933530448	−22409.145559838
First remainder excluding bridges	−1109.532628910	2145.686470716

36.4 Global boosts, Ward identities, and a second gauge

Under the source's global t–x boost, $\gamma = 5/3$, $\gamma v = 4/3$, the incorrect total changes. The corrected class sums and total are invariant. The relevant covariance test is on the full amplitude.

An independent gravitational Compton test sets one graviton polarization pure gauge. For rational source momenta, $C_4 = 64/25$, $A_4 = 0$, the old $B_4 = 16/25$, and corrected $B_4 = -16/25$, so

$$C_4 + 4B_4 + 4A_4 = 0 \tag{36.6}$$

holds only with the correct contraction. Five gauge-parameter choices pass the reported test. For conserved sources, changing de Donder gauge to $\partial^\nu h_{\mu\nu} = 0$ also changes the source to

$$H^{\text{new}}_{\mu\nu} = H^{\text{dd}}_{\mu\nu} - \frac{k_\mu k_\nu}{2k^2}\,\mathrm{tr}_\eta\, H^{\text{dd}}. \tag{36.7}$$

In both configurations, the class contributions change while the total remains invariant. This checks the relative tree coefficients; it does not establish nonlinear BRST closure.

36.5 Ten scalar-bridge channels and particle identity

Six external legs admit $\frac{1}{2}\binom{6}{3} = 10$ unordered three-three partitions, with nine bridge diagrams per channel. Define a four-point current with one off-shell scalar leg:

$$J_I(P_I) = \sum_{a\in I} T(p_a, -P_I) : H_{I\setminus\{a\}}, \qquad C_I = \frac{J_I(P_I)J_{\bar{I}}(-P_I)}{P_I^2}. \tag{36.8}$$

In one channel of the first configuration, $P_I^2 = -144$ and $J_I = J_{\bar{I}} = 196$, giving $-2401/9$. This factorization is propagation of the original scalar and is unchanged by the seagull correction.

The original two-derivative core's $\phi \mapsto -\phi$ symmetry makes elementary vertices even in scalar legs. A 3|3 one-particle pole of the six-scalar tree exchanges the original scalar, and a 2|4 pole exchanges the original graviton. A denominator $1/P^2$ does not rename the internal particle a photon. The stripped four-scalar amplitude $\widehat{\mathcal{M}}_4 = tu/s + su/t + st/u$ is $441/25$ at analytic factorization point $(4, -4/5, -16/5)$, with scalar residue $(441/25)^2 = 194481/625$. This is crossed analytic kinematics, not the probability of producing a real photon from three future null momenta at arbitrary finite angles.

36.6 Off-shell ambiguity and three-body equations

In specified de Donder gauge and field variables, define $R^{\text{dd}}_{\text{bridge}} = \widehat{\mathcal{M}}_6 - C = 4B + 4A$, equal to $170952403830483317/79672592507625$ in the first configuration. It has no explicit three-three scalar-bridge denominator but retains long-range gravitational denominators. It is neither a local six-scalar contact term nor automatically the full three-body irreducible kernel.

The vertex identity

$$(p+q)^\mu T_{\mu\nu}(p,q) = p^2 q_\nu + q^2 p_\nu \tag{36.9}$$

shows that off-shell currents may contain longitudinal terms vanishing at physical poles. For example, $J_I \mapsto J_I + P_I^2\lambda$, $R \mapsto R - \lambda J_{\bar{I}}$ preserve the total amplitude and isolated residue. Taking $\lambda = 3/7$ in a particular channel raises the bridge by 84 and lowers the remainder by 84. This explicitly demonstrates nonuniqueness of off-shell splitting from on-shell data; changing a single channel is not yet a complete local field redefinition preserving all crossing relations.

The formal equation

$$G_3 = G_{0,3} + G_{0,3}(K_{12} + K_{23} + K_{31} + K_{123}^{\mathrm{irr}})G_3 \tag{36.10}$$

becomes a solvable three-body theory only after specifying amputation, time ordering and energy integration, spectator factors, subtraction of two-body iterations, common counterterms, and infrared treatment. The three-scalar subspace couples to graviton-containing and higher-particle channels; eliminating them generally produces an energy-dependent kernel. Substituting one six-point tree value into a single-channel eigenvalue equation guarantees neither unitarity nor gauge consistency. A new-particle search needs a legitimate probe operator and the complete physical spectrum with positive residues.

37 Quantum Effective Theory and Model Selection

37.1 From classical geometry to effective field theory

For normal $c > 1$ sectors, a real Lorentz metric, standard gauge constraints, and specified matter statistics, a low-energy perturbative expansion can be built around a background. Gravity's non-renormalizability does not prevent its use as a low-energy EFT; local higher-dimensional coefficients encode unknown high-energy information.[U]

Equation (26.1) is the leading two-derivative action. Its quantum effective action can also contain symmetry-allowed curvature-squared terms, $R\phi^2$, $\phi^2 H^\dagger H$, mixed response potentials, and higher derivatives. Their coefficients must be matched or forbidden by a stated symmetry.

Running couplings and field normalization can modify the tree relation (29.2). Preserving it requires control of counterterms, beyond the common classical ansatz.

37.2 Critical modules and negative-kinetic regions

At $c = 1$, constraint brackets may change rank across backgrounds, complicating global phase space, path-integral measure, and vacuum perturbation theory. The consistent classical truncation $w = 0$ yields a simpler normal low-energy submodel, not a full construction of critical physical states and quantum fluctuations.

For $c < 1$, at least one kinetic direction is negative. With the present real fields, local two-derivative action, and conventional positive Hilbert interpretation, this creates serious energy or unitarity problems. Multiplying a field by i cannot remove them without changing the reality structure and interactions. A special integration contour or different quantum interpretation must be tested as a complete alternative model.

37.3 Ultralight masses and protection by dark symmetries

The v_4 shift can forbid an ordinary potential, but the masses of ϕ and S and their Higgs portals each need protection arguments. Discrete parity generally permits quadratic mass terms and does not explain tiny masses. Choosing $m_4 = 10^{-19}\,\mathrm{eV}$ defines a phenomenological example without settling naturalness in a supplied high-energy completion.

Stability requires the parity to survive in the full action and vacuum, with S the lightest actual odd eigenstate. Lighter states or symmetry breaking in a high-energy extension would change the conclusion.

37.4 Earlier versions and additional dark-universe modules

The orientation and common-cone branches have different targets, field counts, and amplitudes. Their couplings belong to distinct actions and cannot be transferred between them without matching.[U]

Four-form strengths can supply vacuum-energy-like terms in four dimensions, but potentials, gauge symmetries, boundaries, and integration constants are added module data and must appear in the total action. Even a "complex-geometric cosmological constant" needs definition: $\mathrm{Re}(\sqrt{-\gamma}\,\Lambda_{\mathbb{C}})$ generally differs from $\sqrt{-g}V_0$. For real $\Lambda_{\mathbb{C}}$, the first contains $\mathrm{Re}\,F^2 = 1-\chi^2$ and is scalar dependent. The V_0 of Eq. (26.2) is explicitly defined on the real readout metric.

37.5 Field-space curvature and a common base for multiple sectors

Section 17 derives the curvature generated by a common scalar base. For the $c = 2, 3$ fields, the five-dimensional result is

$$\mathcal{R}_{(2,3)} = -\frac{5}{f^2} + \frac{37}{48f^2}\,\mathrm{sech}^2\,x. \tag{37.1}$$

The cross contribution is $-2\tanh^2 x/f^2$ relative to two isolated targets. Adding $c = 4$ gives Eq. (17.10). These terms reflect the same shared geometry that produces cross-sector decay.[U]

Negative curvature is compatible with a positive kinetic metric. It enters background perturbations, amplitudes, and cutoff estimates. The positively curved $\mathbb{RP}^3$ branch has different stability questions.

37.6 Requirements for phenomenological tests

A phenomenological model must specify its responses, mass eigenstates, parities, critical-sector treatment, portals, gauge kinetics, matching scale, and initial data. Together these determine the sources, spectra, and thermal history.

Particle tests also have levels. An independent photon needs a physical two-form operator overlapping a massless finite-helicity state, positive residue, two propagating helicities, and a stable continuum limit; electromagnetism additionally requires Gauss constraints, charged matter, and measurable charge. Fermions need Spin representations, statistics, and positive quantum states; color fields need complete non-Abelian constraints and propagation. Dark matter needs lifetime, abundance, perturbations, and a production history compatible with cosmology. Mathematical relations and these further tests are placed in one framework to direct subsequent calculations at specific gaps.

Quantum structure and internal observation. The preceding models supply classical phase spaces and, where specified, independent quantum fields. We now examine whether response geometry and physical operations can account for quantum structure itself. The studies collected in R proceed from Pauli and projective constructions to environments, records, nonlinear evolution, and internal clocks. Their central test is whether measurable operations admit a closed quantum state description.

The starting symplectic forms are classical. A quantum observable algebra, positive states, and a universal action scale need further justification. In particular, any field-dependent bracket must satisfy the Jacobi condition (13.3).

38 Pauli, Clifford, Dirac, and Schrödinger Structures

38.1 A Pauli frame generated by the response curve

The traceless part of the response matrix defines

$$T_3 = \frac{\sigma_3 - \chi\sigma_1}{\rho}, \qquad T_1 = -\rho^2\partial_\chi T_3 = \frac{\sigma_1 + \chi\sigma_3}{\rho}, \qquad T_2 = \mathrm{i}T_1T_3 = \sigma_2. \tag{38.1}$$

Direct multiplication gives $T_iT_j = \delta_{ij}I + \mathrm{i}\epsilon_{ijk}T_k$. The response family and its derivative therefore define a Pauli frame. With $\theta = \arctan\chi$ and $U = \exp(\mathrm{i}\theta\sigma_2/2)$, it is $T_i = U\sigma_iU^{-1}$.

This derivative is taken in a fixed response basis. A varying basis $S(\chi)$ introduces connection terms and requires a covariant derivative. The connection $U^{-1}\mathrm{d}U$ here is pure gauge and has zero curvature.

38.2 Clifford factorization and the Dirac operator

For $g = \mathrm{diag}(-,+,+,+)$, choose

$$\Gamma^0 = T_1 \otimes I, \qquad \Gamma^k = \mathrm{i}T_2 \otimes \sigma_k, \qquad \{\Gamma^\mu, \Gamma^\nu\} = -2g^{\mu\nu}I. \tag{38.2}$$

In a constant flat frame, $\mathcal{D} = \mathrm{i}\Gamma^\mu\partial_\mu$ then satisfies $\mathcal{D}^2 = \Box_g$. Curved backgrounds need a spin connection and curvature terms in the square, so the flat identity is not literal there. A complex conformal metric has the four-dimensional transformation law on a chosen local branch

$$\mathcal{D}_\gamma = F^{-5/4}\mathcal{D}_g F^{3/4}. \tag{38.3}$$

Setting $\psi = F^{3/4}\Psi$ makes a constant mass in the complex-metric equation appear as $m\sqrt{F}$. This relates geometry to a first-order operator while raising questions of real inner product, mass reality, and the physical spin module. Conventional Weyl/Dirac modules on real g are available as explicit model components.

The construction establishes Pauli and Clifford algebras and a Dirac operator. Quantum preparation, composition, Born probabilities, and fermionic anticommutators require a physical state and measurement theory in addition to these identities.

38.3 Inputs to the nonrelativistic limit and uncertainty relations

For a real Dirac equation with positive mass, a positive-energy choice, and the usual inner product, remove $e^{-\mathrm{i}mc^2_{\mathrm{light}}t/\hbar}$. The lower component obeys $\eta \simeq (\boldsymbol{\sigma}\cdot\boldsymbol{p})\varphi/(2mc_{\mathrm{light}})$, and the upper component satisfies the Schrödinger equation. The limit uses the supplied wavefunction space, inner product, and $\hbar$.

Likewise, with $\hat{p} = -\mathrm{i}\hbar\nabla$ in an L^2 representation, Cauchy–Schwarz and $[\hat{x}, \hat{p}] = \mathrm{i}\hbar$ give $\Delta x\,\Delta p \geq \hbar/2$. Pauli multiplication similarly gives spin uncertainty relations. Positivity of a Gram determinant on classical target space differs from operator covariance in quantum states.

38.4 A static conformal Schrödinger approximation

The four-dimensional complex scalar wave operator itself satisfies $\Box_\gamma = F^{-1}\Box_g + F^{-2}\nabla F\cdot\nabla$. This is separate from choosing a real modulus metric; replacing F by ρ requires specifying the action.

The source's auxiliary real metric $\widehat{\gamma} = \rho g$ gives a static scalar wave equation. Taking $\Phi = \rho^{-1/2}e^{-\mathrm{i}mc^2_{\mathrm{light}}t/\hbar}\psi$ gives the envelope equation

$$\mathrm{i}\hbar\dot{\psi} = \left[-\frac{\hbar^2}{2m}\Delta + \frac{mc^2_{\mathrm{light}}}{2}(\rho-1) + \frac{\hbar^2}{4m}\Delta\ln\rho + \frac{\hbar^2}{8m}|\nabla\ln\rho|^2\right]\psi + \frac{\hbar^2}{2mc^2_{\mathrm{light}}}\ddot{\psi}. \tag{38.4}$$

Dropping the final term requires envelope frequencies much smaller than $mc^2_{\mathrm{light}}/\hbar$: total energy must lie close to the chosen carrier frequency. Low spatial momentum alone is insufficient. For constant ρ_0, the exact dispersion is

$$E^2 = c^2_{\mathrm{light}}p^2 + m^2c^4_{\mathrm{light}}\rho_0, \quad E = mc^2_{\mathrm{light}}\sqrt{\rho_0} + \frac{p^2}{2m\sqrt{\rho_0}} + O(p^4). \tag{38.5}$$

The potential $\frac{1}{2}mc^2_{\mathrm{light}}(\rho-1)$ is consequently a weak-deviation approximation near $\rho = 1$. At $\rho_0 = 4$, it gives a rest-energy shift of $1.5mc^2_{\mathrm{light}}$, compared with the exact mc^2_{light}.

A massless second-order wave equation usually has $\omega^2 = c^2_{\mathrm{light}}k^2$, not a finite-mass Schrödinger dispersion. For positive A, a classical wave can be rewritten using $\Psi = (A^{1/4}q + \mathrm{i}A^{-1/4}p)/\sqrt{2}$ as $\mathrm{i}\dot{\Psi} = \sqrt{A}\,\Psi$, but this is a frequency decomposition. Historically, the Bargmann metric $\mathrm{d}\boldsymbol{x}^2 + 2\mathrm{d}t\mathrm{d}s - 2U\mathrm{d}t^2$ with fixed null-fiber momentum derives the Schrödinger equation exactly. It adds a coordinate and momentum choice and therefore does not complete the present four-dimensional program.

39 Complex Fibers and Projective Reduction

39.1 Natural complex structure at fixed shared scalar

For normal sectors set $s_n = \sqrt{n^2-1}\rho$ and take

$$J_n = \frac{1}{s_n}\begin{pmatrix}\chi & -(n\rho-1)\\ n\rho+1 & -\chi\end{pmatrix}, \qquad J_n^2 = -I. \tag{39.1}$$

The two-form compatible with G_n is

$$\Omega_n(X,Y) = G_n(J_nX,Y), \qquad \Omega_n = s_n\,\mathrm{d}u_n\wedge\mathrm{d}v_n. \tag{39.2}$$

It defines a Kähler structure on a two-dimensional fiber at fixed ϕ, with

$$G_n(X,X)G_n(Y,Y) - G_n(X,Y)^2 = \Omega_n(X,Y)^2. \tag{39.3}$$

The identity relates fiber length, area, and complex structure. The field-theory symplectic form is instead $\int(\delta u\wedge\delta\pi_u + \delta v\wedge\delta\pi_v)$ and usually gives $\{u,v\} = 0$. Identifying these two geometries would require a physical reduction.

The seven-dimensional normal target also admits contact forms, for example

$$\alpha = \mathrm{d}(\phi/f) + \sum_{n=2}^{4} \frac{s_n}{2f^2}(u_n \mathrm{d}v_n - v_n \mathrm{d}u_n), \qquad \alpha \wedge (\mathrm{d}\alpha)^3 = \frac{6 s_2 s_3 s_4}{f^7} \mathrm{d}\phi \wedge \bigwedge_{n=2}^{4} (\mathrm{d}u_n \wedge \mathrm{d}v_n). \tag{39.4}$$

The displayed top form proves nondegeneracy. Neither uniqueness of this contact form nor quantization of an action unit follows, especially on the noncompact fibers.

39.2 Two-oscillator $\mathbb{CP}^1$ and the action scale

For two equal-frequency oscillators, set

$$z_j = \frac{\sqrt{\omega}\, q_j + \mathrm{i} p_j/\sqrt{\omega}}{\sqrt{2}}, \quad H = \omega \sum_j |z_j|^2, \quad \Omega = \mathrm{i} \sum_j \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j. \tag{39.5}$$

Fixing $I_0 = \sum |z_j|^2$ gives a three-sphere, and reduction by common phase $U(1)$ gives $S^3/U(1) \simeq \mathbb{CP}^1$. The corresponding form and component moment maps are

$$\Omega_{\mathrm{red}} = \frac{I_0}{2} \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\varphi, \quad \int_{\mathbb{CP}^1} \Omega_{\mathrm{red}} = 2\pi I_0, \quad \frac{|z_1|^2}{I_0} = \cos^2 \frac{\theta}{2}. \tag{39.6}$$

Thus classical projective reduction already produces the shape of a Born overlap. Whether common phase is unobservable depends on the instruments and coupled dynamics; the nonlinear tests below address that question.

Adding geometric quantization imposes $I_0/\hbar = k \in \mathbb{Z}$ and gives $\dim H^0(\mathbb{CP}^1, \mathcal{O}(k)) = k + 1$. Obtaining two states requires $k = 1$. The quantization rule, this value, and the probability interpretation are additional choices.

39.3 Uniform measures and overlap probabilities

With uniform measure on $\mathbb{CP}^1$, overlap x with a fixed pure state is uniform on $[0, 1]$, so $x \leq q$ has probability q. On $\mathbb{CP}^{d-1}$ the overlap has $\mathrm{Beta}(1, d-1)$ distribution. Depending on the cap definition, probabilities can have powers such as q^{d-1} rather than a universal linear form.

The sources also retain a classical sphere model of a single qubit: hidden direction $\lambda \in S^2$, preparation density $\rho_{\boldsymbol{s}}(\lambda) = (\boldsymbol{s} \cdot \lambda)_+/\pi$, and measurement along $\boldsymbol{m}$ with hemisphere $\boldsymbol{m} \cdot \lambda > 0$ yield $(1 + \boldsymbol{s} \cdot \boldsymbol{m})/2$. This simulates single-qubit probabilities without fixing the underlying dynamics. Extension to composite systems would also have to address locality, measurement independence, and contextuality.

40 Probability Measures and First-Record Statistics

40.1 Overlap kernels, effect additivity, and an undetermined measure

The Pauli frame gives projectors $P_+(\chi)$ with overlap kernel

$$K_{\mathrm{CR}}(\chi_1, \chi_2) = \mathrm{Tr}[P_+(\chi_1) P_+(\chi_2)] = \cos^2 \frac{\theta_1 - \theta_2}{2} \tag{40.1}$$

of the standard pure-state form. Invariance under only this rotation family still permits inner product $H = I + \epsilon\sigma_2 > 0$. Requiring two nonparallel T_3 to be self-adjoint in the same background-independent inner product instead forces H to be scalar.

Testing only binary orthogonal projectors also does not fix all probabilities. For example,

$$p_*(P_{\boldsymbol{n}}) = \frac{1}{2}[1 + (\boldsymbol{s}\cdot\boldsymbol{n})^3] \tag{40.2}$$

is positive and antipodally normalized but not Born. Extending by $p(tP) = tp(P)$ to a trine POVM $E_i = \frac{2}{3}P_{\boldsymbol{n}_i}$ with $\boldsymbol{s} = \boldsymbol{n}_0$ gives total probability $5/4$. A representation $p(E) = \mathrm{Tr}(\varrho E)$ requires equal probabilities for different realizations of the same effect and additivity over all allowed POVMs. The effect set and operational noncontextuality are substantive premises.

The same geometry admits different preparation measures. The squared-norm ratio of two equal-variance two-dimensional Gaussian vectors is uniform, whereas an imposed density $6z(1-z)$ gives threshold probability $3q^2 - 2q^3$. For coordinate variances σ_2^2, σ_3^2,

$$p(z) = \frac{\sigma_2^2\sigma_3^2}{[z\sigma_3^2 + (1-z)\sigma_2^2]^2}, \qquad \Pr(z \le q) = \frac{q\sigma_3^2}{q\sigma_3^2 + (1-q)\sigma_2^2}. \tag{40.3}$$

A probability calculation therefore needs a specified preparation measure.

40.2 Canonical Gibbs energy fractions

Normal-sector canonical kinetic energy is $\mathcal{K}_n = \frac{1}{2}p_n^T G_n^{-1} p_n$. At fixed configuration, the Gibbs factor $e^{-\beta\mathcal{K}_n}$ is whitened using $G_n^{-1/2}p_n$. In two dimensions, $\mathcal{K}_n$ has $\mathrm{Exp}(\beta)$ distribution; for d_n quadratic degrees of freedom it has $\mathrm{Gamma}(d_n/2, \beta)$ distribution. At common temperature, the ratio of two independent two-dimensional sectors,

$$z = \frac{\mathcal{K}_2}{\mathcal{K}_2 + \mathcal{K}_3} \tag{40.4}$$

is uniform. Three such energy shares have $\mathrm{Dirichlet}(1,1,1)$ distribution. The result uses canonical kinetic energies and does not require equal variances of the original response coordinates.

The issue becomes dynamical preservation. When the shared scalar varies,

$$\dot{\mathcal{K}}_n = -\frac{1}{2}\dot{\phi}\,\dot{y}_n^T G_n' \dot{y}_n - \dot{y}_n \cdot \nabla_{y_n} V, \tag{40.5}$$

the exchange term at the origin is $\lambda_n \dot{\phi}\,\dot{U}_n \dot{V}_n$. An initial Gibbs distribution need not survive driving, filtering, or conditioning. Under free Liouville flow, many different $h(z)$ can be invariant; exchange symmetry requires only $h(z) = h(1-z)$. For example, $h_\epsilon = 1 + \epsilon(6z^2 - 6z + 1)$ gives

$$\Pr(z \le q) = q + \epsilon q(1-q)(1-2q). \tag{40.6}$$

$19/64$, not $1/4$, at $q = 1/4, \epsilon = 1/2$.

40.3 Martingales, stationarity, and absorption

If the probability coordinate obeys a regular absorbing diffusion $\mathrm{d}q = b(q)\mathrm{d}t + \sqrt{a(q)}\mathrm{d}W$, its probability $h(q)$ of eventual absorption at 1 satisfies

$$bh' + \frac{a}{2}h'' = 0, \qquad h(0) = 0, \quad h(1) = 1. \tag{40.7}$$

With $a > 0$ in the interior and almost-sure absorption, $h(q) = q$ requires zero drift $b = 0$. The bias $b = \epsilon q(1-q)(2q-1)$, with $a = \kappa q(1-q)$, gives

$$h(q) = \frac{\int_0^q \exp[(2\epsilon/\kappa)s(1-s)]\,\mathrm{d}s}{\int_0^1 \exp[(2\epsilon/\kappa)s(1-s)]\,\mathrm{d}s}. \tag{40.8}$$

A uniform stationary distribution does not imply a martingale. A component probability in spherical diffusion has $\mathrm{d}q = D(1-2q)\mathrm{d}t + \sqrt{2Dq(1-q)}\,\mathrm{d}W$; A four-dimensional OU construction of the energy fraction likewise has restoring drift. More strongly, a square-integrable stationary martingale has only trivial increments, since $E[(Z_t - Z_s)^2] = E[Z_t^2] - E[Z_s^2] = 0$. Nontrivial measurement absorption cannot also be explained as remaining in the same stationary distribution forever.

The Dirac current gives another conditional argument. If $s = \psi^\dagger\psi$ and $\boldsymbol{v} = \boldsymbol{j}/s$ obey continuity, and a second local density is assumed to have form $P = f(s)$ transported by the same flow, then $(f - sf')\nabla\cdot\boldsymbol{v} = 0$. A sufficiently rich family of flows gives $f = Cs$. This supports compatibility of Born density with that flow, but locality of the density function, choice of flows, and a configuration-space distribution are already premises.

40.4 Complete first-record statistics from competing random thresholds

In MST I, accumulated channel work is $W_i(t) = c_i q_i F(t)$, with a monotone accumulation function and $F(0) = 0$. Thresholds K_i are independent $\mathrm{Exp}(\beta_i)$. Set $r_i = \beta_i c_i q_i$ and $R = \sum_i r_i$. Then

$$p(i,t) = r_i F'(t) e^{-RF(t)}, \quad P_\varnothing(T) = e^{-RF(T)}, \quad P(i \mid \text{a record occurs before the cutoff}) = \frac{r_i}{R}. \tag{40.9}$$

If all $\beta_i c_i$ coincide, normalized channel probabilities are q_i. The model also gives waiting times, no-click probabilities, and possible permanent no-click outcomes with finite available energy, going beyond a single Born-shaped probability.

The result assumes independent thresholds, physical encoding of q_i, and equal hazard slopes. For $S_i(k) = e^{-\beta_i k^\alpha}$, the same construction instead gives probabilities proportional to $\beta_i c_i^\alpha q_i^\alpha$. Positive additivity under operational splitting and merging can select $r(q) = \kappa q$. For general threshold densities, $p(i,t) = f_i(W_i)\dot{W}_i \prod_{j\neq i} S_j(W_j)$.

A first-record law does not specify the state after a click or no-click outcome. An instrument theory must also predict conditional updates and subsequent measurements. This motivates the next stages of MST.

41 Continuous Records and the Two-Channel Environment

41.1 Threshold races and quantum instruments

Let $u = F(t)$ and $\sum_i q_i = 1$. Threshold-race model A has survival function

$$S_A(u) = \exp[-u\sum_i q_i\lambda_i], \tag{41.1}$$

while standard diagonal continuous quantum instrument B gives

$$S_B(u) = \sum_i q_i e^{-\lambda_i u}, \qquad p_B(i,t) = q_i\lambda_i F'(t) e^{-\lambda_i F(t)}. \tag{41.2}$$

The latter is constructed using $M_0 = e^{-F\Gamma/2}$, $M_i(t) = \sqrt{\lambda_i F'(t)}P_i e^{-F\Gamma/2}$, $\Gamma = \sum_i \lambda_i P_i$. If every $\lambda_i > 0$ and observation time is infinite, B has total channel probability q_i without equal detection rates; A gives $q_i\lambda_i / \sum q_j\lambda_j$.

Jensen's inequality gives $S_B \geq S_A$, with leading small-u difference $u^2\,\mathrm{Var}_q(\lambda)/2$. B's no-click posterior weights

$$q_i(u) = \frac{q_i e^{-\lambda_i u}}{\sum_j q_j e^{-\lambda_j u}} \tag{41.3}$$

change the hazard, with $h'_B(u) = -\operatorname{Var}_{q(u)}(\lambda)$. The complete QND conditional process, including clicks and no clicks, is

$$\mathrm{d}q_i = \sum_j (\delta_{ij} - q_i)[\mathrm{d}N_j - \lambda_j q_j \,\mathrm{d}u], \tag{41.4}$$

The martingale property depends on retaining both click jumps and no-click drift.

Model A's nonlinearity can conflict with affine mixing. For example, $\lambda_1 : \lambda_2 = 2 : 1$ gives $p_1 = 2q/(1+q)$ gives 0.4 at $q = 1/4$, while mixing definite preparations gives 0.25. The discrepancy concerns unconditional preparation mixing; postselected statistics need a separately defined comparison. Model B can itself be simulated by sampling $i \sim q_i$ and then its exponential waiting time, but that procedure assumes the weights q_i.

Quantum jump operators $L_i = \sqrt{\lambda_i} P_i$ give Lindblad dephasing rate $(\lambda_i + \lambda_j)/2$, a conditional quantum completion. The limits $\lambda_i \to 0$ and infinite observation time may fail to commute; ideal dark channels need separate treatment.

41.2 The pump approximation and missing stability terms

The physical calculation chooses a local pump $\bar{U}_n = v_n t$, giving at lowest order $-g_n \phi \dot{V}_n$, with $g_n = \lambda_n v_n$. Second-order terms of the full matrix simultaneously change the scalar mass to

$$M_{\mathrm{eff}}^2 = M_\phi^2 - \sum_n \frac{n v_n^2}{2(n+1) f^2}. \tag{41.5}$$

This mass shift is essential to bath stability, especially for a massless shared scalar under timelike pumping. Null traveling pumps have $(\partial \bar{U})^2 = 0$ and avoid some shifts, but require a different derivative analysis. A finite-energy experiment must also replace the infinite homogeneous pump with a specified source or wave packet.

The Hamiltonian of a linear oscillator pair,

$$H = \frac{1}{2} p_\phi^2 + \frac{1}{2}(p_V + g\phi)^2 + \frac{1}{2}\omega_\phi^2 \phi^2 + \frac{1}{2}\omega_V^2 V^2 \tag{41.6}$$

is positive for positive squared frequencies. Its mixed frequencies are

$$\Omega_\pm^2 = \frac{1}{2}\left[A \pm \sqrt{A^2 - 4\omega_\phi^2 \omega_V^2}\right], \quad A = \omega_\phi^2 + \omega_V^2 + g^2. \tag{41.7}$$

A resonant expansion may give $V \sim t \sin \omega t$ and energy $\sim t^2$, while the exact positive oscillator system exhibits bounded beating. Such secular terms indicate the finite range of the expansion.

41.3 Memory kernels and noise from eliminating the common field

Take local modes $V_i(t, \boldsymbol{x}) = Q_i(t) h_i(\boldsymbol{x})$ with normalized Gaussian

$$h_i = (\pi a^2)^{-3/4} e^{-|\boldsymbol{x} - \boldsymbol{x}_i|^2/(2a^2)}, \qquad \omega_{0i}^2 = m_{V_i}^2 + \frac{3}{2a^2}. \tag{41.8}$$

The retarded scalar solution is the sum of a free initial field and a forced term. Eliminating it gives

$$\ddot{Q}_i + \omega_{0i}^2 Q_i + \sum_j \int_0^t \gamma_{ij}(t-s) \dot{Q}_j(s) \,\mathrm{d}s = \xi_i(t). \tag{41.9}$$

The initial field determines $\xi_i = g_i \dot{\Phi}_i^{\mathrm{hom}}$. It becomes stochastic only after a distribution of initial data is supplied. With consistent Fourier conventions,

$$\gamma_{ij}(t) = g_i g_j \int \frac{\mathrm{d}^3 k}{(2\pi)^3} h_i^*(\boldsymbol{k}) h_j(\boldsymbol{k}) \cos(\omega_k t), \tag{41.10}$$

$$J_{ij}(\omega) = \frac{\pi \omega g_i g_j}{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} h_i^* h_j \delta(\omega - \omega_k), \tag{41.11}$$

$$\gamma(t) = \frac{2}{\pi} \int_0^\infty \frac{J(\omega)}{\omega} \cos(\omega t)\, \mathrm{d}\omega. \tag{41.12}$$

Gaussian profiles separated by L give

$$\gamma_{ij}(t) = \frac{4 g_i g_j a^3}{\sqrt{\pi}} \int_0^\infty k^2 e^{-a^2 k^2} \operatorname{sinc}(kL) \cos\!\left(\sqrt{k^2 + M^2}\, t\right) \mathrm{d}k, \tag{41.13}$$

At coincident points use $L = 0$; also $\gamma_{ij}(0) = g_i g_j e^{-L^2/(4a^2)}$.

For $\omega > M$, with $k = \sqrt{\omega^2 - M^2}$, the two-channel spectrum is

$$J(\omega) = 2\sqrt{\pi} a^3 \omega^2 k e^{-a^2 k^2} \begin{pmatrix} g_2^2 & g_2 g_3 \operatorname{sinc}(kL) \\ g_2 g_3 \operatorname{sinc}(kL) & g_3^2 \end{pmatrix}. \tag{41.14}$$

This is a positive-semidefinite Gram matrix. A negative cross term does not make total dissipated power $P = \frac{\omega}{2} Q^\dagger J Q$ negative. Cross-radiation energy cannot arbitrarily be assigned to one independent recording channel.

41.4 Bright/dark modes, frequency dependence, and numerical benchmarks

Equal couplings admit bright/dark combinations. Near threshold with $M > 0$, $J_B \sim k$ and $J_D \sim k^3$; for a massless field they scale as k^3, k^5. A dark mode suppresses low-frequency radiation, but for $g_2 \neq g_3$, generally

$$[J(\omega_1), J(\omega_2)] \propto g_2 g_3 (g_2^2 - g_3^2)[\operatorname{sinc}(k_2 L) - \operatorname{sinc}(k_1 L)] \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0. \tag{41.15}$$

so one fixed rotation cannot diagonalize the environment at every frequency. A massless low-frequency ω^3 spectrum is not a default white-noise Ohmic bath.

Take $f = a = M_\phi = 1$, $v_2 = v_3 = 1/4$, $L = 3$. Then

$$g_2 = \frac{1}{4\sqrt{3}}, \quad g_3 = \frac{1}{4\sqrt{8}}, \quad M_{\mathrm{eff}}^2 = \frac{367}{384}. \tag{41.16}$$

Direct evaluation of the analytic kernel gives:

Quantity	Value
$\gamma_{22}(0), \gamma_{33}(0)$	0.02083333, 0.00781250
$\gamma_{23}(0)$	0.00134466
Normalized instantaneous cross-correlation $e^{-9/4}$	0.10539922
ω at $k = 1$	1.39847387
J_{22}, J_{33}, J_{23}	0.05313466, 0.01992550, 0.00153060
Mean power for $q_2 = 1/4, q_3 = 3/4$	$0.01973791 + 0.00092686 \cos\delta$

Ignoring the phase cross term for the moment, $\beta = 4, T = 50$ give accumulated hazards $1.85768592, 2.08989666$. Model A gives $(P_2, P_3, P_\varnothing) \simeq (0.461505, 0.519193, 0.019301)$, with channel-2 probability 8/17 conditional on a click. Model B at the same parameters gives $(0.249852, 0.703774, 0.046374)$. Unequal couplings make the models plainly noninterchangeable.

Adjusting the pumps to $v_2/v_3 = \sqrt{3/8}$ sets $g_2 = g_3$. For $v_3 = 1/4$, $M_{\text{eff}}^2 = 31/32$, with A's no-click probability about 0.0599405 and $(P_2, P_3) \simeq (0.235015, 0.705045)$, hence conditional weights $(1/4, 3/4)$. This recovery uses pump calibration. Equal couplings admit a fixed bright/dark basis, but the measured channels still need to be identified.

41.5 Thermal noise, long memory, and elimination limits

Classical thermal initial data give the fluctuation–dissipation relation $\langle \xi_i(t)\xi_j(s)\rangle = k_B T\, \gamma_{ij}(t-s)$. Classical noise vanishes as $T \to 0$; exponential thermal-energy thresholds also vanish, which does not automatically produce a stable zero-temperature latch. For equal coupling to a massive field, long-time tails are

$$\gamma_B(t) = 2g^2 a^3 (2M/t)^{3/2} \cos(Mt + 3\pi/4) + O(t^{-5/2}), \tag{41.17}$$

$$\gamma_D(t) = \frac{g^2 a^3 L^2}{4} (2M/t)^{5/2} \cos(Mt + 5\pi/4) + O(t^{-7/2}). \tag{41.18}$$

Long dark-mode energy lifetime and faster decay of the dark memory-kernel tail describe different objects.

The Laplace-domain equation is

$$[s^2 I + \Omega_0^2 + s\widetilde{\gamma}(s)]\widetilde{Q} = [sI + \widetilde{\gamma}(s)]Q_0 + \dot{Q}_0 + \widetilde{\xi}. \tag{41.19}$$

The initial-memory term is part of the response. Integrating out a heavy scalar can produce the local term $J^2/(2M_\phi^2)$, but does not by itself produce white noise. Likewise, a vanishing real massless three-point vertex leaves off-shell and driven processes possible. The absence of a direct $\bar{\psi}\psi\, y_n$ term in the baseline leaves indirect gravitational channels intact.

42 Operational States and Statistical Closure

42.1 Covariant statistical protocols

Let preparation be a measure μ_P on initial data and recording a map R_M. If a Lorentz transformation L satisfies

$$\mu_{LP} = L_* \mu_P, \qquad R_{LM}(Ls) = R_M(s), \tag{42.1}$$

Covariance follows when the preparation measure, state, and apparatus transform together. It leaves the choice of μ_P and the form of equilibrium on a given background open.

The common candidate compression is

$$\Pi\mu = E_\mu[P(b)], \qquad P(b) = \frac{bb^\dagger}{b^\dagger b}. \tag{42.2}$$

It yields positive trace-one matrices and is affine under mixing. Operational completeness requires preparations with the same $\Pi\mu$ to yield the same subnormalized output after *every allowed instrument and every record conditioning*. In record-kernel notation K_r,

$$\Pi K_r \mu = \mathcal{I}_r \Pi\mu \quad \Longleftrightarrow \quad \Pi\mu_1 = \Pi\mu_2 \Rightarrow \Pi K_r \mu_1 = \Pi K_r \mu_2. \tag{42.3}$$

The trace of the subnormalized output must give the record probability. Complete positivity additionally concerns consistency with spectator composite systems and does not follow from single-system positivity.

42.2 Instrument simulation and physical implementation

For supplied effects E_i, the isometry $Wb = \bigoplus_i \sqrt{E_i} b$ makes output norms $b^\dagger E_i b$. A uniform threshold then simulates the POVM. Given Kraus operators, branch-norm sampling and the update $K_\zeta b / \|K_\zeta b\|$ similarly simulate a sequence. Implementing these operations with CR fields is a further dynamical problem.

The same candidate density matrix can hide different higher-order structure. For example, sphere measures $\mu_z \propto |n_z|$ and $\mu_x \propto |n_x|$ both give $\Pi\mu = I/2$, but $E_{\mu_z} n_z^2 = 1/2$ and $E_{\mu_x} n_z^2 = 1/4$. An instrument reading second direction moments distinguishes their operational states. Single-shot agreement also does not guarantee repeated measurements: two identical tests without hidden-state reset often give $\int q^2 \, \mathrm{d}\mu$, while ideal projection requires the appropriate conditional reset for standard sequence probabilities.

42.3 Bogoliubov terms and projective closure

For canonically normalized two-mode kinetics, some reference variables give

$$\mathsf{G} = \begin{pmatrix} 1 & -h \\ -h & 1 \end{pmatrix}, \quad h = \frac{\sinh(\phi/f)}{\sqrt{n^2 \cosh^2(\phi/f) - 1}}, \quad K = \frac{I + h\sigma_x}{1 - h^2}. \tag{42.4}$$

With fixed reference complex coordinates $b = (q + \mathrm{i}p)/\sqrt{2}$, the linear flow is

$$\dot{b} = Ab + B\bar{b}, \qquad A = -\frac{\mathrm{i}}{2}(K + I), \quad B = \frac{\mathrm{i}}{2}(K - I). \tag{42.5}$$

For $B \neq 0$, a common phase changes the relative phase of the two terms. The source gives equal projectors with Bloch derivatives 0 and $-2h/(1-h^2)$ in one convention. Thus this fixed projective description already fails to close under the quadratic flow.

At fixed h, frequency-adapted normal modes give unitary oscillation. For time-dependent $h(t)$, adapted complex coordinates still generate the antilinear term $-\dot{\Omega}/(2\Omega)\,\bar{b}$. No fixed positive quadratic norm makes every different $K(\phi)$ flow unitary. Restricted controls often generate only $\mathrm{Sp}(2,\mathbb{R}) \times \mathrm{Sp}(2,\mathbb{R})$, insufficient for arbitrary mixing of all modes. This controllability limit concerns the analyzed controls, not every possible CR operation.

42.4 Anomalous moments, environmental correlations, and conditioning

For supplied quadratic coefficients, retaining $C = E[bb^\dagger]$ and $M = E[bb^T]$ closes the equations:

$$\dot{C} = AC + CA^\dagger + BM^\dagger + MB^\dagger, \tag{42.6}$$

$$\dot{M} = AM + MA^T + BC^T + CB^T. \tag{42.7}$$

The Nambu state (C, M) closes for the supplied quadratic system. With $H = \omega|z|^2 + \kappa|z|^4$, however, $\frac{\mathrm{d}}{\mathrm{d}t} E[z^2]$ depends on $E[|z|^2 z^2]$. Source ensembles with $E|z|^2 = 1$, $Ez^2 = 0$ and fourth-order moments $0, 2/3$ have initial derivatives differing by $-8\mathrm{i}\kappa/3$. Nonlinearity therefore reopens the moment hierarchy.

For a linear system evolving with its environment, the system covariance also contains

$$\Sigma_x(t) = A\Sigma_x A^T + B\Sigma_y B^T + A\Sigma_{xy} B^T + B\Sigma_{yx} A^T. \tag{42.8}$$

Knowing only the two marginals without initial cross-correlations does not predict later system statistics. A non-Markovian bath makes this particularly explicit.

Threshold conditioning also takes Gaussian states outside the Gaussian family. For two-dimensional thermal energy $K \sim \mathrm{Exp}(\beta)$, conditioning on no click, $K > W$, gives $K = W + \mathrm{Exp}(\beta)$, but component momentum has

$$E[p_1^2] = W + \beta^{-1}, \quad E[p_1^4] = \tfrac{3}{2}(W^2 + 2W/\beta + 2/\beta^2). \tag{42.9}$$

values $0.75, 0.9375$ at $W = 1/2, \beta = 4$, while a Gaussian with the same variance has fourth moment 1.6875. Even quadratic unconditional dynamics does not ensure covariance-only closure of post-record states.

42.5 Stronger requirements for fully time-resolved records

With cumulative hazard $\Lambda_i = -\ln S_i(W_i)$, $p(i,t) = \dot{\Lambda}_i e^{-\sum_j \Lambda_j}$. On an absolutely continuous interval with initially zero hazards and positive survival, requiring $p(i,t) = q_i p_{\mathrm{click}}(t)$ at every time is equivalent to

$$\Lambda_i(t) = q_i \sum_j \Lambda_j(t). \tag{42.10}$$

This criterion characterizes the specified time-factorized record class. It is stronger than obtaining fixed proportions only after integrating to a cutoff. Quantum model B with unequal rates has time-dependent proportions.

43 Physical Records and Latching Dynamics

43.1 From response to persistent records

A linear oscillator response does not by itself produce two persistent record states. Nor do gauge-equivalent Higgs signs define distinguishable memories. For a static scalar field in three flat spatial dimensions, with positive gradient energy and nonnegative relative potential, scaling gives

$$E(\lambda) = \lambda^{-1} T + \lambda^{-3} U, \tag{43.1}$$

and a Derrick-type obstruction to nontrivial finite-energy static stationary points. This does not exclude gauge fields, gravity, boundaries, time-periodic states, or structures outside the premises.

There is also a zero-seed invariant subspace: when the relevant potential derivatives and external sources vanish, $V = \phi = 0$ with zero velocities can persist despite nonzero U. Smooth finite-time flow bounds a seed ϵ by $C_T e^{LT}\epsilon$, and its energy by the corresponding squared scale. A fixed threshold may then remain uncrossed; instability-driven onset can depend on $\ln(1/\epsilon)$.

43.2 Charge as a record variable

In a region D, take

$$Q_D = \int_D j^0 \,\mathrm{d}^3x, \qquad \dot{Q}_D = -\oint_{\partial D} \boldsymbol{j} \cdot \boldsymbol{n} \,\mathrm{d}S. \tag{43.2}$$

which Maxwell equations also express as a weighted electric flux. Suppressing boundary current can preserve charge, a more calibratable record than an arbitrary field sign. Classical field-amplitude splitting nevertheless naturally gives $(e|a|^2, e|b|^2)$ rather than an exclusive $(e, 0)$ or $(0, e)$ in each run.

An extended gauge factor $Z_\gamma(\phi) = Z_0 + z_1\phi$ produces $-a_F\phi\mathcal{F}^2$, with $a_F = z_1/(4g_\gamma^2)$. The corresponding matter force contains

$$\xi_i^{\mathrm{mat}} = -g_i a_F \frac{\mathrm{d}}{\mathrm{d}t}\int h_i\,[G_{\mathrm{ret}} * \delta(\mathcal{F}^2)]\,\mathrm{d}^3x. \tag{43.3}$$

A vacuum plane wave has $\mathcal{F}^2 = 0$, but in a background field $\delta(\mathcal{F}^2) = 4\boldsymbol{B}_0\cdot\delta\boldsymbol{B} - 4\boldsymbol{E}_0\cdot\delta\boldsymbol{E}$ can be nonzero. This channel requires the added parameter $z_1 \neq 0$ and is absent from the $Z_\gamma = 1$ baseline.

43.3 Double-well latches and their statistics

Assign an effective double-well potential to a charge-difference pointer X_D:

$$V_D = \frac{E_b}{4}(X_D^2 - 1)^2 - \mathcal{F}X_D. \tag{43.4}$$

At zero bias, the barrier is $E_b/4$, the minimum curvature $2E_b$, and saddle curvature $-E_b$. A local well disappears at critical bias $|\mathcal{F}| = 2E_b/(3\sqrt{3})$. In the high-barrier overdamped approximation,

$$k_{\mathrm{flip}} \simeq \frac{\sqrt{2}E_b}{2\pi\gamma}e^{-E_b/(4k_BT)}. \tag{43.5}$$

$E_b = 8, k_BT = 0.1, \gamma = 1$ give $k_{\mathrm{flip}} \simeq 3.71\times10^{-9}$, lifetime about 2.69×10^8 and flip probability over retention time 1000 about 3.71×10^{-6}. These are conditional estimates for the stipulated thermal double well. An actual CR bath with algebraic long memory need not reduce to one γ.

For bias $\mathcal{F} = \alpha(2q-1)$, the equilibrium two-state approximation gives

$$P_+(q) = \frac{1}{1+\exp[-2\alpha(2q-1)/(k_BT)]}. \tag{43.6}$$

which differs from q in general. At $q = 1/4$, the ratios $\alpha/(k_BT) = 2$ and 4 give 0.1192029 and 0.0179862, respectively, correcting the shifted numerical cases in S5. Exact equality would require $\mathcal{F} = (k_BT/2)\ln[q/(1-q)]$, divergent at the endpoints. Finite-temperature dark counts also persist without input.

43.4 Results of the XI–XV apparatus studies

The XI–XV studies examined finite-volume $Q \to \phi \to a$ transfer, traps, Maxwell/Penning confinement, finite baths, and latch chains. Boundary conditions, recurrence, external control, and backreaction prevented a general derivation of a robust binary instrument from the baseline action. R reports these outcomes; the individual numerical programs are not reproduced here.

The studies separate three problems: storing a stable record, obtaining single-run probabilities, and closing the state description under future operations. Subsequent work tests the last problem through nonlinear flow and controlled approximations, then returns to physical clocks and probes.

44 Nonlinear Feedback and Moment Closure

44.1 Common-phase feedback

Retain a finite-mode Hamiltonian for a normal pair and the shared scalar:

$$H = \frac{1}{2}p^T K_n(\phi)p + \frac{1}{2}q^T q + \frac{1}{2}\pi_\phi^2 + \frac{1}{2}M^2\phi^2. \tag{44.1}$$

Its exact equations are

$$\dot{q} = K_n p, \quad \dot{p} = -q, \quad \dot{\phi} = \pi_\phi, \quad \dot{\pi}_\phi = -\frac{1}{2}p^T K_n' p - M^2\phi. \tag{44.2}$$

Take $\phi(0) = \pi_\phi(0) = 0$ and $q = (\cos\theta, -\sin\theta)$, $p = (\sin\theta, \cos\theta)$. Then the projector $P(b)$ of $b = e^{\mathrm{i}\theta}(1, \mathrm{i})^T/\sqrt{2}$ is independent of θ. Nevertheless,

$$\ddot{\phi}(0) = -\frac{\lambda_n}{2}\sin 2\theta, \qquad \phi^{(3)}(0) = \lambda_n \cos 2\theta. \tag{44.3}$$

Histories with the same projective state source the shared scalar differently. After feedback into the normal modes, the Bloch-component difference for $\theta = \pm\pi/4$ has

$$\Delta n_z = -\frac{\lambda_n^2}{6}t^3 + O(t^4). \tag{44.4}$$

with coefficient $-1/18$ at $n = 2, f = 1$. The dynamics itself makes phase information discarded by the projector visible again.

Also, $\dot{N} = q^T(K_n - I)p$, $N = b^\dagger b$, is generally not conserved. In the full nonlinear Hamiltonian the original continuous common-phase group usually leaves only overall signs $\alpha = 0, \pi$. The free two-oscillator $U(1)$ reduction therefore does not extend unconditionally to all coupled histories.

44.2 Fixed-complex-structure rigidity and its dimension restriction

For a linear symplectic flow in complex dimension $d \geq 2$, phase-independent projective evolution of every nonzero vector strongly constrains a continuous flow to complex-linear/unitary structure. The source rigidity claim needs the explicit $d \geq 2$ condition. For $d = 1$, $\mathbb{CP}^0$ is one point and every real symplectic flow projects trivially; for example, $\mathrm{diag}(e^t, e^{-t}) \in \mathrm{Sp}(2, \mathbb{R})$ is not in $U(1)$. The actual two-mode test has $d = 2$ and is unaffected.

For varying $K(\phi)$, differences of linear generators contain a nonzero nilpotent block $\left(\begin{smallmatrix} 0 & K-I \\ 0 & 0 \end{smallmatrix}\right)$. which cannot be anti-self-adjoint in any fixed positive inner product: anti-self-adjoint matrices are diagonalizable, unlike nonzero nilpotent matrices. If complex structure depends only on ϕ, the transport condition $J'(\phi)\pi_\phi = [A(\phi), J(\phi)]$ for arbitrary π_ϕ forces $J' = 0$, returning to the fixed-structure obstruction. History-, momentum-, or enlarged-state-dependent constructions lie outside this exclusion.

Forcing ϕ to zero cannot arbitrarily retain the full two-mode space. In the relevant parity-invariant subspace, $\phi = \pi_\phi = V = p_V = 0$ leaves one oscillator and $\mathbb{CP}^0$. Keeping both modes with $\phi = 0$ generally requires $p_U p_V = 0$ throughout the history, forcing the corresponding single-mode restriction in the equal-frequency model.

44.3 An exactly closed single-trajectory $\mathbb{Z}_2$ quotient

Let

$$Q = qq^T, \quad P = pp^T, \quad C = qp^T. \tag{44.5}$$

These are blocks of the rank-one matrix xx^T for $x = (q, p)$, determining x up to overall sign. For Eq. (44.1),

$$\dot{Q} = KC^T + CK, \qquad \dot{P} = -C - C^T, \qquad \dot{C} = KP - Q, \tag{44.6}$$

$$\dot{\phi} = \pi_\phi, \qquad \dot{\pi}_\phi = -\tfrac{1}{2}\operatorname{Tr}(K'P) - M^2\phi. \tag{44.7}$$

These equations close exactly on a single trajectory modulo overall sign. Equivalently, define

$$D = bb^\dagger = \tfrac{1}{2}[Q + P + \mathrm{i}(C^T - C)], \quad F_b = bb^T = \tfrac{1}{2}[Q - P + \mathrm{i}(C + C^T)]. \tag{44.8}$$

The relations $Q = \Re(D + F_b)$, $P = \Re(D - F_b)$, and $C = \Im F_b - \Im D$ recover the quadratic variables completely. Anomalous information F_b cannot be discarded arbitrarily.

Ensemble averaging need not preserve this closure: $E[K(\phi)C]$ is not determined by $E[K]$ and $E[C]$. For finitely many expectations $E[f_a]$ to predict all point preparations and their mixtures exactly, $\operatorname{span}\{1, f_a\}$ must be invariant under the Liouville generator. This gives affine expectation dynamics. A smaller physically preparable class could weaken the requirement, but would need justification.

44.4 Thermal integration retains geometric information

In the canonical finite-mode model, Gibbs integration over normal momenta introduces $\sqrt{\det \mathsf{G}_n} = \cosh(\phi/f)$. If the remaining ϕ potential is $M^2\phi^2/2$, the marginal density is

$$p(\phi) \propto \cosh(\phi/f) e^{-\beta M^2 \phi^2/2}. \tag{44.9}$$

exactly an equal mixture of two Gaussians with variance $\sigma^2 = (\beta M^2)^{-1}$, means $\pm\sigma^2/f$, and total variance $\sigma^2 + \sigma^4/f^2$. Setting $\phi = 0$ before integration would lose this calculable thermal broadening.

45 Controlled Low-Energy and Long-Time Limits

45.1 Uniform finite-time bounds

For total energy $H \le E_*$ in Eq. (44.1) and $M > 0$, $|\phi| \le \sqrt{2E_*}/M$. On this compact interval, set $\delta_* = \sup \|K(\phi) - I\| = O(\sqrt{E_*})$. Frequency-adapted free evolution $U_0(t)$ and the physical projective trajectory admit a bound of the form

$$\|P(b(t)) - P(U_0(t)b_0)\|_1 \le 2\omega_{\max}\delta_* \, t \tag{45.1}$$

for nonzero response amplitude and consistent normalization. If additionally $\phi_0 = \pi_{\phi 0} = 0$, the scalar is first built by a quadratic response source, giving a stronger bound of type CE_*T^2. Constants depend on K', G, and M over the energy region.

The estimates hold uniformly for the specified preparation classes and finite times, despite exact projective nonclosure. They exclude zero response, where $P(b)$ is undefined, and do not extend to arbitrary instruments or duration.

45.2 Weak-probe distinguishability and competing timescales

For a fixed instrument with a uniform Lipschitz response near zero amplitude, different directional inputs with energy $O(E)$ have contrast at most $O(\sqrt{E})$. An even response with bounded second derivatives improves this to $O(E)$. The statement excludes external amplification varied with E, infinite waiting, and independently powered instruments.

If the free-approximation error has sufficient bound CET^2 and the proposed record rate is E^α, choosing $T = E^{-\beta}$ to accumulate records while sending that bound to zero requires $\alpha < \beta < 1/2$. This is the compatibility range of the two assumed estimates.

The physical source contains difference and sum frequencies. With $D_{12} = b_1\bar{b}_2$ and $F_{12} = b_1 b_2$, difference-frequency terms in $p_1 p_2$ depend on D_{12}, while sum-frequency terms depend on F_{12} carrying twice the common phase. Resonance produces secular growth; when the scalar frequency equals the sum frequency, for example, $\phi^{(1)} \sim -\lambda E\, t \sin(Mt)/(4M)$. Nonresonant expansions contain $(\cos \nu t - \cos Mt)/(M^2 - \nu^2)$ and cannot be bounded using a fixed denominator near resonance.

A finite-duration L^1 filter has an entire Fourier transform. If that transform vanishes on a frequency interval, it vanishes everywhere. Exact band rejection is therefore unavailable to a nonzero finite-time filter, although isolated zeros and approximate or asymptotic suppression are possible.

45.3 Convergence of nonlinear intensity fractions

S7 instead studies averages of actual nonlinear intensity histories:

$$J_i(T) = \frac{1}{T}\int_0^T q_i(t)^2\,\mathrm{d}t, \qquad R_T = \frac{J_1(T)}{J_1(T) + J_2(T)}. \tag{45.2}$$

Normalize free frequency to one, take $N_0 = \|b_0\|^2 > 0$, and let $\eta_E = \sup_{|\phi| \le \sqrt{2E}/M} \|K(\phi) - I\|$, $\Delta_T = e^{\eta_E T} - 1$. Duhamel and Gronwall estimates give

$$\|b(t) - e^{-\mathrm{i}t} b_0\| \le \sqrt{N_0}\Delta_T. \tag{45.3}$$

Finite-window residuals of free quadratic oscillations are $O(1/T)$. Define

$$\epsilon_T = \frac{1}{T} + 2\Delta_T(2 + \Delta_T); \tag{45.4}$$

For $\epsilon_T < 1$,

$$\left| R_T - \frac{|b_{01}|^2}{N_0} \right| \le \frac{2\epsilon_T}{1 - \epsilon_T}. \tag{45.5}$$

In a general energy class, $T \sim E^{-1/4}$ gives $O(E^{1/4})$; for the class $\phi_0 = \pi_{\phi 0} = 0$, use $\Delta_T \le e^{CET^2} - 1$ and $T \sim E^{-1/3}$ to obtain $O(E^{1/3})$. The measured continuous fraction thus approaches the projective overlap at a controlled rate.

The theorem concerns continuous intensities. Discrete clicks, state updates, and composite probabilities require additional analysis. It also distinguishes two averaging procedures:

$$E_\mu[R_T] \;\longrightarrow\; \mathrm{Tr}[E_\mu P(b_0)\, P_1]\,, \tag{45.6}$$

Pooling intensities from all runs before taking the ratio instead approaches

$$\frac{E_\mu[N_0 P_{11}(b_0)]}{E_\mu[N_0]}. \tag{45.7}$$

For equally probable preparations of pure channels 1 and 2 with intensities $1, 3$, averaging the run-wise ratio gives $1/2$, while pooling intensities gives $1/4$. The protocol must specify the order of averaging.

46 Parity, Dynamic References, and Readout

46.1 Parity-invariant records

Normal sectors possess the exact discrete transformation

$$\mathcal{P}: \quad \phi \mapsto -\phi, \quad U_n \mapsto U_n, \quad V_n \mapsto -V_n, \qquad Z = \mathrm{diag}(1,-1), \tag{46.1}$$

with the corresponding canonical momentum transformation. If potentials, all retained modules, and boundaries respect it, $\mathsf{G}_n(-\phi) = Z\mathsf{G}_n(\phi)Z$, and the full flow commutes with $\mathcal{P}$. If records and intermediate instrument kernels are invariant too, complete record histories cannot distinguish μ from $\mathcal{P}_*\mu$.

In an already supplied two-dimensional quantum description, this requires effects $E_r = ZE_rZ = a_rI + b_r\sigma_z$, excluding σ_x, σ_y readout with these operations. Symmetry-preserving gravity leaves the indistinguishability intact. A calibrated odd reference changes the accessible effects.

Eliminating the shared scalar generates a Hamiltonian of type $-J^2/(2M^2)$, with $J = \sum_n \lambda_n p_{U_n} p_{V_n}$. It contains cross-sector interactions while remaining globally parity even. In an assumed two-qubit description, the commutant of total $Z \otimes Z$ can be $M_2 \oplus M_2$. This exhibits symmetry-restricted correlated operations without independently generating a composite Hilbert space.

46.2 Odd references and feedback

For a known odd reference s, covariance requires $E_r(-s) = ZE_r(s)Z$. Its I, σ_z coefficients are even, and σ_x, σ_y coefficients odd. A uniform Lipschitz bound gives contrast $\le 2L|s|$. Total energy gives $|s| \le \sqrt{2E}/M$, so transverse contrast for a fixed instrument vanishes if the reference itself is taken to zero energy. External reference resources lie outside this argument.

The crucial inverse-kinetic term is

$$K_{n,12}(s) = \frac{\sinh(s/f)}{\sqrt{n^2-1}\cosh^2(s/f)} = \lambda_n s + O(s^3). \tag{46.2}$$

It both enables relative-phase readout and causes nonunitary feedback. The source provides a stronger counterexample with equal initial projectors: $q_A = q_B = (1,0)$, $p_A = (1,0)$, $p_B = (-1,0)$, At common $\phi = s, \pi_\phi = 0$, the two states have equal energies and $P(b_A) = P(b_B)$, yet $\Delta n_x = 2K_{12}t + O(t^2)$. The reference that enables readout also changes the dynamics of the projective state.

For frozen K, a canonical normal-mode transformation restores unitary evolution; dynamic K restores squeezing. For one mode, $H = k(t)p^2/2 + \omega_0^2 q^2/2$, take $z = (\alpha q + \mathrm{i}p/\alpha)/\sqrt{2}$, $\alpha = \sqrt{\omega_0}k^{-1/4}$. Then

$$\dot{z} = -\mathrm{i}\omega_0\sqrt{k}\, z - \frac{\dot{k}}{4k}\bar{z}. \tag{46.3}$$

Even without response backreaction, the reference generally oscillates by its own field equation and cannot be treated as an indefinitely fixed control.

46.3 Continuity at zero amplitude and exactly scale-invariant readout

If a fixed physical instrument has probability $p(b)$ continuous at $b = 0$, and for every nonzero b exactly obeys $p(b) = b^\dagger E b/(b^\dagger b)$, its directional limits force $E = cI$. Nontrivial exact Born readout must therefore use a normalized-state domain, nonzero-intensity conditioning, external amplification, or a suitable order of limits.

Common phase can also enter even records. For example, the source's two weak-energy initial phases yield $\phi_A \sim \lambda E t^3/6$, $\phi_B \sim -\lambda E t^2/4$, and leading $\Delta\phi^2$ of order $\lambda^2 E^2 t^4/16$, equal to $E^2 t^4/48$ at $n = 2, f = 1$. Parity invariance therefore does not erase all hidden-phase information, although an actual baseline instrument must still be shown to read this quantity.

47 Composite Systems, Contextuality, and Bell Bounds

47.1 Local rays and entangled states

For two already supplied two-level quantum systems, write a pure state $a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, It is separable precisely when $ad - bc = 0$; normalized pure-state concurrence is $2|ad - bc|$. Geometrically, local-ray products $\mathbb{CP}^1 \times \mathbb{CP}^1$ maps into $\mathbb{CP}^3$ by the Segre embedding and covers only separable states. Two response spheres therefore leave the entangled directions unspecified.

With standard tensor products and quantization assumed, scalar-mediated cross-sector interactions can generate entanglement. For $H_{\text{int}} = gA \otimes B$ and an initial product state,

$$1 - \operatorname{Tr} \varrho_A(t)^2 = \frac{2g^2 t^2}{\hbar^2} \operatorname{Var}(A) \operatorname{Var}(B) + O(t^3). \tag{47.1}$$

For example, $\sigma_x \otimes \sigma_x$ sends $|00\rangle$ to $\cos\theta|00\rangle - \mathrm{i}\sin\theta|11\rangle$, with concurrence $|\sin 2\theta|$. The calculation assumes tensor-product composition, positive states, and Born readout. Classical correlations and mixed target curvature supply interactions but do not fix these quantum rules.

47.2 Noncommuting controls, matrix identities, and physical contextuality

Two classical control flows can have order-dependent outcomes and nonzero commutators without physical readouts satisfying the Pauli algebra. Once Pauli matrices are assumed, they form a Peres–Mermin square

$$\begin{pmatrix} X \otimes I & I \otimes X & X \otimes X \\ I \otimes Y & Y \otimes I & Y \otimes Y \\ X \otimes Y & Y \otimes X & Z \otimes Z \end{pmatrix}. \tag{47.2}$$

with row products I and column products $I, I, -I$. The resulting noncontextual bound is 4, whereas the quantum value is 6. An experiment must realize the same observable in each context and verify compatibility and repeatability; reprogramming classical outputs between contexts would not meet that requirement.

A Weyl spinor $\psi = (u, v)$ has bilinear current

$$J^0 = \psi^\dagger \psi, \quad J^i = \psi^\dagger \sigma_i \psi, \quad (J^0)^2 - \sum_i (J^i)^2 = 0. \tag{47.3}$$

This identity relates a supplied Weyl spinor to a null current and a pure Bloch vector. Fermionic statistics and multisystem composition remain separate input.

47.3 Bell bounds and quantitative setting-dependence audits

A hidden-variable model with local responses and setting-independent measure,

$$P(a, b|x, y) = \int P(a|x, \lambda) P(b|y, \lambda) \, \mathrm{d}\mu(\lambda) \tag{47.4}$$

obeys $|S_{\mathrm{CHSH}}| \le 2$. Allow setting-dependent measures μ_{xy} and let $\bar{\mu}$ average over four settings, with $\delta_{xy} = d_{\mathrm{TV}}(\mu_{xy}, \bar{\mu})$. Then

$$|S| \le 2 + 2\sum_{xy} \delta_{xy}, \qquad \bar{\delta} \ge \frac{|S| - 2}{8}. \tag{47.5}$$

For uniform settings, Pinsker's inequality gives an information bound in bits:

$$I(\Lambda : XY) \ge \frac{(|S| - 2)^2}{32 \ln 2}. \tag{47.6}$$

At $S = 2\sqrt{2}$, the lower bound is about 0.0309409 bit. A particular explicit construction in the source costs about 0.04627 bit. This is a countermodel's information cost, not a CR prediction for experimental settings.

Lifts of pure preparations can also be strongly preparation contextual. In S6's four-axis approximate-pure-state construction, if each pure-state error is at most ϵ, two mixtures both compressing near $I/2$ can have

$$d_{\mathrm{TV}}(\mu_z, \mu_x) \ge \max(0, 1 - \tfrac{15}{2}\epsilon). \tag{47.7}$$

The hidden measures can thus differ substantially. Their operational distinction depends on which instruments are physically available.

47.4 Antibunching and no-signaling

If two click probabilities are conditionally independent given one classical random intensity $I \ge 0$ and both increase monotonically with I, $\mathrm{Cov}[p_A(I), p_B(I)] \ge 0$, so the corresponding HBT ratio is at least one. Antibunching excludes this common-intensity class, not every classical apparatus with memory, conditional updates, or other hidden variables.

No-signaling alone permits more than quantum theory: the local CHSH bound is 2, the quantum bound $2\sqrt{2}$, and the general no-signaling bound 4. An internal-observer reconstruction must explain the full allowed correlation set.

48 Self-Gravitating Clocks and Relational Readings

48.1 Matter clocks in the coupled action

The XXXIII–XXXIV studies model clocks and probes with the existing fields $g, \phi, y_n, H, \Psi, A_\mu$. They successively test gravitating matter solutions, monotone clock readings, signal transfer, persistent records, and nonclassical operational states. The first three can be investigated directly in finite-dimensional truncations.

With the baseline potential, set $\phi = 0, V_n = 0, w = 0$, place other matter in specified reference vacua, and retain only normal mode $U = U_2$ with mass $\mu > 0$ in flat FLRW, $\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2 \mathrm{d}\boldsymbol{x}^2$. This consistent homogeneous truncation obeys

$$3M_P^2 H^2 = \frac{1}{2}\dot{U}^2 + \frac{1}{2}\mu^2 U^2, \quad \dot{H} = -\frac{\dot{U}^2}{2M_P^2}, \quad \ddot{U} + 3H\dot{U} + \mu^2 U = 0. \tag{48.1}$$

The clock sources the geometry. On the expanding branch, $\dot{\rho}_U = -3H\dot{U}^2 \le 0$, so its motion includes gravitational backreaction.

48.2 A strict monotonicity condition for unwrapped phase

Let

$$X = U, \qquad Y = -\dot{U}/\mu, \qquad \Theta = \operatorname{unwrap}\operatorname{atan2}(Y, X). \tag{48.2}$$

A nonzero oscillator history cannot first reach the origin in (X, Y) at finite time, since uniqueness of the smooth ODE would make the entire solution trivial. Its phase derivative is

$$\dot{\Theta} = \frac{X\dot{Y} - Y\dot{X}}{X^2 + Y^2} = \mu - \frac{3HXY}{X^2 + Y^2}, \qquad \mu - \tfrac{3}{2}|H| \le \dot{\Theta} \le \mu + \tfrac{3}{2}|H|. \tag{48.3}$$

Since $\dot{H} \le 0$ on the expanding branch, $H(0) < 2\mu/3$ gives throughout the corresponding solution interval $\dot{\Theta} \ge \mu - \frac{3}{2}H(0) > 0$. One can invert $t = t(\vartheta)$ and define relational quantities

$$\mathcal{O}_H(\vartheta) = H(t(\vartheta)), \qquad \frac{\mathrm{d}\mathcal{O}_H}{\mathrm{d}\vartheta} = -\left.\frac{\dot{U}^2}{2M_P^2\dot{\Theta}}\right|_{\Theta=\vartheta}. \tag{48.4}$$

Phase unwrapping needs cross-cycle historical information; an instantaneous angle gives only a reading modulo 2π. One homogeneous clock also supplies no complete spatial event labels. Local reference coordinates additionally require rods or other matter reference fields.

The independent integration uses $M_P = \mu = 1, U_0 = 0.15, \dot{U}_0 = 0, a_0 = 1$, $H_0 = 0.15/\sqrt{6}$, and DOP853 for Eq. (48.1) over $0 \le t \le 120$, yielding

Test quantity	Recalculated value
Accumulated phase cycles	19.0815068
Minimum phase speed at 24001 samples	0.9105852
$H(120)$	0.00511062264
$\rho_U(120)$	7.8355391×10^{-5}
Maximum sampled absolute Friedmann residual	About 1.1×10^{-17}

These values agree with the S9 benchmark. The analytic bound establishes monotonicity, while the residual checks the homogeneous numerical solution.

48.3 Internal three-field signal transfer and parity indistinguishability

Retaining (ϕ, U_2, V_2) gives, in reference canonical variables,

$$\ddot{y} + 3H\dot{y} + \mathsf{G}_2^{-1}\mathsf{G}_2'\dot{\phi}\,\dot{y} + \mathsf{G}_2^{-1}\nabla_y V = 0, \tag{48.5}$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = \tfrac{1}{2}\dot{y}^T\mathsf{G}_2'\dot{y}. \tag{48.6}$$

A suitable linearization in ϕ, V_2 about $\bar{U}_2$ gives

$$\ddot{V}_2 + 3H\dot{V}_2 + \mu_V^2 V_2 = \lambda_2(\dot{\phi}\,\dot{\bar{U}}_2 - \mu_U^2\phi\bar{U}_2). \tag{48.7}$$

Thus the shared scalar can transmit information to another matter mode. S9's full three-field integration uses $(\phi_0, \dot{\phi}_0, U_{20}) = (0.018, 0.010, 0.15)$, $(M_\phi, \mu_U, \mu_V) = (0.8, 1, 1.2)$ and reports $\ddot{V}_2(0) \simeq -0.00127256$ and peak response energy about 1.2568×10^{-6}. These three-field values are quoted from S9.

Parity partners $(\phi, V_2) \mapsto (-\phi, -V_2)$ have equal energy readings. This demonstrates internal signal transfer while leaving localized storage and binary recording to be constructed.

48.4 Dispersion at a common clock event

The XXXIV counterexample includes gravity and clocks to avoid an insufficient comparison at equal coordinate times. Retain U_2, U_3, with $M_P = 1, \mu_2 = 1, \mu_3 = 1.3$. Then

$$\ddot{U}_j + 3H\dot{U}_j + \mu_j^2 U_j = 0, \quad 3H^2 = \tfrac{1}{2}\sum_{j=2,3}(\dot{U}_j^2 + \mu_j^2 U_j^2), \quad \dot{H} = -\tfrac{1}{2}\sum_{j=2,3}\dot{U}_j^2. \tag{48.8}$$

On a three-torus of unit reference comoving volume, probe canonical variables are $q = U_2, \pi_2 = a^3\dot{U}_2$. Fix clock data $U_3(0) = 0.12, \dot{U}_3(0) = 0$, with initial clock energy 0.012168. Choose nine equally weighted probe samples

$$q_0 = 0.06 + \sigma\xi, \qquad \pi_{20} = 0.035 + \sigma\zeta, \qquad \xi, \zeta \in \{-1, 0, 1\}. \tag{48.9}$$

Each sample sets its own H_0 from its energy to satisfy Friedmann, so background feedback is not frozen. Initially $\Delta q_0 \Delta\pi_{20} = \frac{2}{3}\sigma^2$.

Read the probe at each history's first event $U_3 = 0, \dot{U}_3 < 0$, namely $\Theta_3 = \pi/2$. A separate event-integration program recalculates all 54 histories, giving

σ	Initial $\Delta q \Delta\pi_2$	Common-clock $\Delta q \Delta\pi_2$
10^{-2}	6.66667×10^{-5}	6.91038×10^{-5}
3×10^{-3}	6.00000×10^{-6}	6.21881×10^{-6}
10^{-3}	6.66667×10^{-7}	6.90974×10^{-7}
3×10^{-4}	6.00000×10^{-8}	6.21876×10^{-8}
10^{-4}	6.66667×10^{-9}	6.90973×10^{-9}
10^{-5}	6.66667×10^{-11}	6.90973×10^{-11}

For small σ, the coefficient is $0.6909733\,\sigma^2$; maximum Friedmann residual at the integration nodes is about 2.4×10^{-17}. The product of standard deviations differs from the square root of the covariance determinant, but both shrink as σ^2.

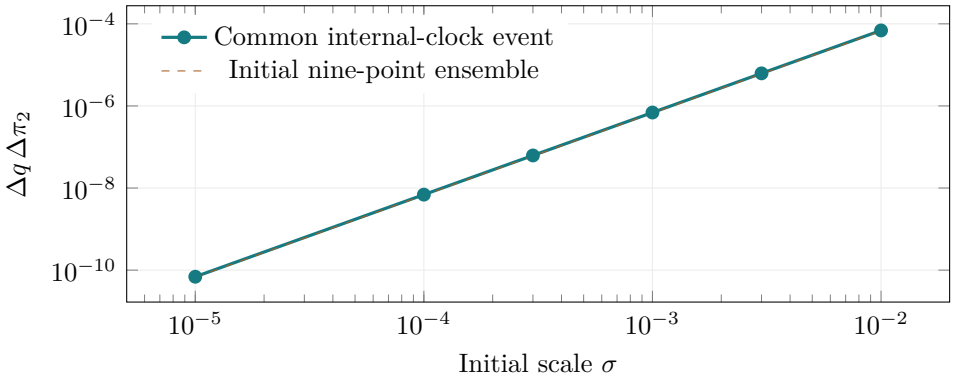


Figure 3: Probe dispersion at a common internal-clock event in the recalculated R benchmark. Both curves scale as σ^2; relational timing supplies no nonzero ensemble-independent lower bound.

An analytic argument also guarantees the limit. Smooth finite-time flows depend smoothly on initial data, and the clock crossing is transverse, $\dot{U}_3 \neq 0$. The implicit function theorem gives smooth event times. Evolving to the same clock event and then reading q, π_2 is therefore locally Lipschitz. Initial diameter $O(\sigma)$ gives each output standard deviation $O(\sigma)$, and their product $O(\sigma^2)$.

The model combines a causal metric, a finite-energy matter clock, gravitational backreaction, and a common relational event, yet admits arbitrarily sharp classical ensembles. A quantum lower bound would require a further restriction on the physical state space.

49 Quantum Positivity, Action Scales, and Loop Phases

49.1 Causal brackets and the action scale

For suitable globally hyperbolic backgrounds, normally hyperbolic operators, and support conditions, let $E = G_{\mathrm{ret}} - G_{\mathrm{adv}}$. For linear field functionals $F_f = \int f\phi \,\mathrm{dvol}_g$, the Peierls bracket is

$$\{F_f, F_h\}_P = E(f,h) \tag{49.1}$$

(up to the overall propagator sign convention). Spacelike-separated supports have zero bracket. As classical functions, $F_f F_h = F_h F_f$ still holds. Replacing this by $[\widehat{F}_f, \widehat{F}_h] = \mathrm{i}\kappa E(f,h) I$ needs a further representation rule.

The same oscillator equation $\ddot{q}+\omega^2 q = 0$ admits a positive-energy representation for any $\kappa > 0$, with ground-state two-point function

$$W_\kappa(t,s) = \frac{\kappa}{2\omega} e^{-\mathrm{i}\omega(t-s)} \tag{49.2}$$

a positive kernel with the same frequency and propagation, while $\Delta q \Delta p = \kappa/2$ varies with κ. Likewise, $S \mapsto \Lambda S$ preserves the classical stationary trajectories but rescales momenta and the symplectic form. Fixing an absolute action scale requires couplings and operational calibration beyond the trajectories and null cone.

Classical $H \geq 0$ does not imply a one-sided spectrum of vacuum-ordered correlators. A real oscillator $q = A\cos(\omega t+\Theta)$ with uniform random phase gives $E[q(t)q(0)] = A^2 \cos \omega t/2$, containing both frequency signs. Adding a positive quantum state with nonzero dynamical fluctuations and one-sided ordered spectrum can exclude the corresponding fully commutative stationary description but still does not fix the particular CCR or κ. A dynamical FLRW background may lack a globally preferred stationary vacuum.

49.2 Weyl-state positivity

If actual canonical operators satisfy $[\hat{q}, \hat{p}] = \mathrm{i}\kappa I$, let $W(\xi) = e^{\mathrm{i}(\xi_q \hat{q} + \xi_p \hat{p})}$ and $\sigma(\xi,\eta) = \xi_q \eta_p - \xi_p \eta_q$. Then

$$W(\xi)W(\eta) = e^{-\mathrm{i}\kappa\sigma(\xi,\eta)/2} W(\xi+\eta). \tag{49.3}$$

A positive quantum state's characteristic function $\Phi(\xi) = \mathrm{Tr}[\varrho W(\xi)]$ must satisfy, for every finite set of test points,

$$M_{jk} = e^{\mathrm{i}\kappa\sigma(\xi_j,\xi_k)/2} \Phi(\xi_k - \xi_j) \succeq 0. \tag{49.4}$$

Together with normalization, continuity, and the usual conditions, this characterizes full positivity of regular Weyl-algebra states. The covariance inequality $\Sigma + \mathrm{i}\kappa J/2 \succeq 0$ is only a low-order consequence; second moments alone do not test non-Gaussian positivity.

The deterministic classical origin has $\Phi_{\mathrm{cl}} = 1$. Choose points $(0,0), (a,0), (0,b)$ with $\theta = \kappa ab/2$. Then

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & e^{\mathrm{i}\theta} \\ 1 & e^{-\mathrm{i}\theta} & 1 \end{pmatrix}, \qquad \det M = -4\sin^2(\theta/2). \tag{49.5}$$

For generic nonzero area, M is not positive semidefinite. Quantum twisted positivity therefore excludes this deterministic classical state, provided that positivity is imposed on the physical algebra.

For an uncorrelated Gaussian characteristic function $\Phi = e^{-(V_q\xi_q^2+V_p\xi_p^2)/2}$, set $r = e^{-V_q a^2/2}, s = e^{-V_p b^2/2}$. The same three-point determinant is

$$\det M = (1-r^2)(1-s^2) - 4r^2s^2\sin^2(\theta/2) = a^2b^2(V_qV_p - \kappa^2/4) + O(a^4b^2 + a^2b^4). \tag{49.6}$$

Positivity at all sufficiently small test separations forces $V_qV_p \geq \kappa^2/4$. A correlated Gaussian instead requires $\det\Sigma \geq \kappa^2/4$. A zero determinant at one fixed test spacing is not a test of full state positivity.

Making κ directly depend on the shared scalar while also imposing $[\phi, \Pi_\phi] = \mathrm{i}\kappa_0 I$, $[q, \Pi_\phi] = [p, \Pi_\phi] = 0$ gives by Jacobi $\kappa_0\kappa'(\phi) = 0$. This excludes the simple variable-scale model without cross-commutators; more elaborate mixed algebras require new analysis.

49.3 Direct phase of the complex factor and full field-space holonomy

The phase $\vartheta = \arctan\chi$ of $F = 1 + \mathrm{i}\chi$ is globally defined for real χ, hence

$$A_\chi = \mathrm{d}\vartheta = \frac{\mathrm{d}\chi}{1+\chi^2}, \qquad \mathrm{d}A_\chi = 0, \qquad \oint A_\chi = 0. \tag{49.7}$$

The response eigenbasis angle $\beta = -\frac{1}{2}\arctan\sinh(\phi/f)$ is also globally continuous and has no winding by itself. By contrast, the seven-dimensional target has mixed curvature (27.5). Target tangent transport, phase-fiber transport, and quantum central extensions act on different spaces.

Quantum displacements are defined by $U_q(a) = e^{-\mathrm{i}a\hat{p}/\kappa}$, and $U_p(b) = e^{\mathrm{i}b\hat{q}/\kappa}$, giving

$$U_q(a)U_p(b)U_q(a)^{-1}U_p(b)^{-1} = e^{-\mathrm{i}ab/\kappa}I. \tag{49.8}$$

Coherent-path comparison can read this central phase. Associativity requires a two-cocycle but fixes no action scale: nonzero values of κ and the trivial cocycle are all compatible with that requirement.

49.4 A classical rotor counterexample

For a logical countertest, introduce a classical angle θ and conjugate action I,

$$S_\theta = \int \left[I\left(\dot{\theta} - \frac{p}{\kappa}\dot{q}\right) - H_\theta(I)\right]\mathrm{d}t. \tag{49.9}$$

with equations giving $\dot{\theta} = (p/\kappa)\dot{q} + H'_\theta(I)$. After subtracting the dynamical phase, a closed loop retains

$$\delta\theta_{\mathrm{geom}} = \frac{1}{\kappa}\oint p\,\mathrm{d}q. \tag{49.10}$$

A rectangular loop $q \to p \to -q \to -p$ gives $-ab/\kappa$, measurable relative to another rotor. The model remains classical and admits arbitrarily sharp q, p distributions. The loop phase alone therefore does not certify quantum state structure.

The same limitation appears in classical estimates of signal width. For massless-wave initial data, $V_\epsilon(0, \boldsymbol{x}) = A_\epsilon F(\boldsymbol{x}/\epsilon)$, $\dot{V}_\epsilon = 0$ has gradient energy

$$E_\epsilon = \frac{A_\epsilon^2\epsilon}{2}\int |\nabla F|^2\,\mathrm{d}^3x. \tag{49.11}$$

so width, amplitude, and energy can decrease together. Within this weak-field estimate, the null cone supplies no amplitude-independent minimum length; global nonlinear existence is a separate question.

50 Identifying the Physical Observable Algebra

50.1 Readouts and generators on the same Hilbert space

The Koopman–von Neumann construction makes the distinction between representation and readout explicit:

$$\mathcal{H}_{\mathrm{KvN}} = L^2(\mathbb{R}^2, \mathrm{d}q\,\mathrm{d}p), \qquad \rho_{\mathrm{cl}} = |\Psi(q,p)|^2. \tag{50.1}$$

with actual classical readings represented by multiplication operators $Q_{\mathrm{read}}\Psi = q\Psi$, $P_{\mathrm{read}}\Psi = p\Psi$, so $[Q_{\mathrm{read}}, P_{\mathrm{read}}] = 0$. For the positive probability density

$$\rho_\epsilon(q,p) = \frac{1}{2\pi\epsilon^2}\exp[-(q^2+p^2)/(2\epsilon^2)] \tag{50.2}$$

one has $\Delta Q_{\mathrm{read}}\Delta P_{\mathrm{read}} = \epsilon^2 \to 0$. A complex Hilbert space, positive probabilities, and unitary Koopman flow do not eliminate these states.

On a common dense Schwartz domain in the same space, separately define auxiliary generators

$$Q = \frac{q}{2} + \mathrm{i}\kappa\partial_p, \qquad P = \frac{p}{2} - \mathrm{i}\kappa\partial_q. \tag{50.3}$$

They obey $[Q,P] = \mathrm{i}\kappa I$ exactly, and appropriate self-adjoint realizations give

$$(U_q(a)\Psi)(q,p) = e^{-\mathrm{i}ap/(2\kappa)}\Psi(q-a,p), \tag{50.4}$$

$$(U_p(b)\Psi)(q,p) = e^{\mathrm{i}bq/(2\kappa)}\Psi(q,p-b), \tag{50.5}$$

with the Weyl central relation $U_q(a)U_p(b) = e^{-\mathrm{i}ab/\kappa}U_p(b)U_q(a)$. For the same normalized vector $\Psi_\epsilon = \sqrt{\rho_\epsilon}$,

$$(\Delta Q)^2 = (\Delta P)^2 = \frac{1}{4}\left(\epsilon^2 + \frac{\kappa^2}{\epsilon^2}\right), \qquad \Delta Q\Delta P \geq \frac{\kappa}{2}. \tag{50.6}$$

The same vector thus supports sharp classical readings and auxiliary generators obeying an uncertainty bound, because the two operator pairs represent different quantities.

A Weyl phase and a positive Hilbert representation are insufficient unless matter probes actually measure the noncommuting generators. Identifying that physical algebra is the central remaining step.

50.2 An internal probe coupling

The response action does supply an internal measurement coupling. In FLRW with unit comoving volume, normal-sector momenta are $\boldsymbol{P} = a^3\mathsf{G}_n\dot{\boldsymbol{y}}$, giving

$$H_{\mathrm{kin}} = \frac{1}{2a^3}\boldsymbol{P}^T K_n \boldsymbol{P}, \qquad H_{\mathrm{cross}} = \frac{\sinh(\phi/f)}{a^3\sqrt{n^2-1}\cosh^2(\phi/f)}P_U P_V. \tag{50.7}$$

At $n=2$, the leading term is $\phi P_U P_V/(a^3 f\sqrt{3})$. Exact inversion of G_n therefore couples the shared scalar to a probe through fields already present in the action.

If P_U, a are controlled over a finite experiment and other potential, pump, and backreaction errors are bounded, the interaction approximates $H_{\mathrm{meas}} = g(t)\phi P_V$. With $G = \int g(t)\mathrm{d}t$, the ideal pulse gives the canonical map

$$(\phi, P_\phi, V, P_V) \longmapsto (\phi, P_\phi - GP_V, V + G\phi, P_V). \tag{50.8}$$

with symplectic Jacobian $A^T J A = J$. The pointer V receives the signal and P_ϕ receives its backreaction. Classical input permits arbitrarily small $\Delta V\,\Delta P_V$, so this interaction alone imposes no quantum error bound. Its local realization with finite resources in full CR remains to be demonstrated.

50.3 Readouts, controls, and quantum states

Object	Available structure	Further conclusion not supplied
Physical classical readings	Real functions on initial data and probability measures	Not automatically a noncommutative observable algebra.
Control and evolution	Noncommuting Hamiltonian flows, history dependence, geometric phase	Do not determine actual measured operators.
Auxiliary representation	Complex Hilbert space and prequantum/Weyl generators	Do not exclude sharp classical readouts.
Physical quantum states	Positive quantum states on the actual observable algebra	Need an origin, preparability, and composite consistency.

If full CR is still interpreted through points λ of a classical initial-data space and every protocol's final reading is a real function $R_i(\lambda)$, ordinary products commute. A multiplicative representation π also satisfies $[\pi(R_i), \pi(R_j)] = 0$. The classical-measure positivity $\int F^* F \, \mathrm{d}\mu \geq 0$ is positivity of a commutative algebra, not the nonzero-κ condition in Eq. (49.4). A nonclassical completion could change this state space; the obstruction applies to the stated classical interpretation.

The next step is to derive preparable and readable operations from the action and determine their algebra. A quantum reconstruction must establish noncommuting physical observables, full state positivity, composition, and a common action scale. The uncertainty inequalities then follow from those structures.

51 Conclusions and Open Problems

51.1 What the postulates determine

Complex spacetime and the null cone provide the setting. P1 connects clock-and-signal observations to the full complex interval; its all-direction compatibility condition gives $\gamma = (1 + \mathrm{i}\chi)g$. P2 identifies real cotangent momentum and its natural complex extension. Scheme A then gives $C_A = F^{-1} C_{\mathrm{R}}$, one regular real constraint, and the canonical free action.

These requirements admit explicit scalar, Maxwell, and local Dirac realizations. They do not select a unique interacting theory, as the positive-weight counterexamples show. Restricted complex curvature and the chosen G_c response provide one concrete dynamical realization.

Its common scalar couples the sectors through a single source equation and a curved target manifold. Section 17 derives the single-sector and shared-base curvature; Section 27 relates the mixed Riemann components to scattering amplitudes.

The table records the principal results and the assumptions under which they hold.

51.2 Results and assumptions

Result	Precise premises	Present significance
$\gamma = (1 + \mathrm{i}\chi)g$	Lorentz real readout and all-real-null-direction compatibility in P1.2	Restricts allowed complex geometry.

Result	Precise premises	Present significance
Positive-mass-shell rigidity	Complete positive-mass shell, fixed quadratic form, direction-independent coefficient	An alternative explicit geometric bridge.
$C_A = F^{-1}C_{\mathrm{R}}$	Conformal form, natural complexification, quadratic mass-coefficient class	Unifies massive and massless free shells.
Real free geodesics	Natural one-form, real multiplier, real variation	Connects actual momentum, clocks, and motion.
Several-spin free baselines	Selected scalar real-part action, independent connection, local spin structure	Proves compatible constructions exist.
Free-field nonuniqueness	Positive-weight action family	Same cone and shell do not fix couplings.
$f^2 = 3M_P^2/2$	Restricted complex EH, volume branch, boundary treatment	Normalizes the healthy geometric scalar.
$\operatorname{spec} G_c = (c \pm 1)\rho$	A particular response constitutive relation	Classifies normal, critical, and negative-kinetic regions.
Critical second-class/contact structure	$c = 1$ model and regular constraint stratum	Defines reduction tasks, not automatic electromagnetism.
Internal group and sixteen generation labels	$E_3 \oplus E_2$, determinant condition, even exterior algebra	Consistent representations, hypercharge, and anomalies.
Cross-sector curvature and dark decay	Shared ϕ, full spectrum, selected vertices	Turns unified geometry into calculable interactions.
Projective and Clifford candidates	Chosen algebra, complex structure, symplectic reduction	Candidate quantum kinematics.
Internal clocks and physical records	Specific normal response, initial states, apparatus, couplings	Operational existence constructions.
Sharp common-clock ensembles and KLM	Allowed classical preparation and actual readout algebra	Locates quantum positivity still to add or derive.

51.3 Shared geometry and the remaining physical questions

The response parameter classifies kinetic structure: $c > 1$ is positive, $c = 1$ is critical, and the negative-sign ranges are fixed by $(c \pm 1)\rho$. Integer carriers account for particular reference weights without selecting all internal representations or restricting the continuous parameter to integers. The gauge and chiral construction adds precisely those internal data.

The common scalar has observable consequences once an action and spectrum are specified. In the dark model it mediates cross-sector interactions and can open three-body decays even when a two-body channel is closed. Exact stability requires a full-spectrum parity argument; cosmological longevity instead requires decay rates and a consistent production history. The scalar's own relic abundance and transverse perturbations enter the same calculation.

Quantum reconstruction remains less complete. Pauli identities, projective geometry, symplectic dynamics, and internal clocks are available, but do not select the algebra of actual readings or its positive states. Nonlinear feedback obstructs a simple exact projective quotient, while controlled

low-energy windows retain approximate closure. Sharp ensembles at a common gravitating-clock event show why internal time alone cannot supply a universal uncertainty bound.

The free reference action is insensitive to χ at fixed g, m; the interacting response action generally is not. Establishing which scalar effects are measurable therefore requires preparation, dynamics, and apparatus to be treated together. This is also where different realizations satisfying P1/P2 can be compared.

51.4 Open problems

Task	Question	Checkable completion criterion
Observers and P1	Which signals and instruments implement all-direction clock/rod comparison and test complex compatibility?	Explicit preparation, records, errors, and distinguishable backgrounds beyond coordinates.
Cross-species action selection	Why does actual matter select a reference coupling and mass rule?	Defined candidate class and selection mechanism tested against positive-weight counterexamples.
Full critical system	How do regular/singular $c = 1$ strata and inhomogeneous constraints evolve together?	Physical count, constraint propagation, principal symbols, energy estimates, well-posed data.
Complete dark cosmology	Can spectra, portals, lifetime, abundance, transverse modes, and geometric relics coexist at one parameter point?	One full parameter set with thresholds, widths, evolution, and observational statistics.
Microscopic matter origins	How are carriers, spinors, generations, and hierarchy selected?	Actual states and transformation laws, not label counting instead of spectra.
Internal quantum reconstruction	Do allowed states, effects, and composition satisfy quantum positivity?	Finite tests on one operational algebra, action scale, updates, composition rules.
Fundamental completion	How do global complex geometry and high-energy/nonperturbative definitions fit physical readout?	Global construction, degrees of freedom, stability, matching limits, falsifiable predictions.

The principal unresolved choices concern the interacting action, the critical spectrum, and the physical quantum state space. Resolving them would turn the geometric compatibility established by P1/P2 into sharper predictions for particles, cosmology, and internal experiments.

A An Optimal Bound for Approximate Null-Cone Compatibility

A.1 Statement and norm convention

In one fixed observer's orthonormal frame, take $g = \mathrm{diag}(-1, 1, 1, 1)$, a real symmetric four-by-four matrix b, and $\chi = \frac{1}{4}(-b_{00} + \mathrm{Tr}\, B)$. Suppose

$$\sup_{|\boldsymbol{n}|=1} |b_{00} + 2\boldsymbol{a}\cdot\boldsymbol{n} + \boldsymbol{n}^T B\boldsymbol{n}| \le \varepsilon, \qquad a_i = b_{0i}. \tag{A.1}$$

Then $\|b - \chi g\|_F \le \sqrt{11}\,\varepsilon/2$, with optimal constant. This improves the coarse bound in S8. The positive component norm refers to the chosen observer frame and is not invariant under boosts.

A.2 Proof of the upper bound

Set $M = B + b_{00} I$ with eigenvalues λ_i. Spatial orthogonal rotations preserve the norm and spherical supremum, so work in an eigenbasis. For $t = \mathrm{Tr}\, M$,

$$E = b - \chi g: \quad E_{00} = \frac{t}{4}, \quad E_{0i} = a_i, \quad E_{\mathrm{spatial}} = M - \frac{t}{4} I. \tag{A.2}$$

and hence

$$\|E\|_F^2 = \sum_i \lambda_i^2 - \frac{1}{4}\Big(\sum_i \lambda_i\Big)^2 + 2\sum_i a_i^2. \tag{A.3}$$

At $\boldsymbol{n} = \pm e_i$, the constraints $|\lambda_i \pm 2a_i| \le \varepsilon$ are equivalent to $|\lambda_i| + 2|a_i| \le \varepsilon$. For $\varepsilon > 0$, set $d_i = \lambda_i/\varepsilon \in [-1, 1]$, giving

$$\frac{\|E\|_F^2}{\varepsilon^2} \le F(\boldsymbol{d}) := \sum_i d_i^2 - \frac{1}{4}\Big(\sum_i d_i\Big)^2 + \frac{1}{2}\sum_i (1 - |d_i|)^2. \tag{A.4}$$

On each fixed-sign orthant, F has Hessian $3I - \frac{1}{2}\mathbf{1}\mathbf{1}^T$, with eigenvalues $3, 3, 3/2$, and is convex. Its maximum on that closed cube is attained at a vertex. Combining the eight orthants requires only the 27 vertices $d_i \in \{-1, 0, 1\}$.

For $0, 1, 2, 3$ nonzero components, the respective attainable maxima are $3/2, 7/4, 5/2, 11/4$. The last is attained with two $+1$ and one -1, or the opposite signs. Thus $F \le 11/4$ proves the bound. At $\varepsilon = 0$, cone rigidity directly gives $E = 0$.

A.3 An explicit matrix attaining the bound

Take

$$b = \varepsilon\, \mathrm{diag}(0, 1, 1, -1), \qquad \chi = \varepsilon/4. \tag{A.5}$$

The null-direction residual is $\mathscr{F}(\boldsymbol{n}) = \varepsilon(1 - 2n_3^2)$, with supremum exactly ε, while

$$E = \frac{\varepsilon}{4}\,\mathrm{diag}(1, 3, 3, -5), \qquad \|E\|_F^2 = \frac{1 + 9 + 9 + 25}{16}\varepsilon^2 = \frac{11}{4}\varepsilon^2. \tag{A.6}$$

so the constant is optimal. The coarse S8 value $\sqrt{23}/2$ remains valid. Finite-direction data require an additional argument to bound the entire-sphere supremum used here.

B Two Closure Theorems

B.1 Why an antilinear generator obstructs the common-phase quotient

At fixed reference complex structure let $\dot{b} = Ab + B\bar{b}$. For the representative $e^{i\alpha}b$ of the same projective state, the antilinear part of the projective derivative carries $e^{-2i\alpha}$. Identical projective derivatives for every α and $b \neq 0$ require

$$(I - P(b))B\bar{b} = 0 \quad \text{for all } b. \tag{B.1}$$

Taking $b = e_i$ gives $Be_i = \beta_i e_i$. Taking $b = e_i + e_j$ gives $\beta_i = \beta_j$, while $b = e_i + ie_j$ gives $\beta_i = -\beta_j$. In complex dimension at least two, every $\beta_i = 0$, so $B = 0$. If the flow also preserves the standard real symplectic form, its complex-linear generator obeys the corresponding anti-Hermitian condition. Dimension one lacks two independent directions, explaining the $\mathbb{CP}^0$ counterexample.

B.2 Why closure for every ensemble requires a Liouville-invariant subspace

For finitely many statistics $m_a(\mu) = \int f_a \, d\mu$, suppose future statistics are determined by m_a for every allowed point mass and mixture. Two measures with equal m_a must then also have equal $\int L_H f_a \, d\mu$, or their derivatives differ immediately. On this domain containing every finite mixture, the output is affine in the measure and is an affine combination in the same finite function space:

$$L_H f_a = c_a + \sum_b A_{ab} f_b. \tag{B.2}$$

Thus $\dot{m}_a = c_a + \sum_b A_{ab} m_b$. A nonlinear system continually generating new function products cannot close exactly in those finite expectations for all measures. A physical restriction to special measures may alter this, but their stability under evolution and record conditioning must first be proved.

C Decay Rates and Sign Conventions

C.1 Two-body decay

Equation (28.6) gives tree amplitude magnitude $|\mathcal{M}| = \lambda_4 |p_S \cdot k_V|$. For $S \to \phi V_4$, $|p_S \cdot k_V| = (m_4^2 - M_\phi^2)/2$, and phase space gives

$$\Gamma = \frac{|\mathcal{M}|^2}{16\pi m_4} \left(1 - \frac{M_\phi^2}{m_4^2}\right). \tag{C.1}$$

Substituting $\lambda_4^2 = 1/(15f^2)$ gives Eq. (28.7). The distinct final particles need no identical-particle factor.

C.2 Cross-sector three-body decay

Let $C_n = \lambda_4 \lambda_n / M_\phi^2$, with three distinguishable massless daughters of momenta k_1, k_2, k_3. Define positive invariant $q^2 = -(k_2 + k_3)^2$. The contact operator of Eq. (29.4) gives

$$|\mathcal{M}|^2 = \frac{C_n^2}{16}(q^2)^2(m_4^2 - q^2)^2, \qquad \frac{d\Phi_3}{dq^2} = \frac{m_4^2 - q^2}{128\pi^3 m_4^2}. \tag{C.2}$$

Hence

$$\begin{aligned}\Gamma &= \frac{C_n^2}{4096\pi^3 m_4^3}\int_0^{m_4^2} dq^2\,(q^2)^2(m_4^2-q^2)^3 \\ &= \frac{C_n^2 m_4^9}{245760\pi^3},\end{aligned} \tag{C.3}$$

The dimensionless integral is $\int_0^1 dx\,x^2(1-x)^3 = 1/60$. This is leading order in heavy-mediator expansion. For insufficiently small m_4/M_ϕ, restore $1/(M_\phi^2 - q^2)$; massive daughters also change thresholds and phase space.

C.3 Annihilation threshold and validity limits

In the present signature, $SS \to V_4V_4$ at threshold has $p_i^2 = -m_4^2$, $k_i^2 = 0$, $p_i \cdot k_j = -m_4^2$, and $t = u = -m_4^2$. Each exchange diagram contributes magnitude $\lambda_4^2 m_4^4/(M_\phi^2 + m_4^2)$; their sum gives Eq. (28.9), with 1/2! for identical final particles supplying its cross-section coefficient.

These are tree-level particle rates. Coherent condensates require perturbation theory about their backgrounds, and additional operators can alter amplitudes through interference. Particle lifetime, condensate stability, and symmetry protection are consequently separate tests.

D Source Map

D.1 Direct sources and thematic integration

The four direct sources are U, R, V, and H. The following map locates their material; historical studies contained in U and R retain their original attribution.

Source range	Retained material	Location
U: geometry, special relativity, sources, GR	Complex bilinearity, real slice, free shells, common-cone lemma, restricted variation, exact solutions, inverse reconstruction	Sections 3–15 and total field equations.
U: $c = 1, 2, 3$ and Maxwell	Response spectra, second-class constraints, composite counterexamples, independent connections, quotient group, electroweak/color	Sections 16–25.
U: $c = 4$, common mediator, cosmology	Generalized masses, two-/three-body decay, dark parity, misalignment, transverse perturbations, observations	Sections 28–31.
U: earlier alternatives	Bounce, plateaus, two four-forms, membrane energy accounting, $\mathbb{RP}^3$, Higgs production, vacuum response	Sections 32–34.
U: particle origins and amplitudes	Free Fock reference, quantum samples, composite-vector spectra, nine-derivative current, seagull correction, kernel limits	Sections 35, 36.

Source range	Retained material	Location
R: parent theories, QB/QP	Rank locking, counterterms, U route, Pauli/Dirac, projective structure, probability nonuniqueness	Sections 19, 21, 38–40.
R: MST I–XV	Thresholds, continuous records, environment spectra, covariant histories, state quotients, charge, latches	Sections 40–43.
R: XVI–XXV	Nonlinear phase feedback, discrete quotient, moment closure, finite-time low-energy bounds, Bell	Sections 44–47.
R: XXVI–XXXII and null cones	Dynamic odd references, history averages, mixed curvature, principal symbols, clock/signal reconstruction	Sections 5, 27, 30 and later statistical sections.
R: XXXIII–XXXVIII	Self-gravitating clocks, sharp common-clock ensembles, KLM, loop phases, KvN, exact probes	Sections 48–50.
V/H: P1/P2 v3.5	New statements, natural momentum, shell rigidity, real symplectic structure, Schemes A/B	Sections 3–9.
V/H: free fields and hundred-round review	Scalar counterexample, exact actions, nonuniqueness, Maxwell, Dirac, Jacobi, historical rounds	Sections 10–13; Appendix G.

D.2 Reading R's internal source labels S1–S9

Historical references retain R's S1–S9: S1 is the chat research chronicle; S2 the full-history synthesis; S3 the MST research continuation; S4 the MST dynamics continuation; S5 statistical foundations and physical records; S6 the XI–XXV synthesis; S7 XXVI–XXXII; S8 the null-cone first-principles study; and S9 XXXIII–XXXVIII. They are summarized by direct source R. The internal-geometry "U route" differs from source code U.

E Conventions, Revisions, and Computational Checks

E.1 Substantive changes from the earlier architecture to v3.5

Item	Treatment adopted here
Postulate names	P1: observers, inertial equivalence, complete common cone. P2: unified free shell, natural complex momentum, real canonical realization. Complex spacetime and the null cone are separate foundations.
Null-cone proof	Earlier consequences of "postulate two" are assigned to P1.2; the geometric definition does not assume $b = \chi g$ first.
Positive mass and imaginary conformal factor	Use $F^{-1}M^2$ with natural momentum; do not generalize the unmodified-mass obstruction to the whole theory.

Item	Treatment adopted here
Reality and symplectic structure	A directly pulls back the real one-form; B needs calibration. Constraint regularity and real multipliers are explicit.
Free fields	Add the naive complex KG counterexample, exact off-shell action identity, positive-weight nonuniqueness, and full variation order.
Dirac mass	Retain the constant-complex-mass result $m_0\sqrt{F}$ and add A's compatible $mF^{-1/2}$; they are different model choices.
Types of connection	Distinguish complex Levi–Civita, B's flat calibration, internal gauge, and quantum Berry connections.
Response involution	Here J_{CR} is divided by ρ and $J_{\mathrm{CR}}^2 = I$. Some H passages use an unnormalized matrix squaring to $\rho^2 I$.
Fundamental theory and EFT	CR retains its candidate fundamental-unification objective; EFT is the calculational and matching level of selected models.
Quantum level	Add Jacobi consistency for $\kappa(\phi)$, connected to internal-clock and KLM studies.
Historical materials	Keep H's hundred-round index distinct from U/R numbering; source syntheses are not independent reruns of all historical programs.

E.2 Computational checks and inherited results

The reconstruction checks Scheme A's shell identity, the real/imaginary constraint relation, the ten-point massive and nine-direction null tests, the nonconformal inverse-metric example, conformal volume weights, and the weighted-field redefinition. The text gives analytic arguments where numerical rank or sampling alone would be insufficient.

The accompanying calculations from the earlier synthesis check response spectra, Pauli multiplication, target curvature, the sharp cone bound, anomalies, decay phase space, the nine-derivative integral, and rational six-point sums. They also integrate the specified single-clock and common-clock examples. The six-point check sums source components rather than regenerating every diagram.

Historical simulation and data-analysis results are retained with their stated assumptions. This revision does not rerun the full diagram calculations, cosmological likelihoods, or apparatus simulations. It preserves the displayed mathematics while reorganizing the curvature derivation and editing the exposition; compilation, cross-references, and page layout are checked separately.

E.3 Self-contained compilation

The TeX is self-contained, including tables, plots, and bibliography. Compile it three times with XeLaTeX and the standard amsmath, titlesec, TikZ, and PGFPlots packages. Numbering belongs to this edition; cross-references update automatically.

F Development of P1/P2 through v3.5

H records the following progression to Scheme A v3.5. The stages mark changes in assumptions and constructions, not independent experimental validations.[H]

Version/stage	Main change	Required qualification
Original	P1: inertial equivalence preserving interval and reality; P2: $G^{-1}(P,P)+M^2=0$	Flat free kinematics; curved backgrounds require real momentum and mass terms to be specified.
v2.0	Strengthen local inertial structure in P1; add full real–complex shell compatibility to P2	Mass-shell compatibility is a new fundamental condition.
v2.1	One complete massive shell suffices; shared geometry can fix cross-species complex coefficients	Species consistency still needs full matter theory.
v2.2	Compress P2b to constancy of a complex quadratic form on one positive-mass shell	Equivalent to conformal rigidity in the specified quadratic class, not an independent derivation.
v3.0	Separate a mass-calibrated bridge and natural complexification into two rigorous levels	Frame covariance differs from active Lorentz invariance of a background.
v3.1	Integrate clock-carrying observers, inertial frames, and the common cone into strengthened P1	Complex-cone compatibility is an independent strengthening, not automatic from round-trip clock readings.
v3.2–v3.3, B	Preserve the original shell's literal form with $P=\sqrt{F}p^{\mathbb{C}}$	The complex canonical one-form needs calibration to preserve real Hamiltonian dynamics.
v3.4, A	Natural $P=p^{\mathbb{C}}$, mass term $F^{-1}M^2$, untwisted real one-form	The original complex shell is rigorously extended in curved backgrounds rather than kept literally unchanged.
v3.5, A	Separate conformal mass shells from real scalar, Maxwell, and spinor free-field realizations	Free-field existence leaves action selection and quantum completion open.

G Index of the Hundred Research Rounds in H

This index preserves H's summaries of rounds 1–100. H reconstructs the first ten from earlier summaries and reports the subsequent sequence directly. The entries refer to the assumptions and qualifications developed in the text, particularly reality conditions, effective propagation speeds, local spinor maps, and the strengthening of P1/P2.[H]

Round	Topic	Result, counterexample, or proof boundary
1	Inertial group and background	Full Lorentz transformations exist in the flat complex model; covariance does not require active invariance of a fixed imaginary metric.
2	Massless shell	If every real null direction is complex null, cone rigidity gives complex conformality; P2 alone does not ensure complete massless states.
3	Massive complex shell	Real $m>0$ momentum is incompatible with an unadjusted complex mass term on nonzero χ.

Round	Topic	Result, counterexample, or proof boundary (continued)
4	Free curves	Complex conformality preserves formal unparameterized null paths; timelike complex geodesics generally differ, requiring a real free action.
5	Holomorphy and analyticity	Real-analytic data admit local holomorphic lifts; arbitrary smooth solutions need not have holomorphic ambient geometry.
6	Restricted gravity	A selected complex EH real part gives Einstein gravity plus a normal real scalar; the original postulates do not uniquely select that action.
7	Response spectrum	Eigenvalues are $(c \pm 1)\rho$; allowed deformations demonstrate nonuniqueness under weak axioms.
8	Critical constraints	Rank-one contact constraints reduce on regular strata; zero strata and extendibility of linearization remain.
9	Internal gauge/dark sectors	Internal $3 + 2$ is compatible but nonunique; off-shell cross-sector decay can spoil isolated dark stability.
10	Quantum readout	Classical Hilbert/symplectic structure and internal measurement do not automatically give the physical Weyl algebra or Born rule.
11	Free fall	A common cone does not guarantee universal composite free fall; background masses can cause fifth forces.
12	Higher-derivative causality	Covariant X^2 interactions can change principal symbols and produce an effective cone beyond background g.
13	GR truncation	Normal-sector gradient sources can excite the scalar at $\phi = 0$; constant-scalar truncation need not be consistent.
14	Critical quantum protection	A small positive correction can raise rank one to two, requiring protection.
15	Complex phase and curvature	Direct $d \arg F$ is flat; seven-dimensional target curvature is nonzero and coordinate invariant.
16	Internal topology	Nontrivial $c_2(E_2)$ can obstruct an everywhere nonzero Higgs section.
17	Gauge degrees of freedom	Induced projected curvature lacks independent quadratic YM kinetics near a constant background.
18	Dark stability	Closed two-body decay does not close off-shell cross-sector three-body decay.
19	Cosmic bounce	A normal free scalar alone cannot bounce smoothly in flat cosmology; curvature and vacuum energy permit conditional bounces.
20	Quantum positivity	Finite Weyl-matrix positivity excludes sharp classical states; Jacobi constrains field-dependent action scales.
21	Original P1	Flat complex Lorentz realization works; covariance differs from active background invariance.
22	Nonconformal countermodel	Weak P1/P2 can admit nonconformal imaginary geometry and restricted free-momentum directions.
23	Unmodified mass obstruction	A complete real massive shell also obeying $q_\gamma + M^2 = 0$ requires $\gamma = g$.

Round	Topic	Result, counterexample, or proof boundary (continued)
24	Massive bridge M	One coefficient A compatible with the full positive-mass shell gives $\gamma^{-1} = Ag^{-1}$.
25	Massless bridge N	One common-null state and full active Lorentz-orbit closure imply a common cone.
26	Nine-direction test	Six axial and three mixed null directions determine $b = \chi g$.
27	Conformal Hamiltonian characteristics	$C_\gamma = F^{-1}C_g$ has compatible characteristic directions on the zero surface; real evolution needs a real representative.
28	Curved connections	Pointwise Lorentz frames and shells do not uniquely specify transport or the gravitational connection.
29	Indistinguishable imaginary scalar	Different constant χ can have identical real free kinematics.
30	Bridge theorems	Separate conditional M/N proofs and identify each added physical premise.
31	Active/passive P1	Equivalence of real Lorentz frames does not require active background invariance.
32	One massive species suffices	One complete positive-mass shell fixes the ratio of complex quadratic forms.
33	Ten-point massive test	Ten specified future massive momenta determine all quadratic coefficients.
34	Direction-dependent A	Arbitrary $A(p)$ compensates nonconformal differences; on-shell direction independence is essential.
35	Cross-species factor	One metric compatible with two complete shells requires the same momentum-independent coefficient.
36	Real Hamiltonian representative	$\mathrm{Re}(\overline{A}C_\mathbb{C})/\lvert A\rvert^2 = C_g$ supplies real canonical dynamics.
37	Postulate/cone equivalence	Shell bridging and complex conformality/common cones are equivalent within the specified quadratic class.
38	Real-inverse countermodel	An imaginary null rank-one perturbation gives correct real parts of metric and inverse but breaks the common cone.
39	Natural complex momentum	A maximally real slice naturally extends real momentum to complex cotangent space.
40	P1/P2 v2.1	Streamline to full local inertial equivalence and complex compatibility of a massive shell.
41	Real observations alone	Real clocks, rods, and geodesics do not uniquely identify the imaginary metric.
42	Same wavefront	One real phase satisfying both shells provides a bridge; its identity is an independent condition.
43	Multiplying complex actions	$S_\mathbb{C} = F(x)S_\mathrm{R}$ adds $S_\mathrm{R}dF$ and depends on an arbitrary phase constant.
44	On-shell constancy	A need only be direction independent on the physical shell; equivalent off-shell extensions remain.
45	Lorentz-orbit constancy	Active invariance of a fixed background forces constant shell calibration; covariance alone does not.
46	Constraint count	The full complex shell must not impose extra independent direction constraints on one particle.

Round	Topic	Result, counterexample, or proof boundary (continued)
47	Off-shell representatives	$C\exp(\beta C/\Lambda^2)$ and C have the same zero surface and on-shell Hamiltonian vector field.
48	Unmeasured constant chi	Identical free worldlines at different χ require matter couplings for discrimination.
49	Local analytic lift	Analytic g, χ admit local holomorphic $G(z)$, without guaranteed global gluing.
50	P1/P2 v2.2	Restate strengthened P2 as constant complex inertial calibration on one massive shell.
51	Full inertial group	Do not define all physical inertial frames by the background stabilizer; retain the passive/active distinction.
52	Original mass obstruction	Natural momentum plus an unadjusted complex mass term forces $\chi = 0$ for $m > 0$.
53	Real-free-experiment counterexample	Observations reading only g cannot exclude nonconformal imaginary geometry.
54	Complex calibration H	Constancy of $q_\gamma(p)$ on a complete positive-mass shell implies conformality.
55	P1 and calibration	Active background Lorentz invariance gives on-shell constancy, unlike ordinary covariance.
56	Open velocities suffice	Exact constancy on a nonempty open velocity region extends across the quadratic mass shell by analyticity.
57	Massive/massless bridges	Massive-shell constancy and complete massless common-cone conditions are equivalent conditional routes.
58	Real Hamiltonian completion	$C_{\mathrm{phys}} = C_g$ generates standard free trajectories.
59	Calibration observability	Free kinematics alone still cannot directly calibrate physical χ.
60	P1/P2 v3.0	Label strengthened real–complex shell compatibility P2b explicitly, not a consequence of old P2.
61	Observer versus frame	Clock-carrying observers may accelerate; local inertial frames describe instantaneous free kinematics.
62	Radar-time limits	Same-clock round trips do not uniquely fix one-way speed or the full directional cone.
63	Clocks, rods, cone	Complete null signals give a real conformal metric class; timelike clocks calibrate scale.
64	Common cone in P1	The strengthened P1 explicitly incorporates complex-null compatibility through observer/frame interaction.
65	Retaining original P2	Natural $P = p^{\mathbb{C}}$ with unchanged $+M^2$ conflicts with nonzero χ and positive mass.
66	B momentum scaling	$P = \sqrt{F}\,p^{\mathbb{C}}$ retains the original complex mass-shell form.
67	Arbitrary matrix lifts	Direction-dependent J can hide nonconformality; equivariance restrictions are needed.
68	Fixed Lorentz equivariance	A real-linear lift commuting with the prescribed natural complexified representation is a complex scalar.
69	B symplectic structure	$\lambda^{-1}P\,dx$ restores the real canonical form; original $P dx$ does not do so automatically.

Round	Topic	Result, counterexample, or proof boundary (continued)
70	P1/P2 v3.2	P1 covers observation/cones; P2 combines the original complex shell and a uniformly twisted momentum real form.
71	Twisted reality involution	B admits an antilinear reality condition differing from the natural cotangent real form.
72	Fixed representations	Equivariance excludes arbitrary J only when the complex Lorentz representation is fixed first.
73	Massive square root	Massive shells require $\lambda^2 = F$; massless shells do not fix normalization of λ.
74	B one-form obstruction	Uncalibrated Pdx is generally not real for $\chi \neq 0$.
75	Calibration connection	$\mathcal{A} = d \ln \lambda$ is a flat single-valued calibration one-form.
76	Complex-action counterexample	Multiplying a real first-order action by nonconstant λ can drive real momentum into complex directions.
77	B real action	Using $\lambda^{-1} Pdx$ recovers the standard g free action exactly.
78	Vertices and transport	Common λ preserves local total-momentum conservation; transport needs a calibrated covariant derivative.
79	Classical A/B equivalence	Matching real canonical structure gives identical real shells and free trajectories.
80	P1/P2 v3.3	B is conditionally consistent; an independent origin of its momentum real form remains open.
81	Natural complex momentum	A pairs complex momentum naturally and really with displacements on a maximally real slice.
82	Compatible mass coefficient	Complex conformality and complete positive-mass-shell compatibility uniquely fix A's $F^{-1}M^2$.
83	Rank-one constraint	Real and imaginary shell parts impose only one real constraint.
84	Original canonical form	A directly restricts Pdz to pdx, without B's calibration.
85	Real multipliers	Choose $e_{\mathbb{C}} = F e_{\mathrm{R}}$ or real representative $\mathrm{Re}(FC_A) = C_g$.
86	Real/complex geodesics	Timelike connections differ for nonconstant F; formal unparameterized null paths are compatible.
87	No direct fifth force	The minimal real free-particle action has no explicit χ; other couplings can generate scalar forces.
88	Complete wave operators	Minimal complex $\Box_\gamma$ has the extra first-order term $\nabla \ln F \cdot \nabla$.
89	Global and quantum measures	F^{-1} has no zeros; identical classical zero surfaces do not determine a quantum measure.
90	P1/P2 v3.4	Structural economy favors A as a baseline; complete free-field actions remained a research task at this stage.
91	HJ and characteristics	Massive WKB Hamilton–Jacobi shells differ from highest-order PDE characteristic null cones.
92	Complex KG for real fields	The imaginary equation requires $\nabla\chi \cdot \nabla\varphi = 0$.
93	Real scalar action	$\mathrm{Re} \int \sqrt{-\gamma}[\gamma^{-1}(d\varphi)^2 + F^{-1}m^2\varphi^2]$ recovers real KG exactly.
94	Action nonuniqueness	Weight $1 + \alpha\chi^2$ preserves principal cone and WKB shell but changes the full field equation.
95	Reference-free-field principle	Fixing the physical readout field and removing direct $\nabla\chi$ coupling selects $w = 1$ within a specified weight family.

Round	Topic	Result, counterexample, or proof boundary (continued)
96	Stress and shared source	The reference scalar has normal energy and on-shell conservation; no direct χ source does not mean no coupling in full CR.
97	Maxwell cancellation	Four-dimensional volume and inverse-metric weights cancel, recovering the standard free Maxwell action.
98	Local Dirac map	$\Psi = F^{-3/4}\psi$, $m_\gamma = mF^{-1/2}$ map to the real Dirac equation.
99	Real field symplectic current	KG has a classical symplectic current; local common cones do not guarantee global hyperbolicity or a unique quantum state.
100	P1/P2 v3.5	A constructs free fields of several spins; unique cross-species action selection remains missing.

H Notation and Model Assumptions

Symbol	Meaning used here
g, γ	Real physical Lorentz metric; restricted complex metric $\gamma = Fg$.
F, χ, ρ	$F = 1 + \mathrm{i}\chi$; real shared variable; $\rho = \lvert F\rvert = \sqrt{1+\chi^2}$.
ϕ, f	Canonical shared scalar $\phi = f \operatorname{arsinh}\chi$; $f^2 = 3M_P^2/2$.
c, n, c_{light}	Continuous response parameter; selected integer sector; speed of light.
G_n, G_n, K_n	Original response; reference-normalized response; G_n^{-1}. S9's h_n is written G_n.
U_n, V_n, λ_n	Reference canonical responses; $\lambda_n = 1/[f\sqrt{n^2-1}]$.
X, Y, w, D, N	Critical coordinates, contact one-form, constraint quantities. $D > 0$ denotes the regular branch.
$\mathcal{K}_n$	Canonical quadratic kinetic-energy random variable, not inverse matrix K_n.
$b, P(b), \varrho$	Reference complex amplitude, rank-one projector, candidate density matrix. Geometric ρ is not a density matrix.
D, F_b	Trajectory-wise normal and anomalous quadratic moments, distinct by context from geometric F and critical constraints.
$\Pi, K_r, \mathcal{I}_r$	Measure-to-state compression, actual record kernel, candidate effective instrument.
$J(\omega), \gamma(t)$	Environment spectrum and memory kernel; time-dependent $\gamma(t)$ here is not the complex metric.
$R_{\mathrm{tar}}, R[g]$	Normal-matter target scalar curvature; spacetime Ricci scalar.
Θ, ϑ	Unwrapped internal-clock phase and its specified reading; the complex factor's phase is defined separately where used.
$\kappa, \hbar$	Candidate quantum action scale requiring explanation; experimental action constant. Natural units do not derive it.
$Q_{\mathrm{read}}, P_{\mathrm{read}}; Q, P$	Actual classical multiplication readouts in the KvN example; auxiliary Weyl generators on the same space.

Spacetime, response targets, and quantum state spaces have different dimensions and roles. Fixed-background, finite-mode, and self-gravitating calculations also have different domains of va-

lidity. Peierls brackets, Weyl algebras, KLM positivity, Koopman–von Neumann theory, geometric quantization, and Bell inequalities are established tools; the CR results here are their applications and tests for the specified actions.

Direct Sources

[U] *CR Unified Geometry and the c Sectors*. Research manuscript. `CR_Unified_Geometry_and_c_Sectors.tex`. Collects the geometric, gauge, dark-sector, cosmological, and microscopic studies cited here.

[R] *CR Integrated Research Synthesis*, 1 November 2026. `CR_Integrated_Research_Synthesis_2026-11-01.tex`. Covers the chat chronicle and full-history synthesis, QB/QP, MST I–XXV, XXVI–XXXII, the null-cone study, and XXXIII–XXXVIII.

[V] *CR P1/P2 v3.5 Review*, 2 November 2026. `CR_P1P2_v3_5_Review_Article_2026-11-02.tex`. P1/P2 v3.5, Schemes A/B, momentum reality, real constraints, and free fields.

[H] *CR Original Postulates and One Hundred Rounds of Research: Comprehensive Report*, 2 November 2026. `CR_P1P2_100_Rounds_Comprehensive_2026-11-02.tex`. Research history, postulate revisions, constructions, and counterexamples. Earlier discussions and screenshots are cited through this account.